CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :

- Dual nature of radiation, Photoelectric effect, Hertz and Lenard’s observations;

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :

- Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations;
- Einstein's photoelectric equation-particle nature of light.

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :

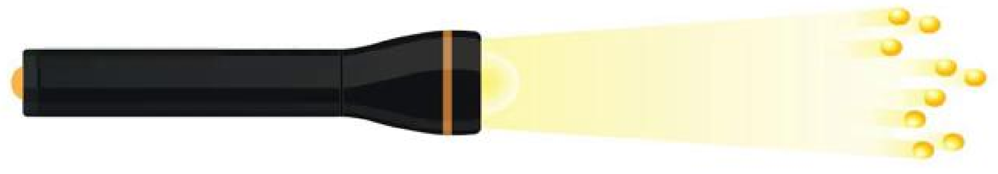
- Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations;
- Einstein's photoelectric equation-particle nature of light.
- Experimental study of photoelectric effect

CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

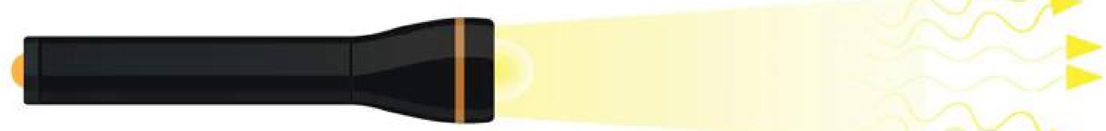
Learning Objectives :

- Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations;
- Einstein's photoelectric equation-particle nature of light.
- Experimental study of photoelectric effect
- Matter waves-wave nature of particles, de-Broglie relation.

DUAL NATURE OF RADIATION AND MATTER

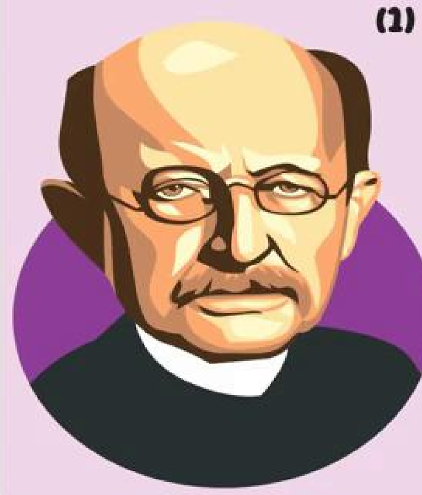


Particles



Waves

PLANCK'S QUANTUM THEORY OF LIGHT



(1) The energy of one photon is proportional to its frequency
(2) $E \propto \nu$. $E = h\nu$
h = Plank's constant
= 6.62×10^{-34} JS
(3) Energy of any light or radiation is one integral multiple of $h\nu$.
$E = Nh\nu$
(4) Energy of one photon.
$E = h\nu = \dfrac{1240}{\pi(\text{nm})}\ \text{eV}$

PROPERTIES OF PHOTONS

(1) Photon is just a packet of energy.
(2) Energy of photon does not change with medium.
(3) Photon can not be deflected by electric field and magnetic field.
(4) Momentum of photon $|\vec{P}| = m \times c = \dfrac{h}{\lambda} = \dfrac{E}{C}$
(5) Intensity of light beam = $\dfrac{\text{Energy}}{\text{area} \times \text{time}}$

PHOTON EMITTED PER SECOND

(1) $E = Nh\nu$
(2) Power, $P = Nh\nu$
$\Rightarrow = \dfrac{P}{h\nu} = \dfrac{P\lambda}{hc}$

Number of photon per second = $\dfrac{\text{Power}}{\text{energy of one Photon}}$

PHOTON FLUX

- Photon flux is no. of photon incident normally to a surface per seconds
- $\phi = \dfrac{n}{A} = \dfrac{P}{h\nu}$

FORCE AND RADIATION PRESSURE EXERTAD BY A LIGHT BEAM

(1) $n = \dfrac{P\lambda}{hc}$
(2) momentum of one photon = $\dfrac{h}{\lambda}$ momentum imparted per second = $n \times \dfrac{h}{\lambda} = \dfrac{P}{C}$
$\Rightarrow$ Force exerted = $\dfrac{P}{C}$
(3) Radiation pressure = $\dfrac{F}{A} = \dfrac{P}{CA} = \dfrac{I}{C}$

PHOTOELECTRIC EFFECT

(1) It is a phenomenon of ejecting electrons by falling light of suitable frequency on a metal
(2) Ejected electrons are called photoelectrons.
(3) Current flowing due to the photoelectrons is called photocentric current

LAWS

(1) No emission takes place below the threshold frequency.
(2) Above threshold frequency, no. of photoelectrons emitted per seconds is directly proportional to intensity of radiation
(3) The emission of photoelectrons is an instantaneous process.
(4) Above threshold frequency, K.E (max) depends on frequency

WORK FUNCTION

- Minimum energy required for getting a free electron away from the metal surface.
- work function $(\phi_o) = h\nu_o$

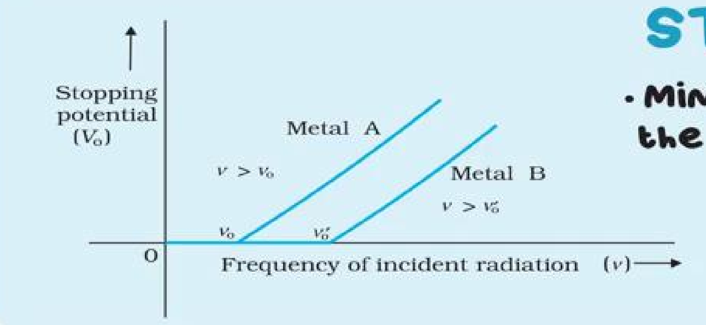
$\nu_o = \dfrac{\phi}{h}$ = threshold frequency

EINSTEIN'S PHOTOELECTRIC EQUATION

- The electron is emitted with maximum K.E
K.Emax = $h\nu - \phi_o$
$h\nu$ = K.Emax + ϕ_o
- Range of K.E.

$0 \le K.E._{\text{photoelectrons}} \le h\nu - \phi_o$

STOPPING POTENTIAL

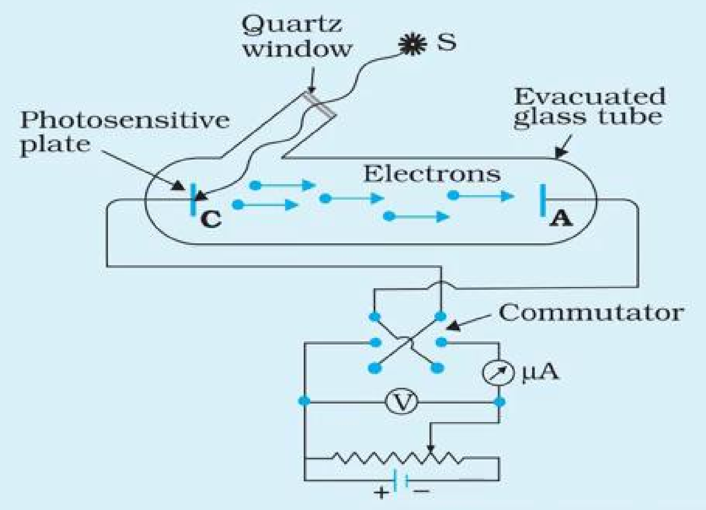


- Minimum negative potential required to stop the electron of maximum K.E.

$V_o = \dfrac{K.E_{max}}{e} = \dfrac{h}{e}(\nu - \nu_o)$ Volts

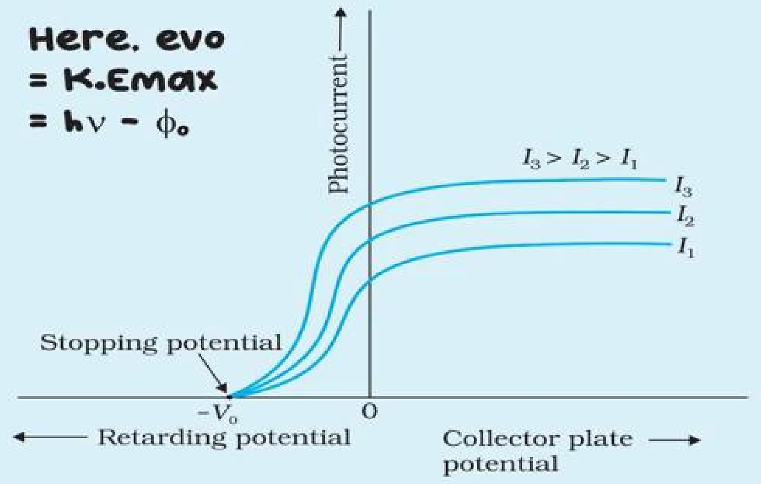
EXPERIMENTAL STUDY

- The emission of electrons causes flow of electric current in the circuit.

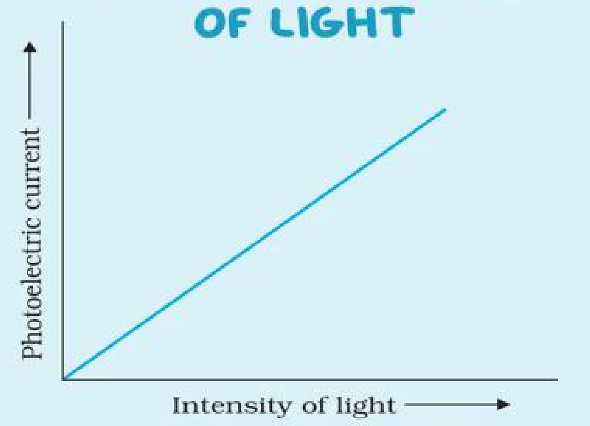


EFFECT OF POTENTIAL

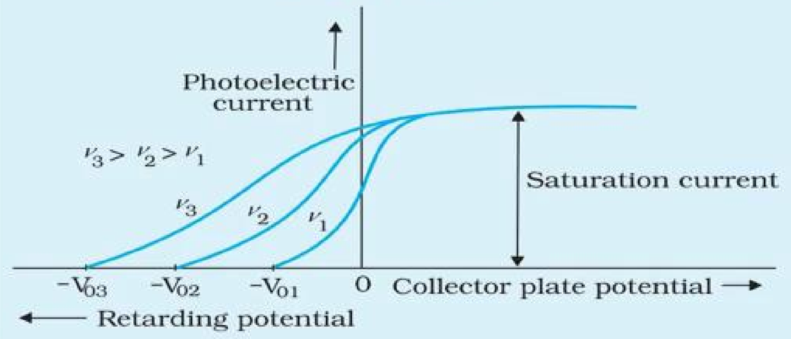
Here, $e\nu_o$ = K.Emax = $h\nu - \phi_o$



EFFECT OF INTENSITY OF LIGHT



EFFECT OF FREQUENCY OF INCIDENT



DUAL NATURE OF LIGHT

de – Broglie wavelength, $\lambda = \dfrac{h}{m\nu}$ & $2\pi r = n\lambda$

- $m\nu r = \dfrac{nh}{2\pi}$

This is Bohr quantisation condition

PARTICLE NATURE OF LIGHT

(1) In interaction of radiation with matter, radiation behaves as if is mode of particles called photons
(2) $E = h\nu$ and $p = \dfrac{h\nu}{c}$
(3) In a photon – particle collision, total energy and total momentum are conserved.

MATTER WAVE THEORY

- de – Broglie wavelength associated with moving particles. $\lambda = \dfrac{h}{p}$
- K.E of particle = $\dfrac{1}{2}m\nu^2 = \dfrac{p^2}{2m}$
- momentum, $p = m\nu$
$\sqrt{2m \times K.E}$

FOR UNCHARGED PARTICLES

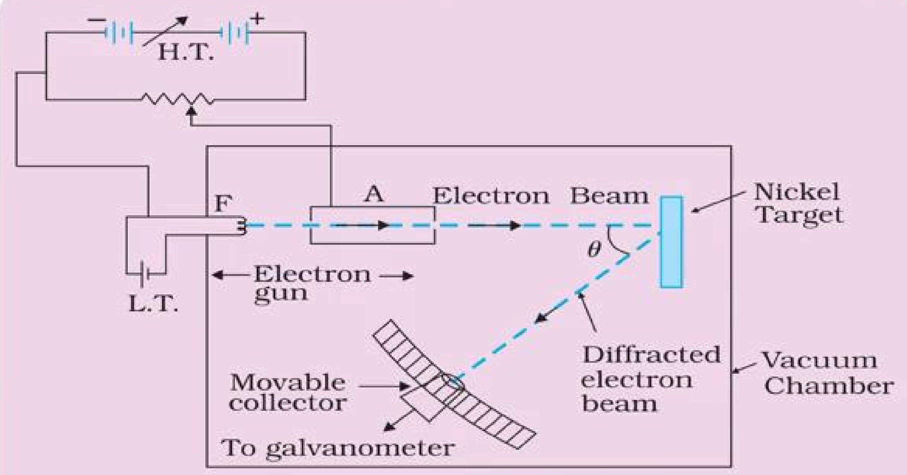
$\lambda = \dfrac{h}{m\nu} = \dfrac{h}{\sqrt{2m \times K.E}}$

FOR ACCELERATED CHARGED PARTICLES

$\lambda = \dfrac{h}{\sqrt{2m \times K.E}}$

V = potential difference

DAVISSON-GERMER EXPERIMENT



- at ϕ = 50° and accelerating potential = 54 V, maxima is obtained
- This experiment confirmed the wave nature of electron.

SPECIAL CASE FOR ELECTRON

$\lambda = \dfrac{1.227}{\sqrt{V}}$ nm

And

$V = \dfrac{150.1}{[\lambda(\text{Å})]^2}$ Volt

FOR GASEOUS MOLECULES

$K.E = \dfrac{3}{2}KT$

$\Rightarrow \lambda = \dfrac{h}{\sqrt{2m \times \dfrac{3}{2}KT}}$

$\Rightarrow \lambda = \dfrac{h}{\sqrt{3mKT}}$

INTRODUCTION

⭐ Wave Nature of Light Was Well Established

* By the end of the 19th century, light was widely accepted as a wave.

INTRODUCTION

⭐ Wave Nature of Light Was Well Established

* By the end of the 19th century, light was widely accepted as a wave.

* Maxwell’s equations predicted electromagnetic waves.

INTRODUCTION

⭐ Wave Nature of Light Was Well Established

* By the end of the 19th century, light was widely accepted as a wave.

* Maxwell’s equations predicted electromagnetic waves.

* Hertz’s experiments (1887) confirmed the existence of electromagnetic waves.

* Thus, the wave theory of light was strongly supported.

INTRODUCTION

⭐ Important Discoveries from Discharge Tube Experiments

Experiments on electric discharge through gases at low pressure (≈ 0.001 mm Hg) led to major discoveries:

INTRODUCTION

⭐ Important Discoveries from Discharge Tube Experiments

Experiments on electric discharge through gases at low pressure (≈ 0.001 mm Hg) led to major discoveries:

(i) X-rays — Roentgen (1895)

* X-rays were discovered during experiments on gas discharge.

* This became an important tool for understanding atomic structure.

(iii) Electron — J.J. Thomson (1897)

* Thomson confirmed cathode rays are made of negatively charged particles (electrons).
* He used crossed electric and magnetic fields to determine:
* Speed of electrons: about 0.1–0.2 times the speed of light
* e/m ratio: 1.76×10^{11} C/kg
* The e/m ratio was the same regardless of the gas or metal used.

✔ This proved electrons are universal constituents of matter.
Thomson was awarded the Nobel Prize in 1906.

INTRODUCTION

⭐ Important Discoveries from Discharge Tube Experiments

Experiments on electric discharge through gases at low pressure (≈ 0.001 mm Hg) led to major discoveries:

(i) X-rays — Roentgen (1895)

* X-rays were discovered during experiments on gas discharge.

* This became an important tool for understanding atomic structure.

(ii) Cathode Rays — Crookes (1870s)

* A yellowish-green fluorescence appeared on the glass opposite the cathode.

* This showed the presence of rays emerging from the cathode → cathode rays.

* Crookes suggested these rays were streams of fast-moving, negatively charged particles.

(iii) Electron — J.J. Thomson (1897)

* Thomson confirmed cathode rays are made of negatively charged particles (electrons).
* He used crossed electric and magnetic fields to determine:
* Speed of electrons: about 0.1–0.2 times the speed of light
* e/m ratio: 1.76×10^{11} C/kg
* The e/m ratio was the same regardless of the gas or metal used.

✔ This proved electrons are universal constituents of matter.
Thomson was awarded the Nobel Prize in 1906.

INTRODUCTION

⭐ Important Discoveries from Discharge Tube Experiments

Experiments on electric discharge through gases at low pressure (≈ 0.001 mm Hg) led to major discoveries:

(i) X-rays — Roentgen (1895)

* X-rays were discovered during experiments on gas discharge.

* This became an important tool for understanding atomic structure.

(ii) Cathode Rays — Crookes (1870s)

* A yellowish-green fluorescence appeared on the glass opposite the cathode.

* This showed the presence of rays emerging from the cathode → cathode rays.

* Crookes suggested these rays were streams of fast-moving, negatively charged particles.

(iii) Electron — J.J. Thomson (1897)

* Thomson confirmed cathode rays are made of negatively charged particles (electrons).
* He used crossed electric and magnetic fields to determine:
* Speed of electrons: about 0.1–0.2 times the speed of light
* e/m ratio: 1.76×10^{11} C/kg
* The e/m ratio was the same regardless of the gas or metal used.

✔ This proved electrons are universal constituents of matter.
Thomson was awarded the Nobel Prize in 1906.

INTRODUCTION

⭐ More Evidence Supporting the Existence of Electrons

* Electrons were found to be emitted when metals were:

* Heated (thermionic emission)

* Irradiated with ultraviolet light (photoelectric emission)

* In every case, the emitted particles had the same e/m as cathode-ray particles.

✔ This confirmed that the same electron is produced in different conditions.

INTRODUCTION

⭐ More Evidence Supporting the Existence of Electrons

* Electrons were found to be emitted when metals were:

* Heated (thermionic emission)

* Irradiated with ultraviolet light (photoelectric emission)

* In every case, the emitted particles had the same e/m as cathode-ray particles.

✔ This confirmed that the same electron is produced in different conditions.

⭐ Millikan’s Oil Drop Experiment (1913)

R.A. Millikan measured the charge of the electron accurately.

Found that: Charge (e) = 1.602×10^{-19} C

Charge appears only in integral multiples of this value → charge is quantised.

Using e and e/m, the mass of the electron was calculated.

ELECTRON EMISSION

⭐ Meaning of Electron Emission

* Metals contain free electrons that are responsible for electrical conductivity.
* These free electrons cannot normally escape from the metal surface.
* When an electron tries to leave, the metal surface becomes positively charged and pulls the electron back.
* An electron can come out of the metal only when it gains sufficient energy to overcome this attractive force.
* The minimum energy required for an electron to escape from the metal surface is called the work function.

⭐ 2. Work Function (ϕ_0)

* Definition: The minimum energy required by an electron to escape from the metal surface.
* Unit: electron volt (eV) $1eV = 1.602 \times 10^{-19}$ J

* The value of ϕ_0 depends on the metal and the nature of its surface.

* Different metals have different work functions.
* Example :

  Cesium (Cs): 2.14 eV

  Platinum (Pt): 5.65 eV — highest among the listed metals.

ELECTRON EMISSION

TABLE 11.1 WORK FUNCTIONS OF SOME METALS

Metal	Work function ϕ_o (eV)	Metal	Work function ϕ_o (eV)
Cs	2.14	Al	4.28
K	2.30	Hg	4.49
Na	2.75	Cu	4.65
Ca	3.20	Ag	4.70
Mo	4.17	Ni	5.15
Pb	4.25	Pt	5.65

ELECTRON EMISSION

⭐ Ways to Supply Energy for Electron Emission

An electron can be emitted from a metal surface when enough energy is supplied by one of the following methods:

ELECTRON EMISSION

⭐ Ways to Supply Energy for Electron Emission

An electron can be emitted from a metal surface when enough energy is supplied by one of the following methods:

(i) Thermionic Emission

* Electrons are emitted by heating the metal.
* Heating gives enough thermal energy for electrons to overcome the work function.
* Used in: vacuum tubes, cathode ray tubes, electron guns

ELECTRON EMISSION

⭐ Ways to Supply Energy for Electron Emission

An electron can be emitted from a metal surface when enough energy is supplied by one of the following methods:

(i) Thermionic Emission

* Electrons are emitted by heating the metal.
* Heating gives enough thermal energy for electrons to overcome the work function.
* Used in: vacuum tubes, cathode ray tubes, electron guns

(ii) Field Emission

* Achieved by applying a very strong electric field ($\approx 10^8\ Vm^{-1}$)
* The strong field pulls electrons out of the metal surface.
* Used in: spark plugs, field emission microscopes.

ELECTRON EMISSION

⭐ Ways to Supply Energy for Electron Emission

An electron can be emitted from a metal surface when enough energy is supplied by one of the following methods:

(i) Thermionic Emission

* Electrons are emitted by heating the metal.
* Heating gives enough thermal energy for electrons to overcome the work function.
* Used in: vacuum tubes, cathode ray tubes, electron guns

(ii) Field Emission

* Achieved by applying a very strong electric field ($\approx 10^8\ Vm^{-1}$)
* The strong field pulls electrons out of the metal surface.
* Used in: spark plugs, field emission microscopes.

(iii) Photoelectric Emission

* Electrons are emitted when light of suitable frequency falls on the metal surface.
* The emitted electrons are called photoelectrons.
* The minimum frequency required is called threshold frequency (ν_0).

PHOTOELECTRIC EFFECT

* When light of suitable frequency falls on a metal surface, electrons are emitted.

* The emitted electrons are called photoelectrons.

* The phenomenon is called the photoelectric effect.

PHOTOELECTRIC EFFECT

(1) Hertz's observations

PHOTOELECTRIC EFFECT

(1) Hertz's observations

* Hertz (1887) discovered the photoelectric effect accidentally during experiments on electromagnetic waves.

* He observed that sparks across a detector loop became stronger when the emitter plate was illuminated with ultraviolet light.

* UV light helped electrons escape from the metal surface, allowing the spark discharge to occur more easily.

* When light shines on a metal, electrons near the surface absorb energy from incident light.

* If the absorbed energy is greater than the work function, electrons escape from the metal surface.

* Conclusion: Light can cause the emission of charged particles (electrons) from a metal surface.

PHOTOELECTRIC EFFECT

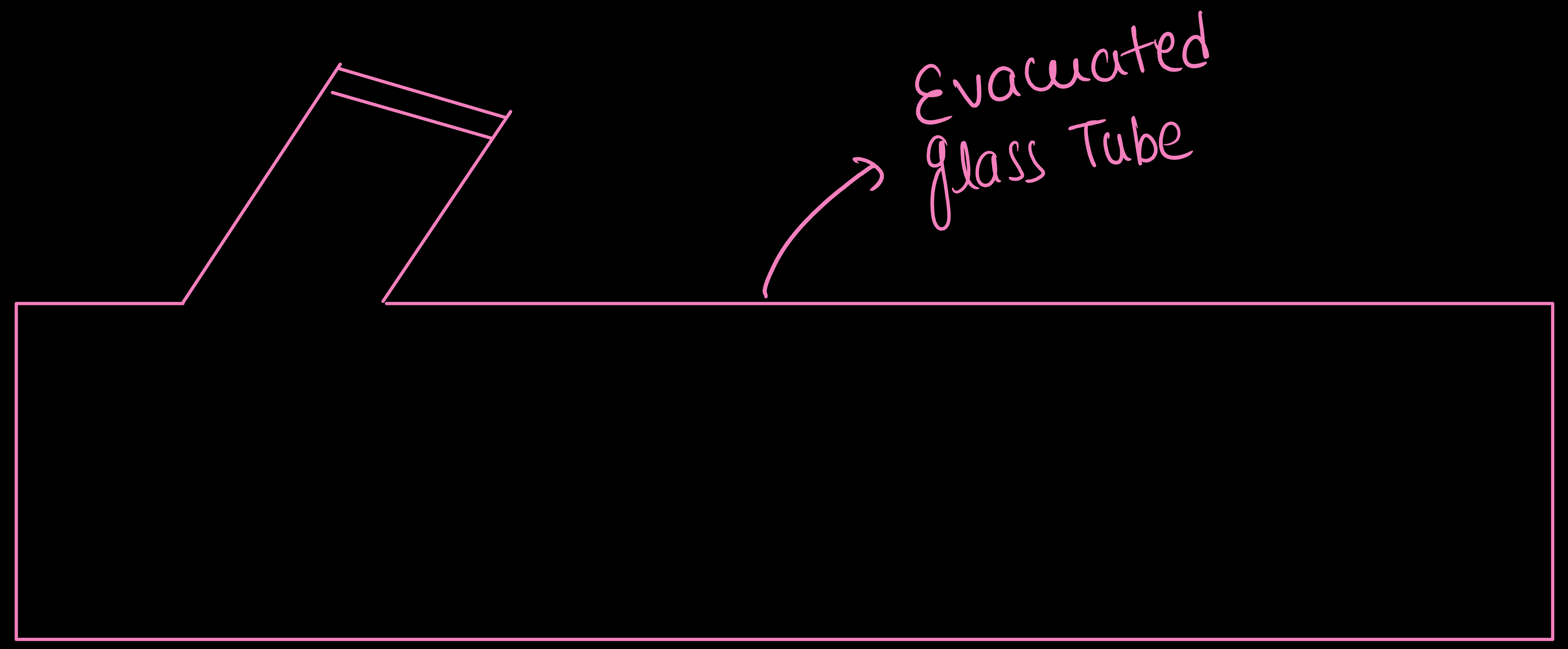
(2) Hallwachs' and Lenard's observations

Lenard's Observations :

PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :



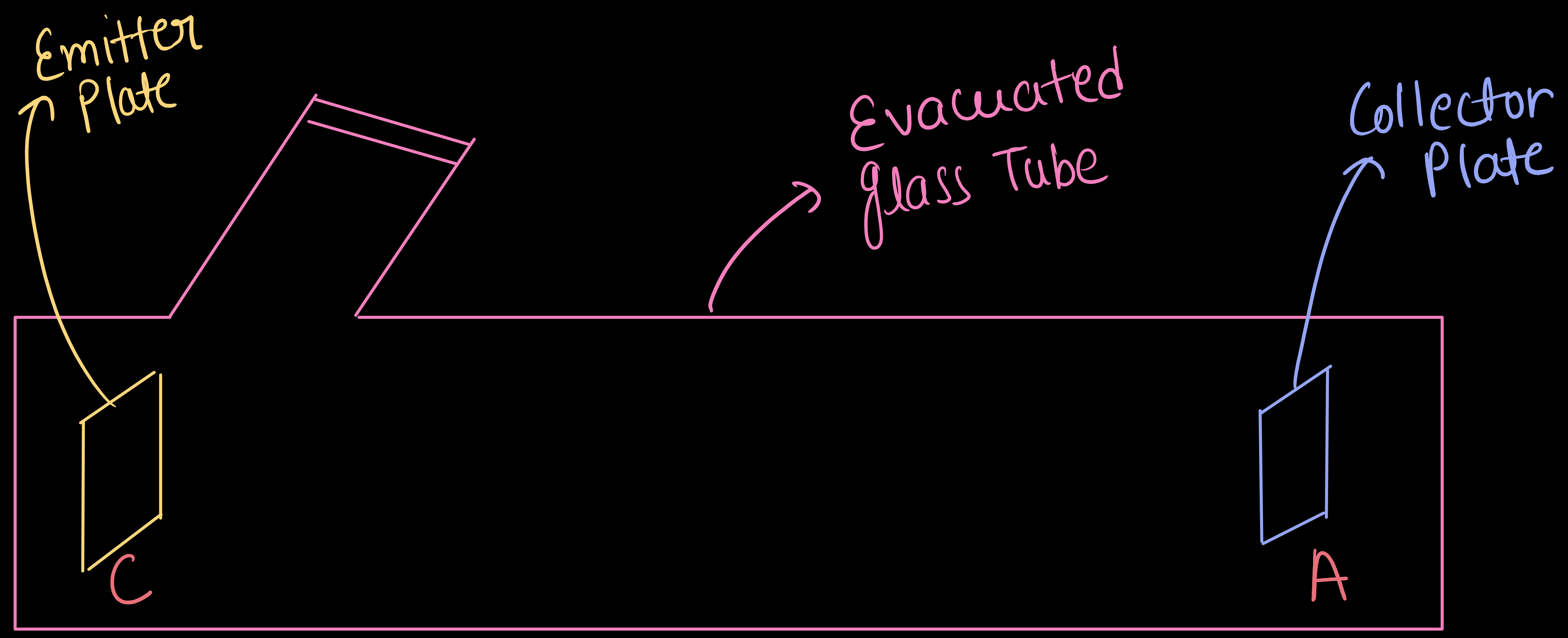
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

* Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)



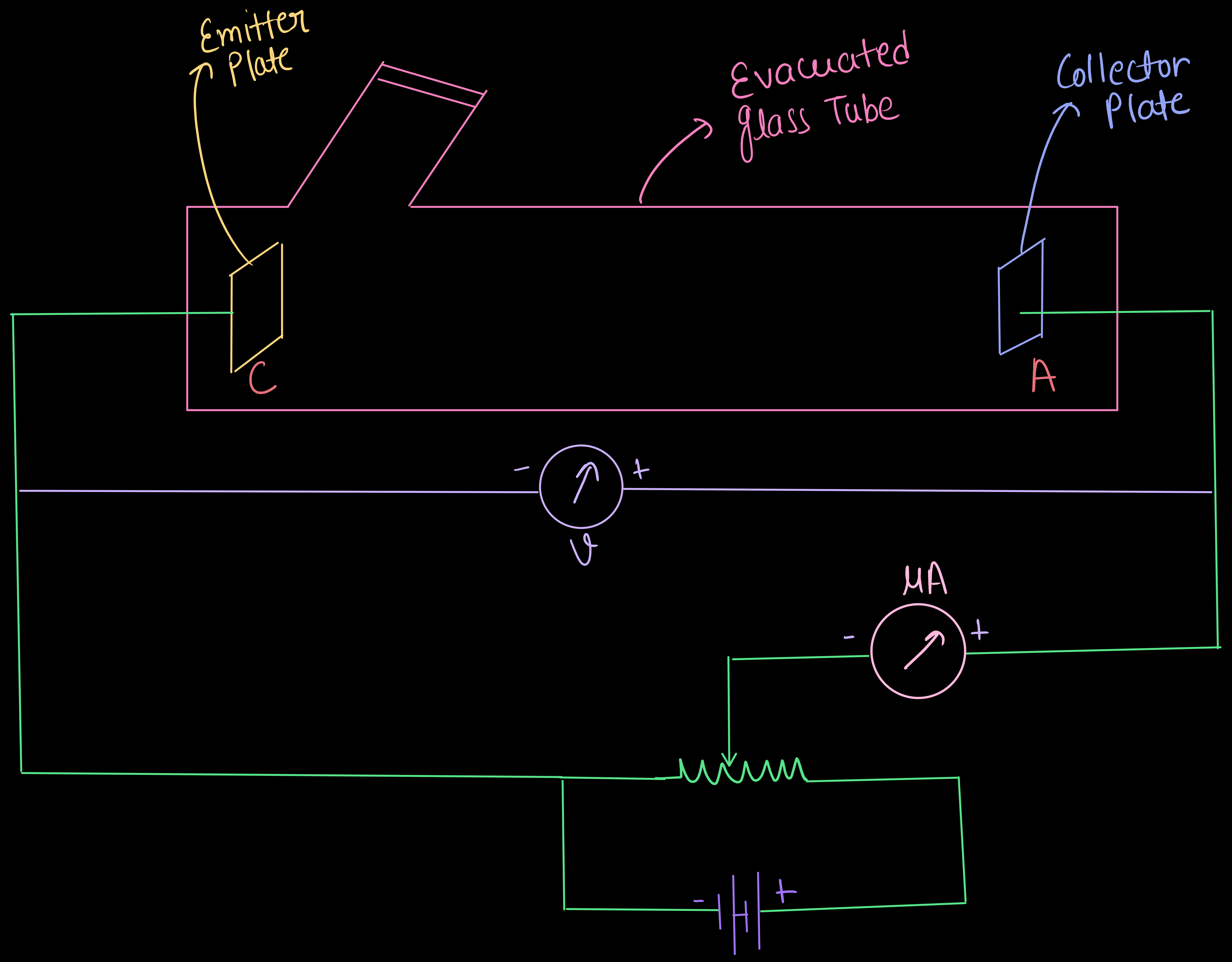
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

* Lenard used an evacuated glass tube with:

  Emitter plate (C)
  Collector plate (A)



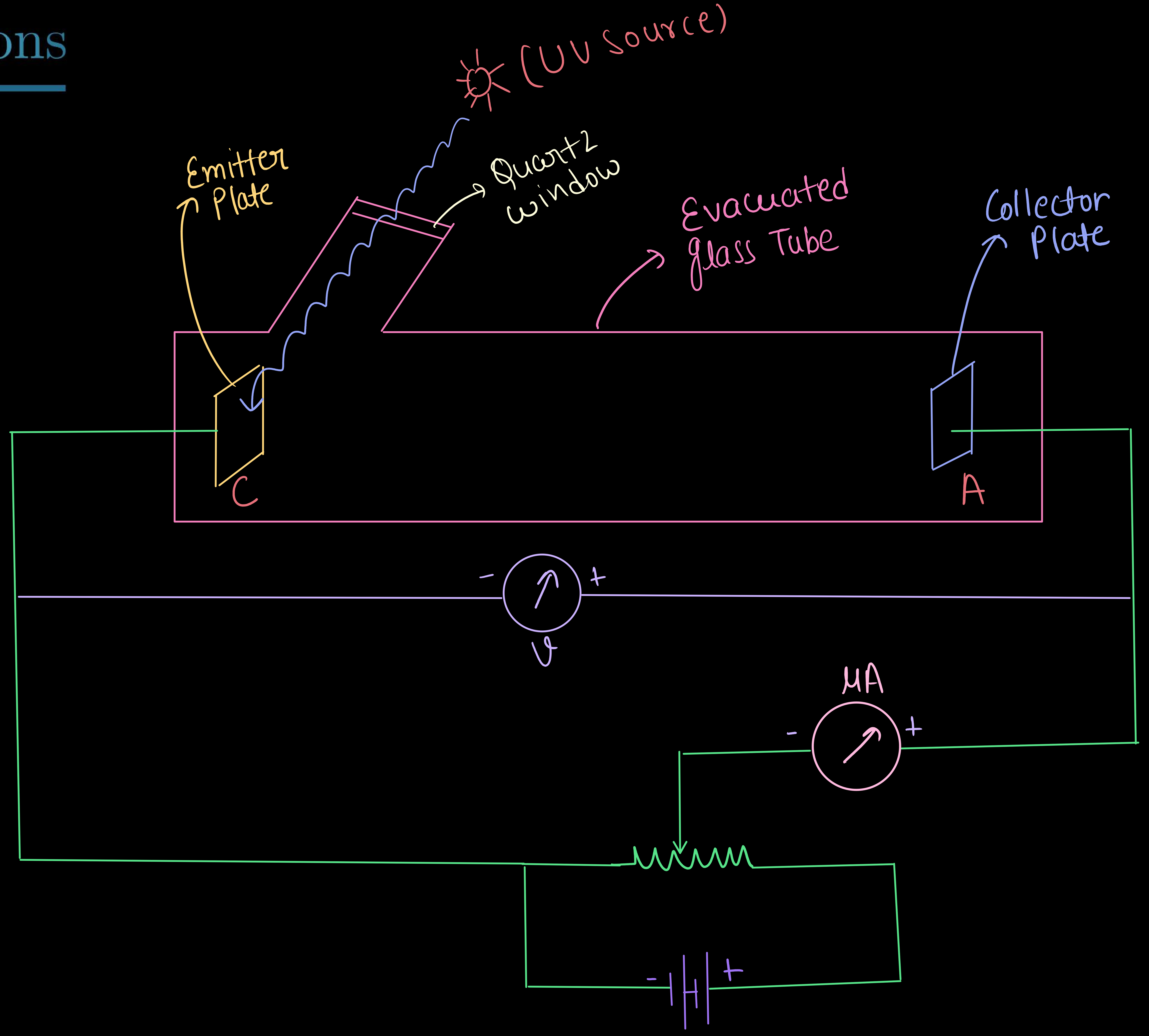
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

* Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

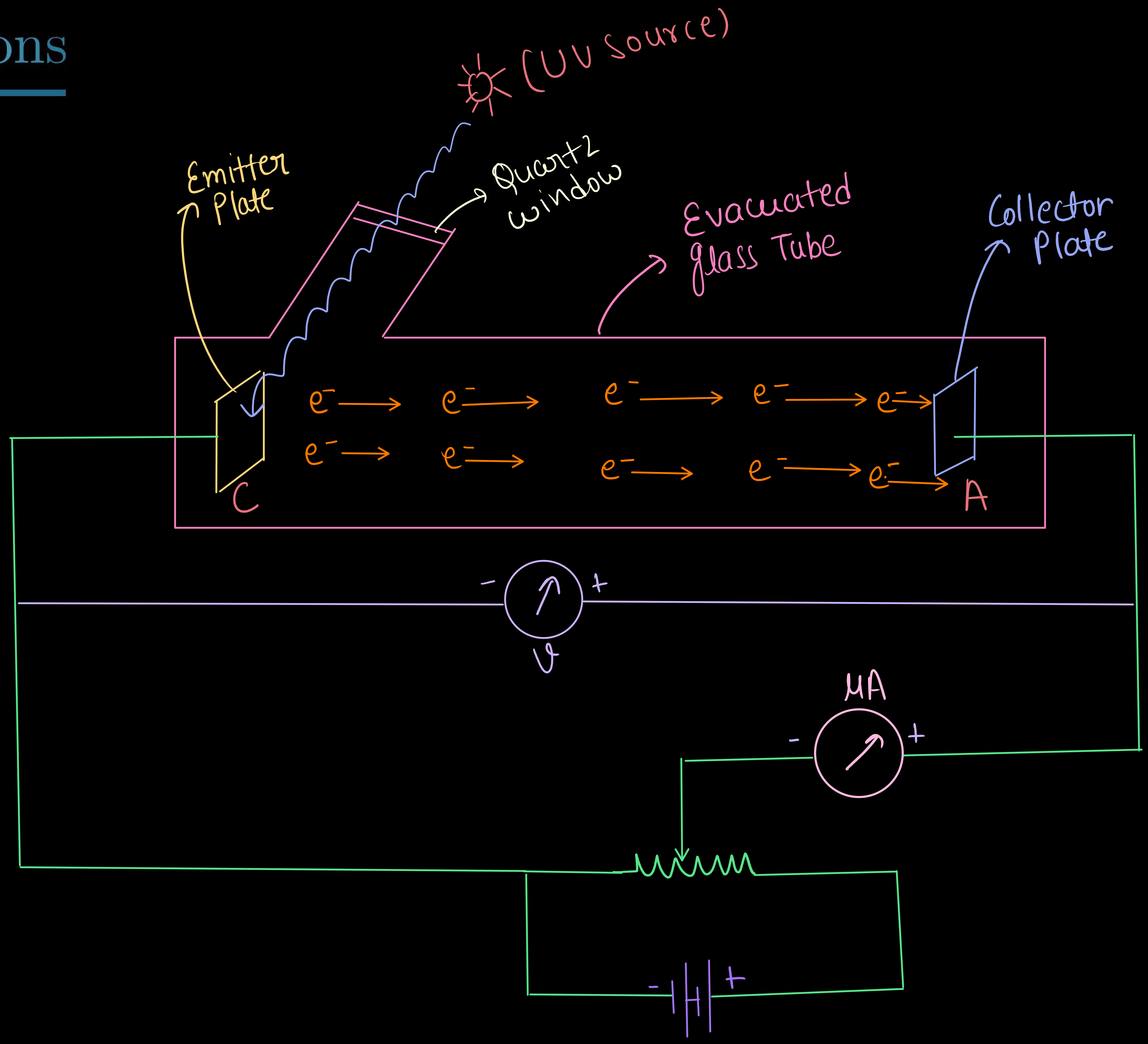
* Lenard used an evacuated glass tube with:

  Emitter plate (C)
  Collector plate (A)

* When UV light fell on plate C:

  * Electrons were emitted and attracted to plate A.

  * A current flowed in the external circuit (photoelectric current).



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

* Lenard used an evacuated glass tube with:

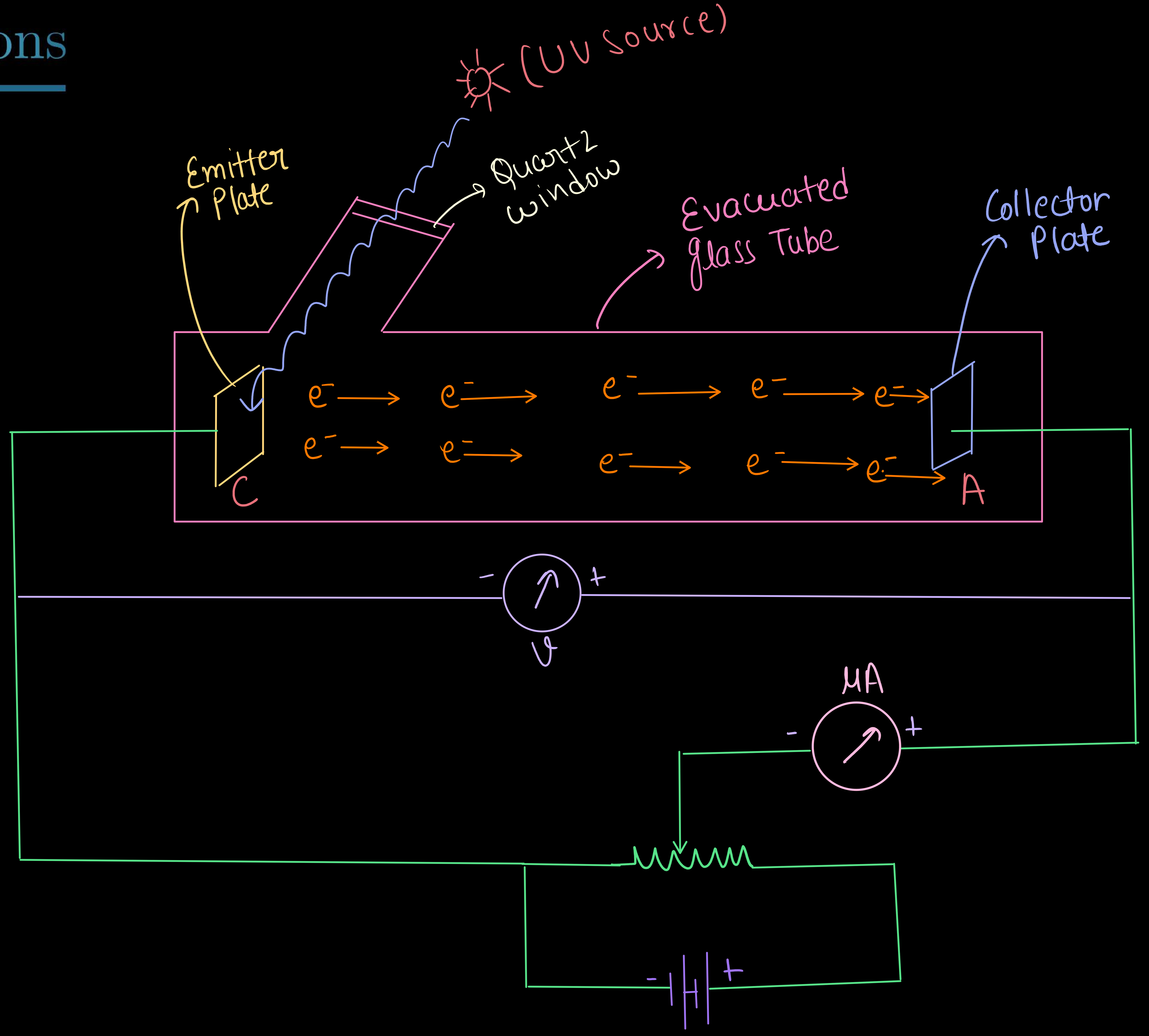
  Emitter plate (C)
  Collector plate (A)

* When UV light fell on plate C:

  * Electrons were emitted and attracted to plate A.

  * A current flowed in the external circuit (photoelectric current).

  * Current stopped immediately when UV light was switched off.



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

Lenard's Observations :

* Lenard used an evacuated glass tube with:

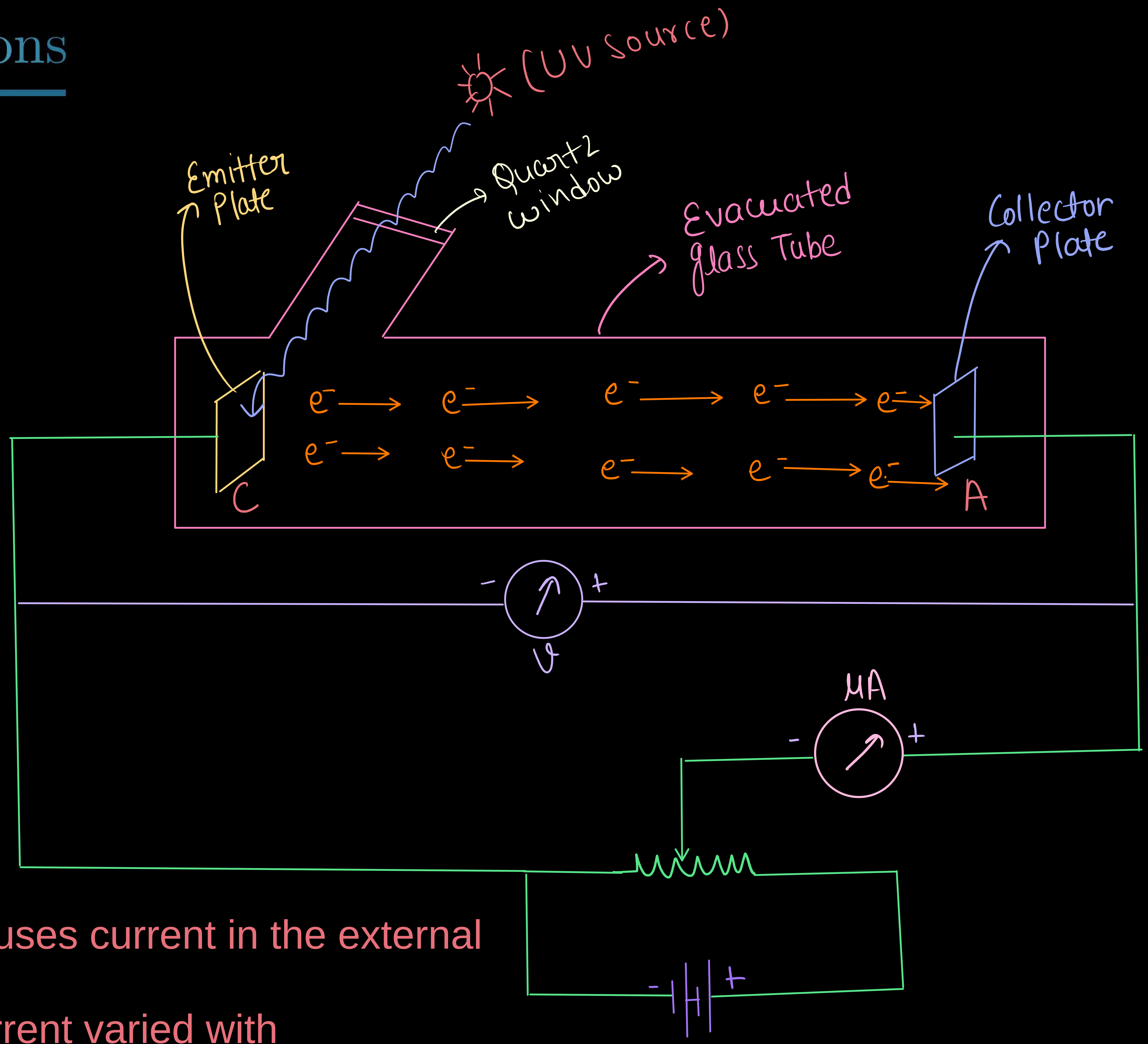
  Emitter plate (C)
  Collector plate (A)

* When UV light fell on plate C:

  * Electrons were emitted and attracted to plate A.

  * A current flowed in the external circuit (photoelectric current).

  * Current stopped immediately when UV light was switched off.

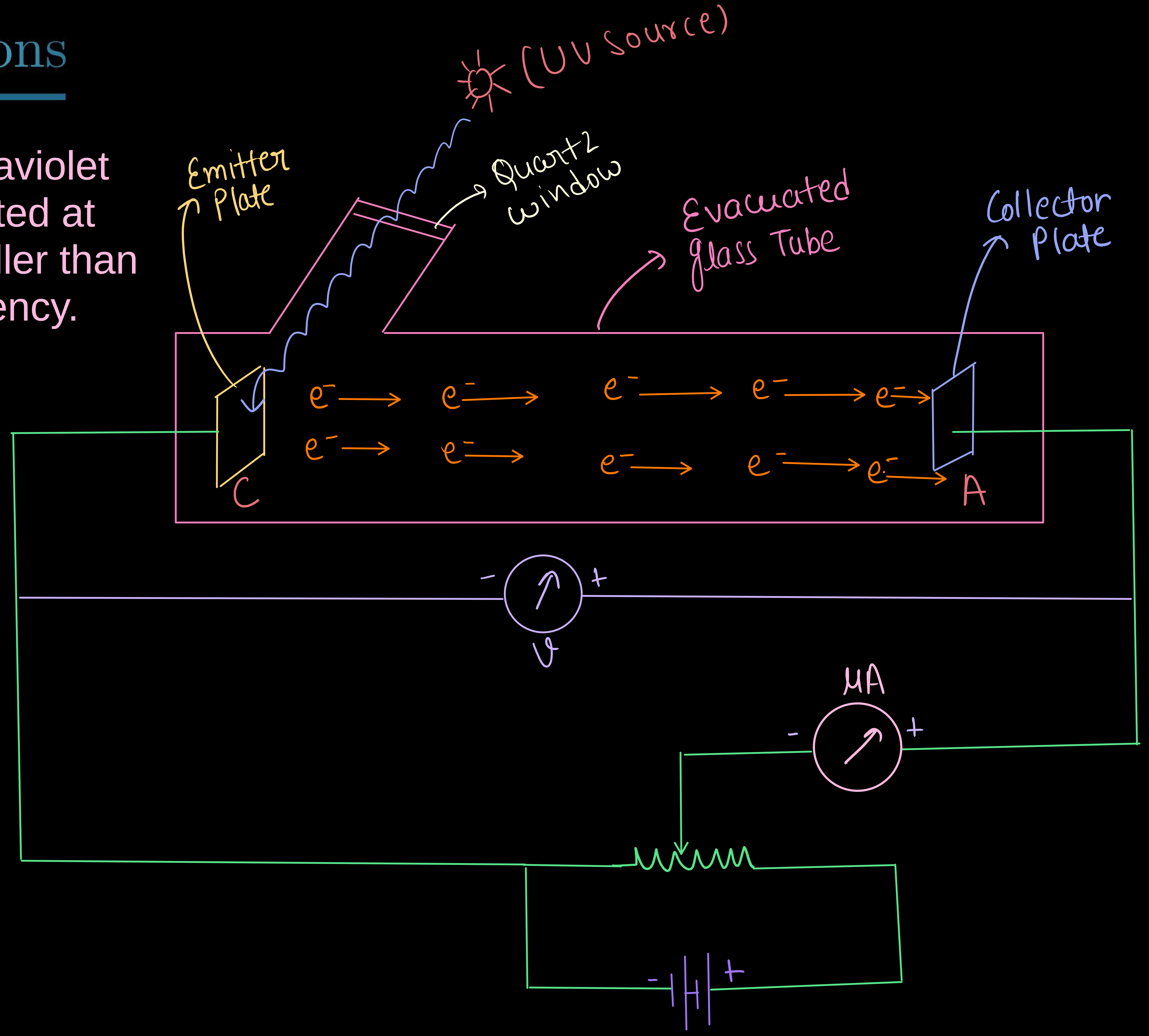


* Thus, light falling on the surface of the emitter causes current in the external circuit.
* Hallwachs and Lenard studied how this photo current varied with collector plate potential, and with frequency and intensity of incident light.

PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency.

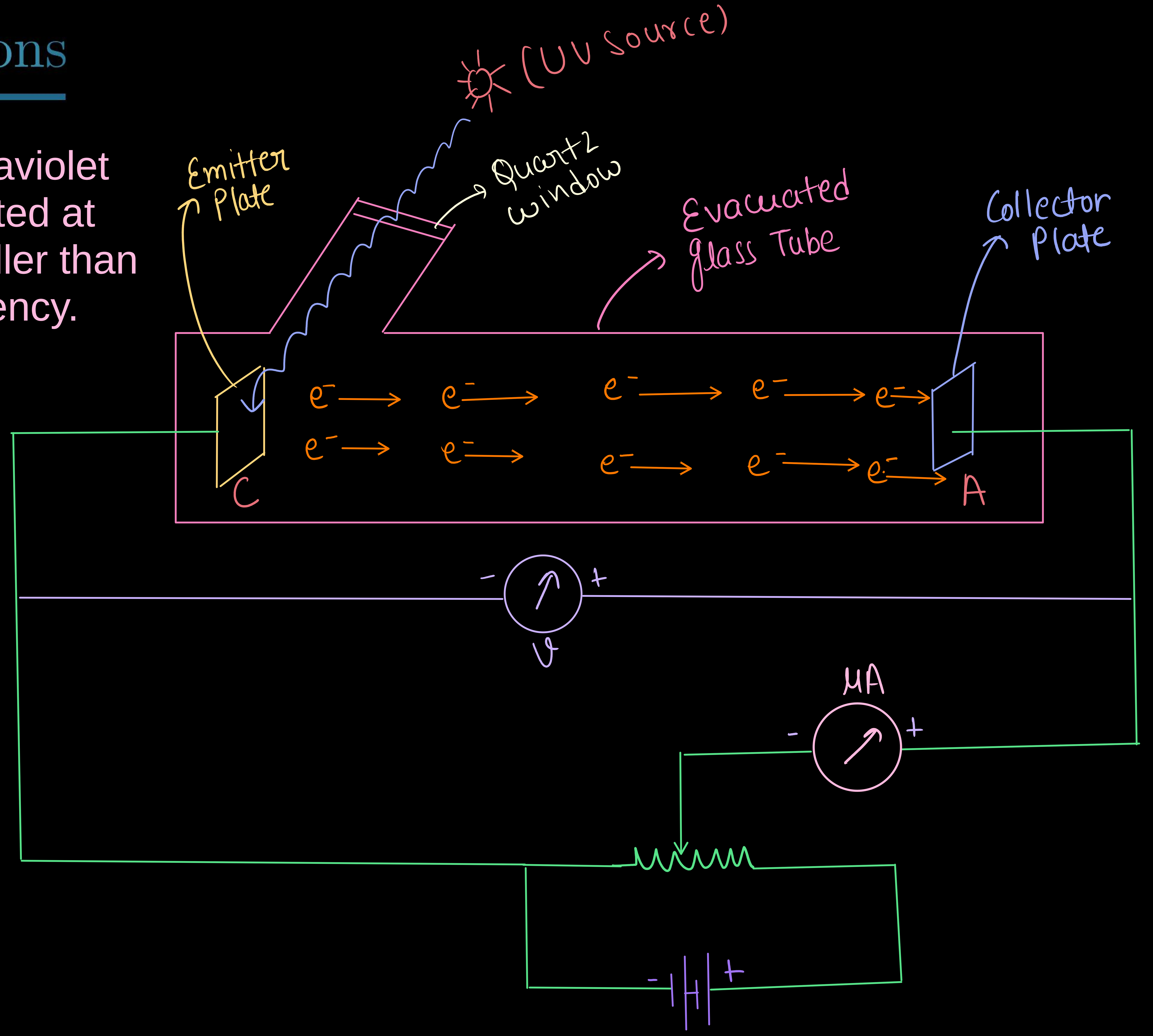


PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency.

* This minimum frequency depends on the nature of the material of the emitter plate.



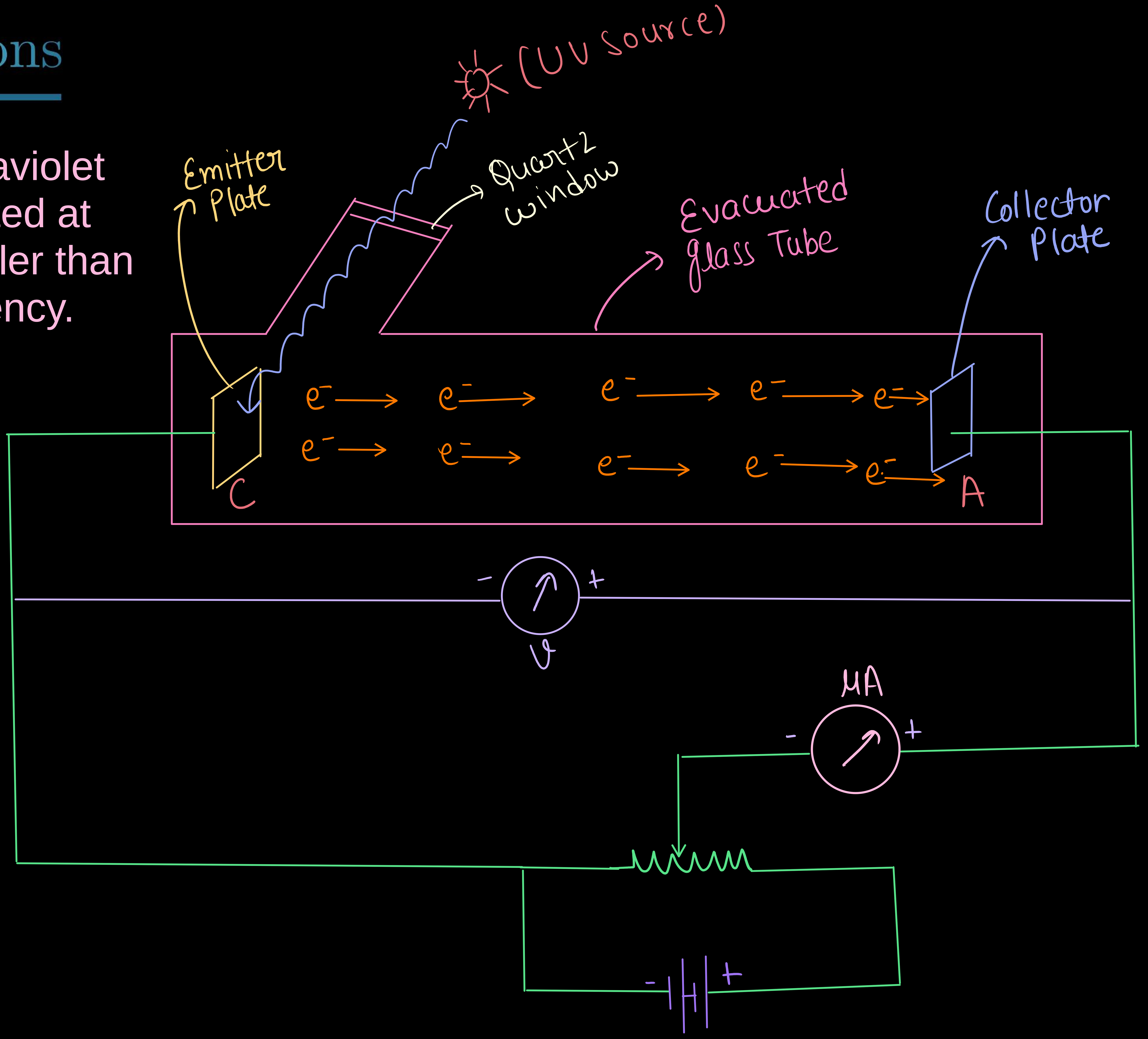
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency.

* This minimum frequency depends on the nature of the material of the emitter plate.

* It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.



PHOTOELECTRIC EFFECT

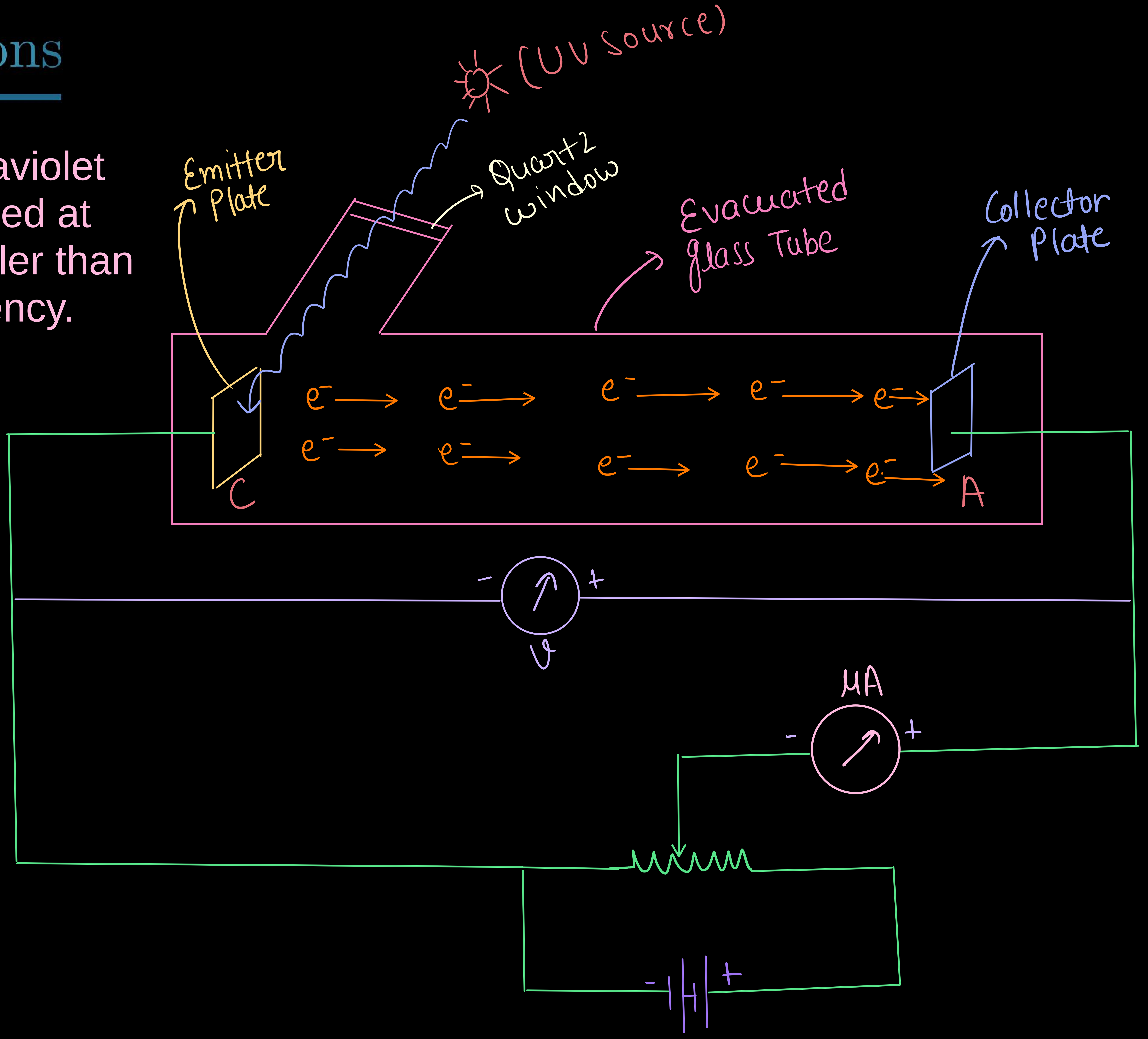
(2) Hallwachs' and Lenard's observations

* Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency.

* This minimum frequency depends on the nature of the material of the emitter plate.

* It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.

* However, some alkali metals such as lithium, sodium, potassium, caesium and rubidium were sensitive even to visible light.



PHOTOELECTRIC EFFECT

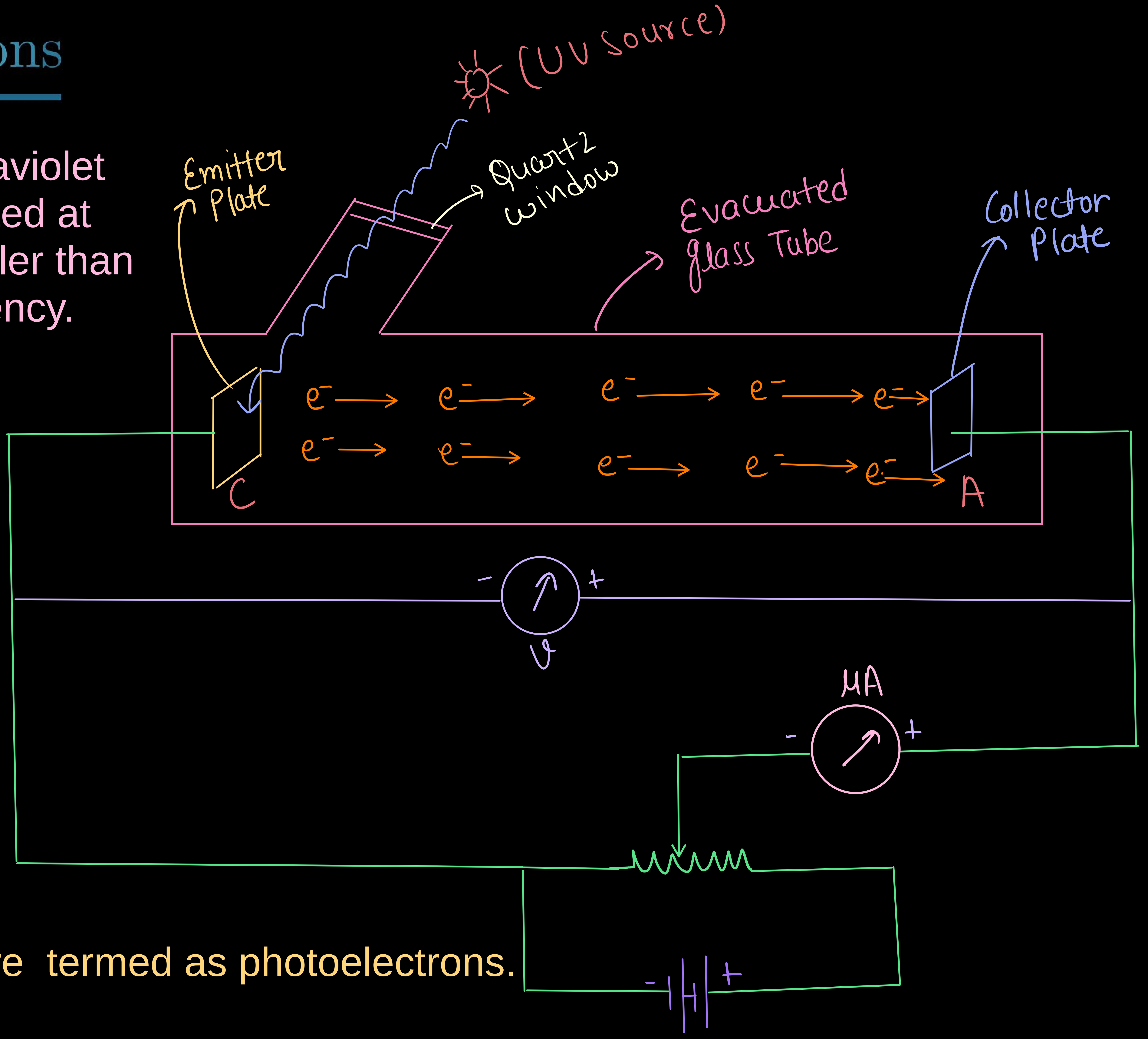
(2) Hallwachs' and Lenard's observations

* Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency.

* This minimum frequency depends on the nature of the material of the emitter plate.

* It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.

* However, some alkali metals such as lithium, sodium, potassium, caesium and rubidium were sensitive even to visible light.

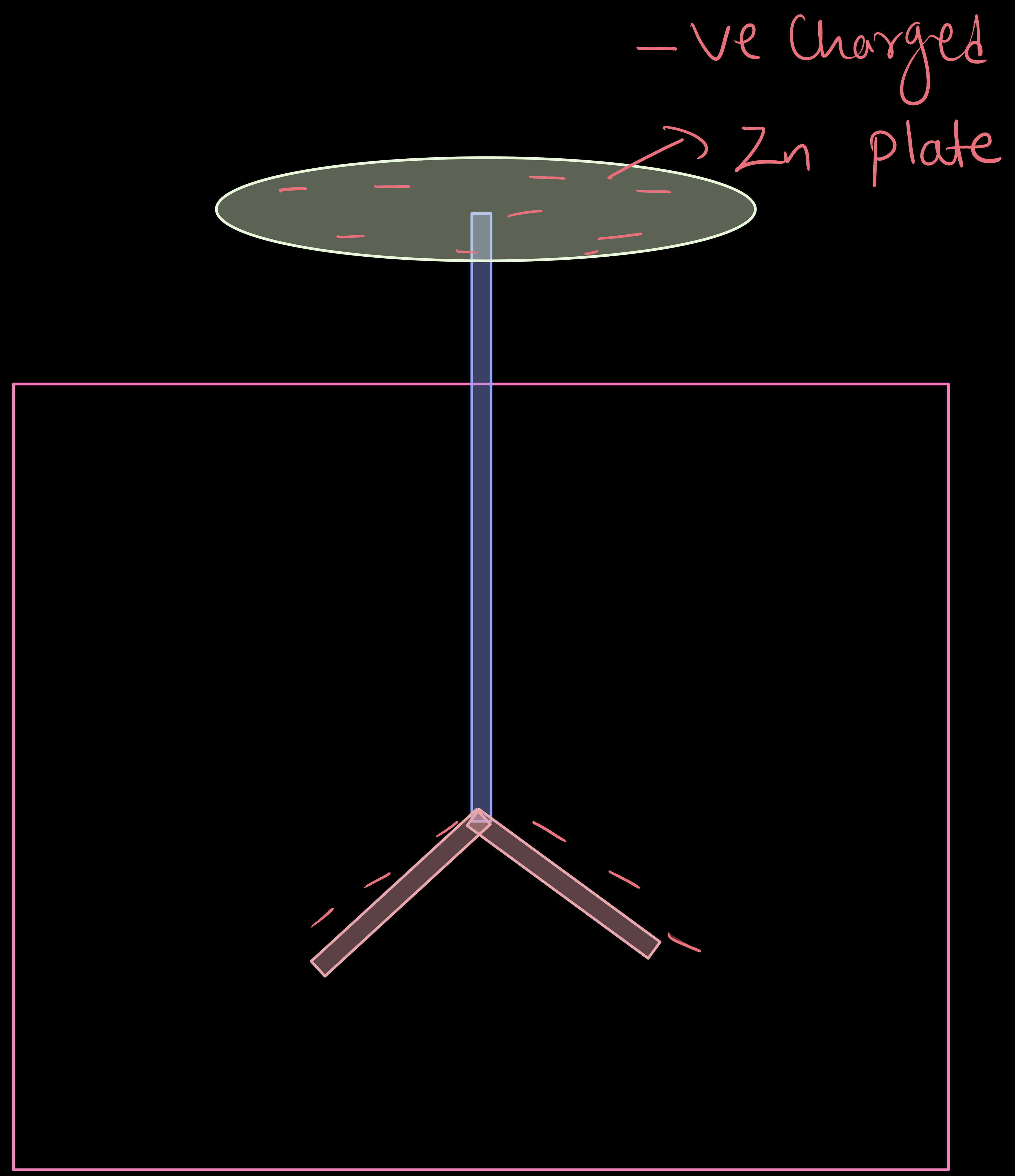


* After the discovery of electrons, these electrons were termed as photoelectrons. The phenomenon is called photoelectric effect.

PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

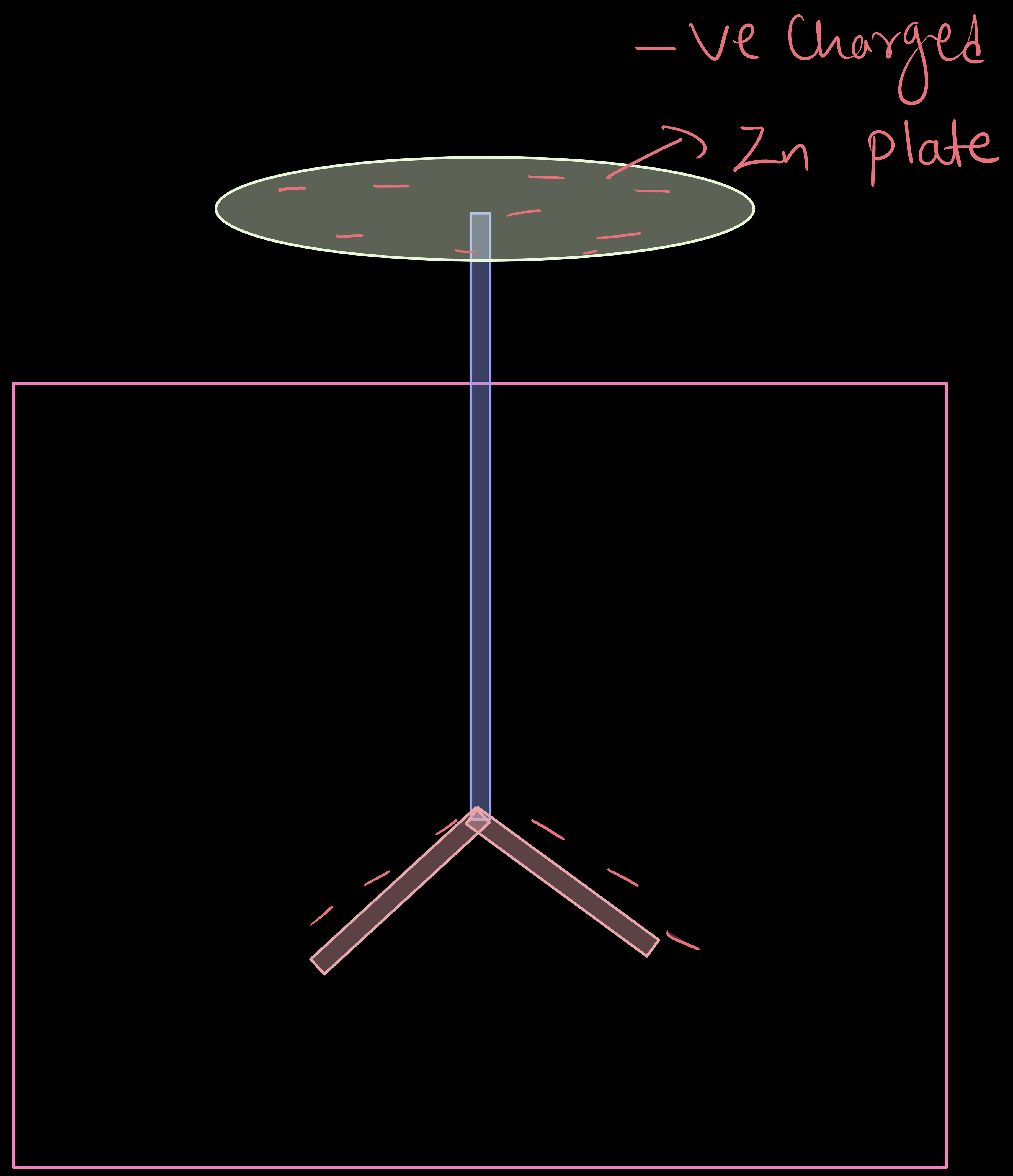
* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

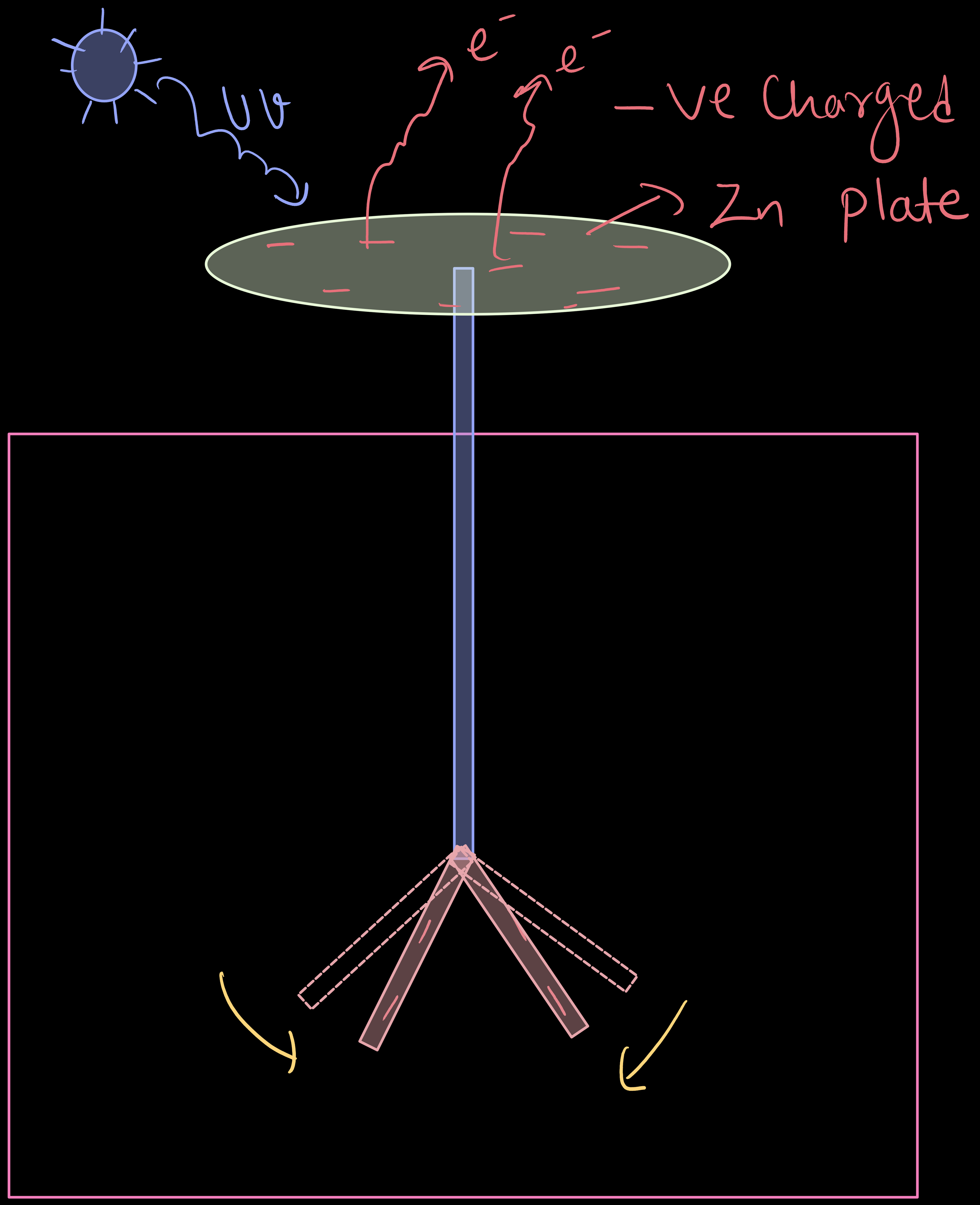
* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

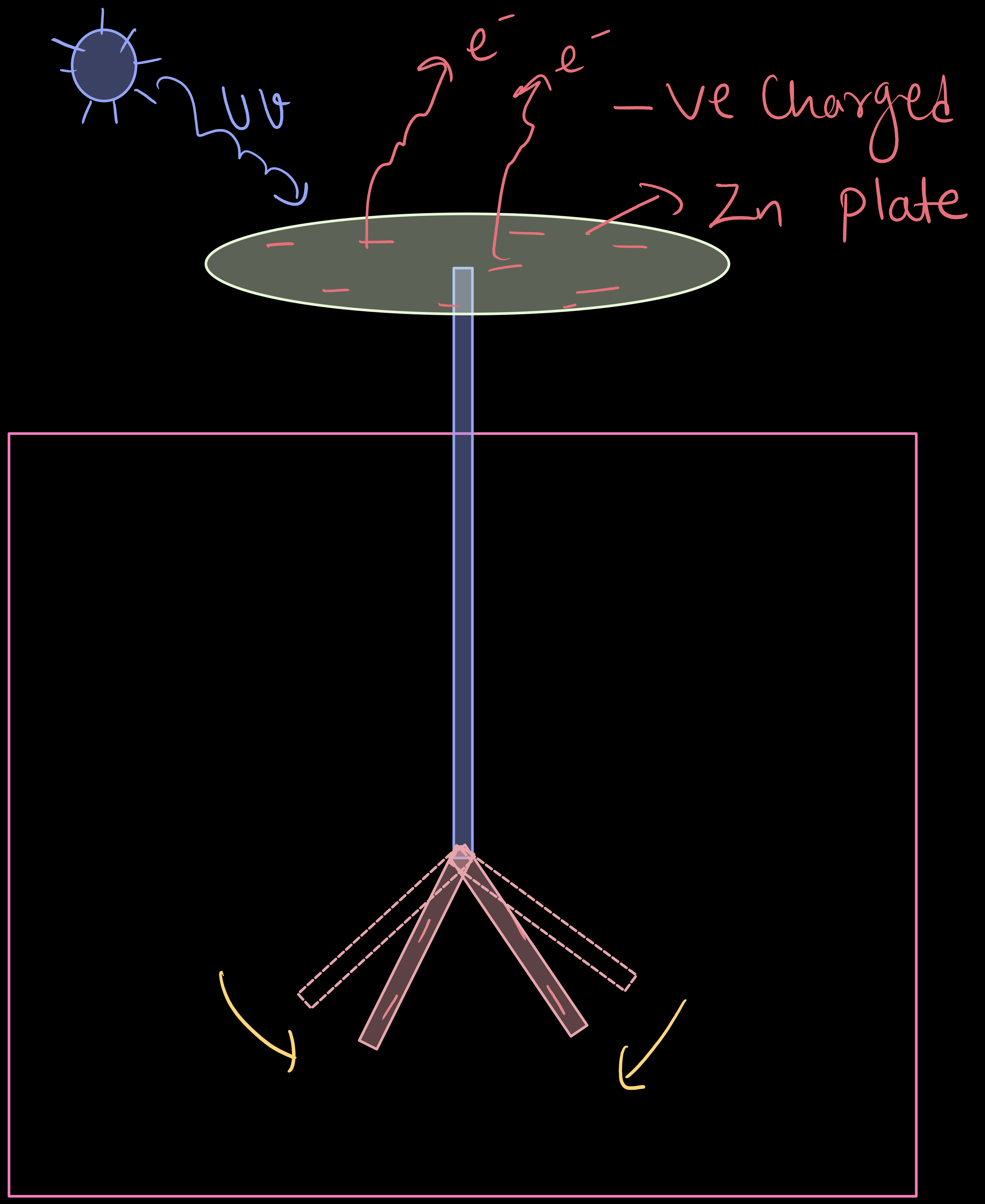


PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* He observed that the zinc plate lost its charge when it was illuminated by ultraviolet light.

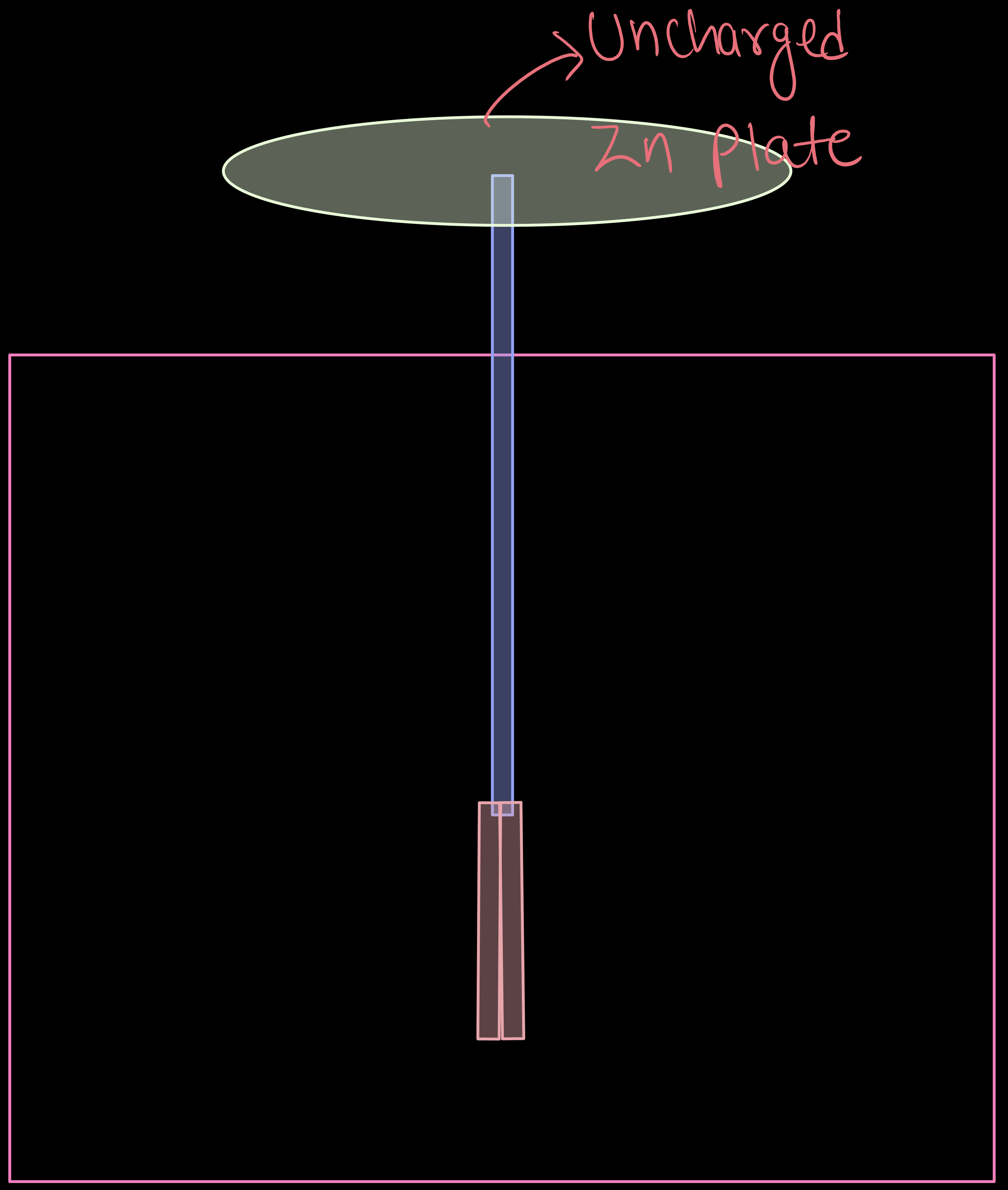


PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.



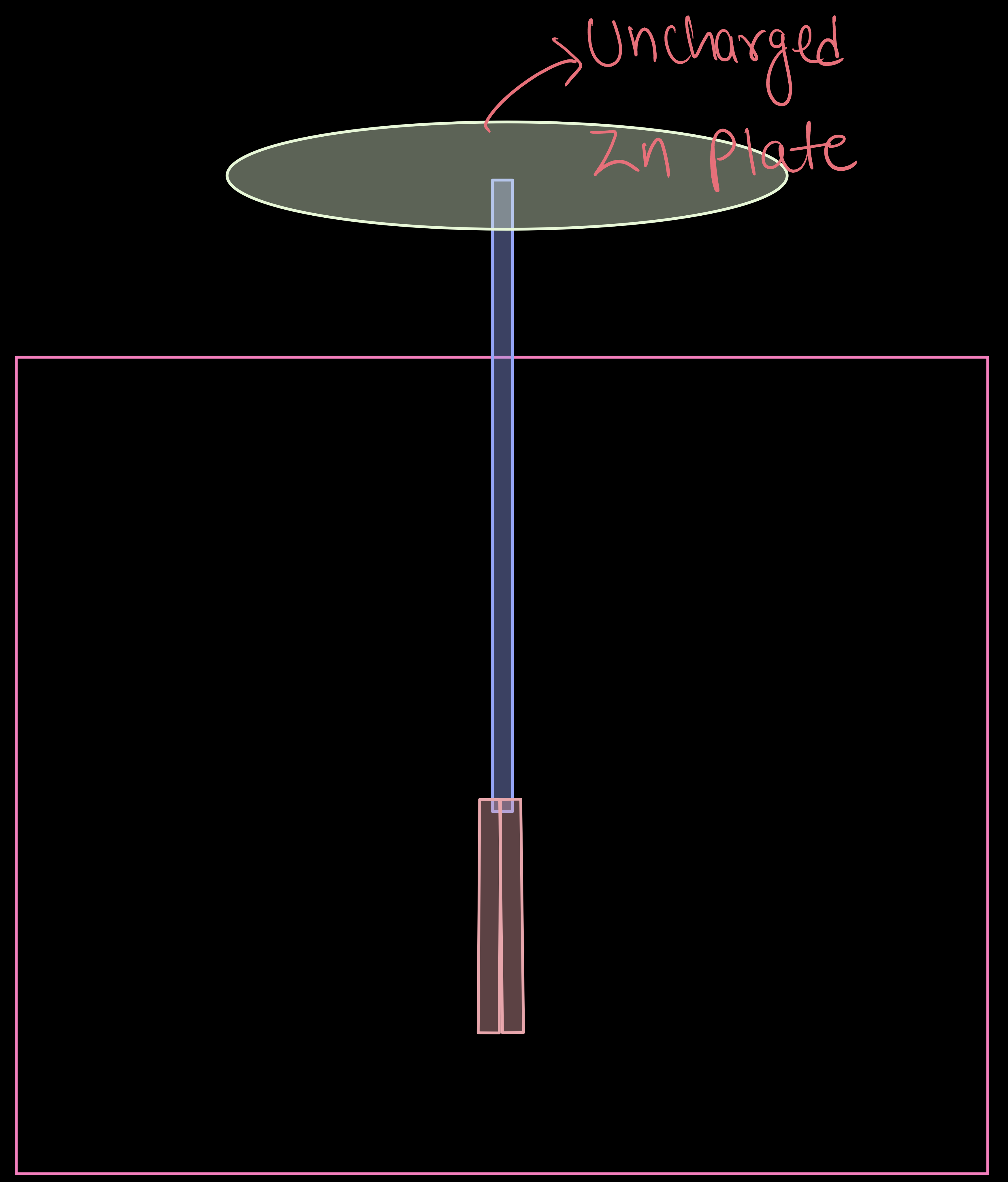
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* He observed that the zinc plate lost its charge when it was illuminated by ultraviolet light.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

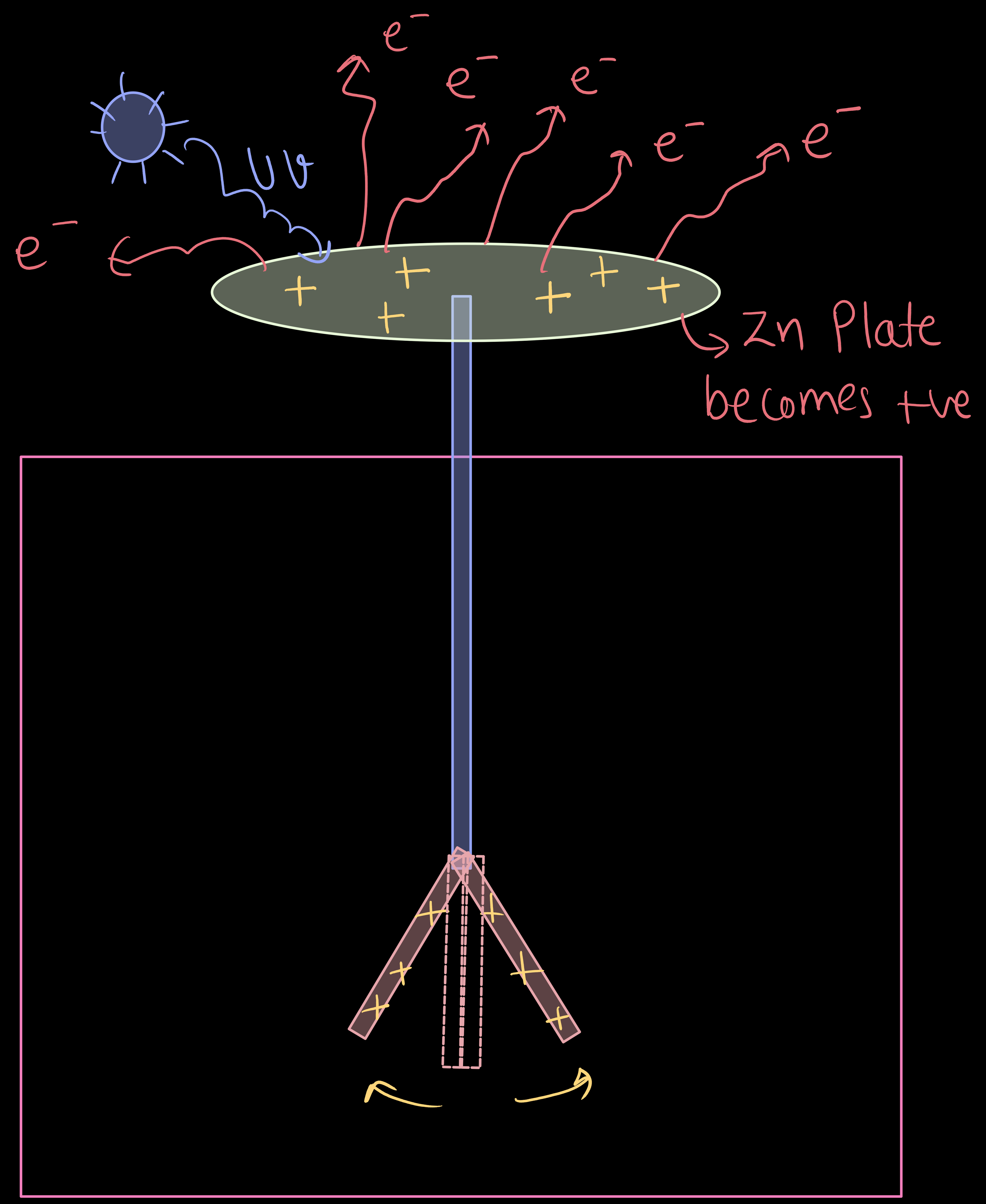


PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.



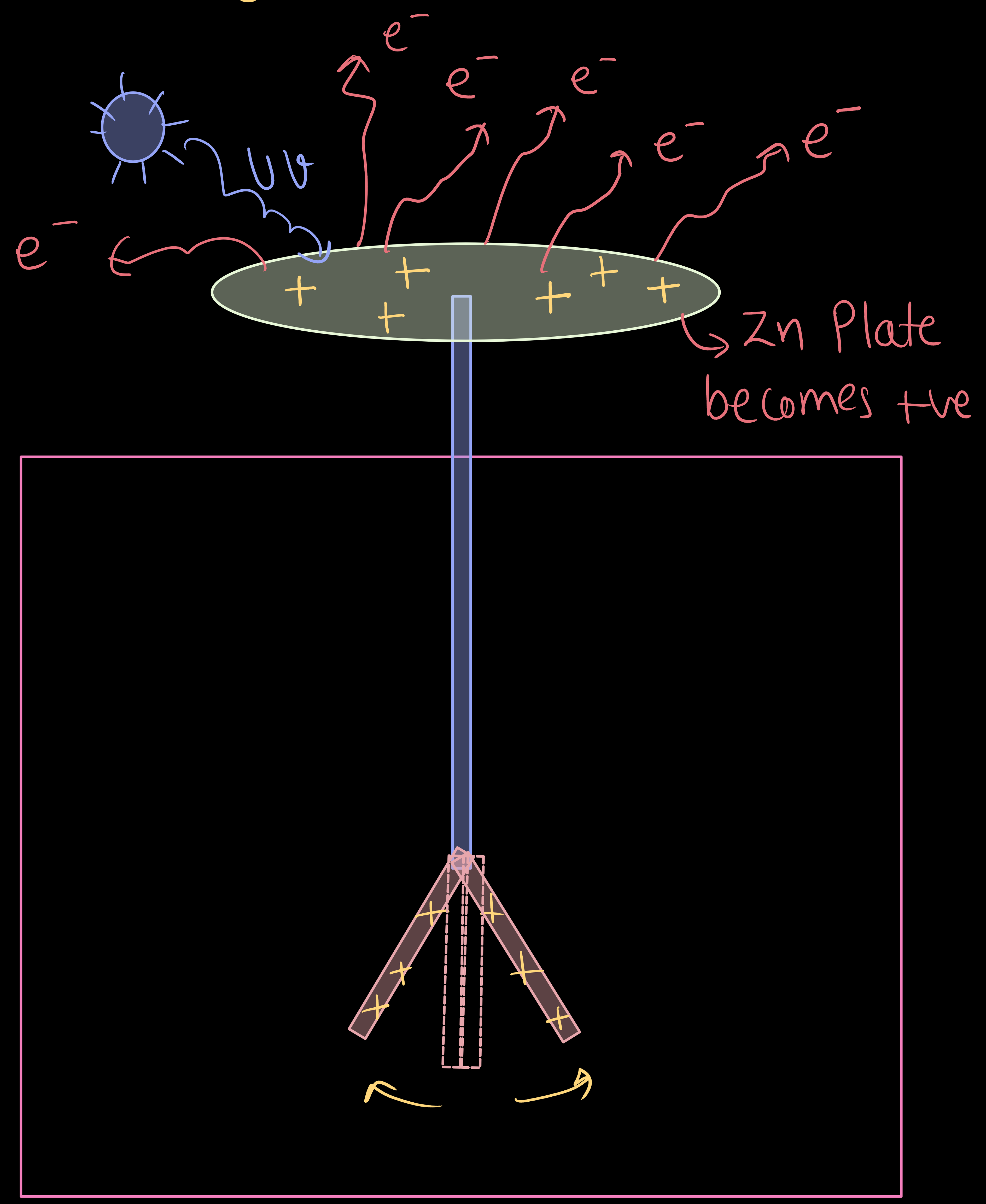
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* He observed that the zinc plate lost its charge when it was illuminated by ultraviolet light.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.



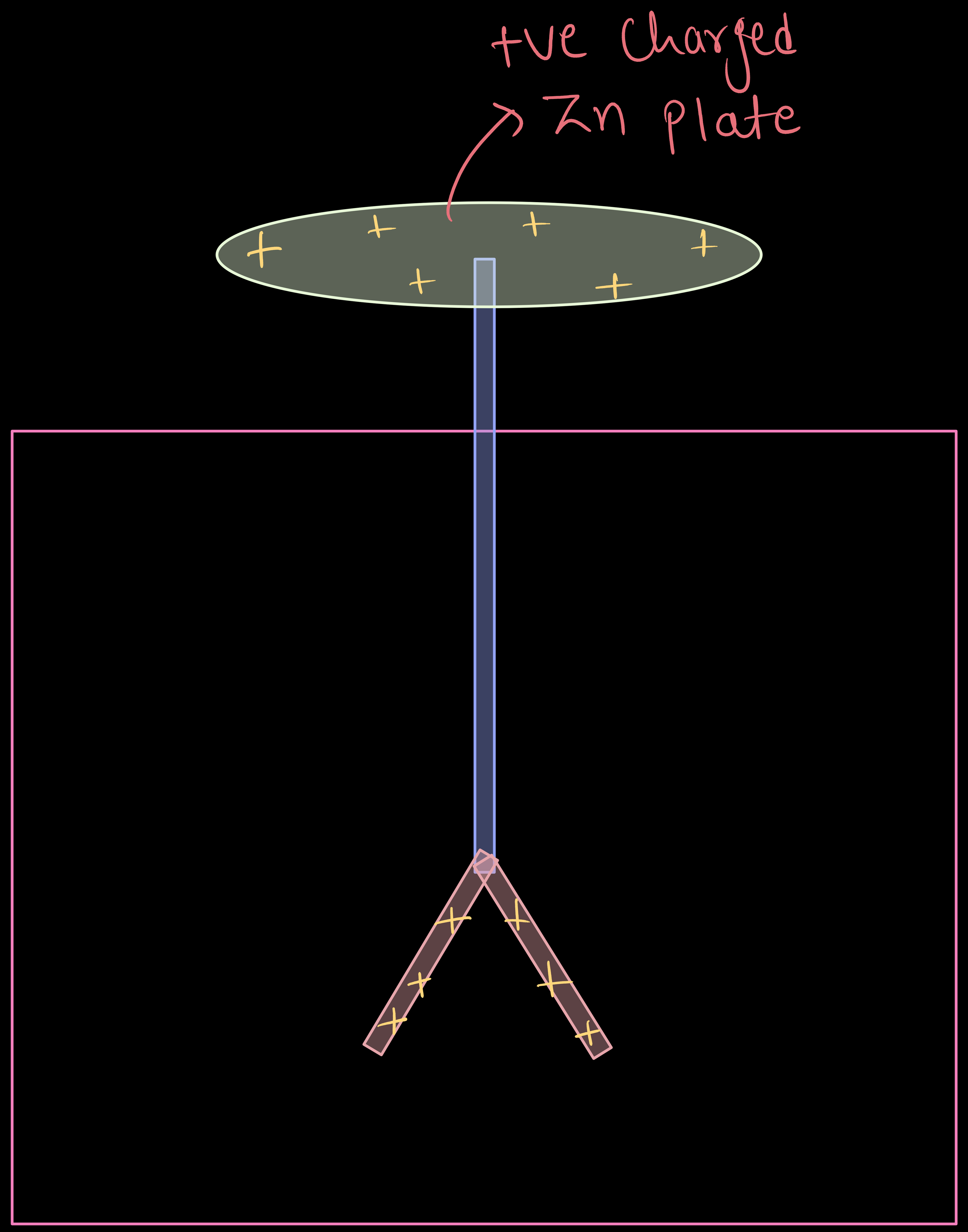
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.



PHOTOELECTRIC EFFECT

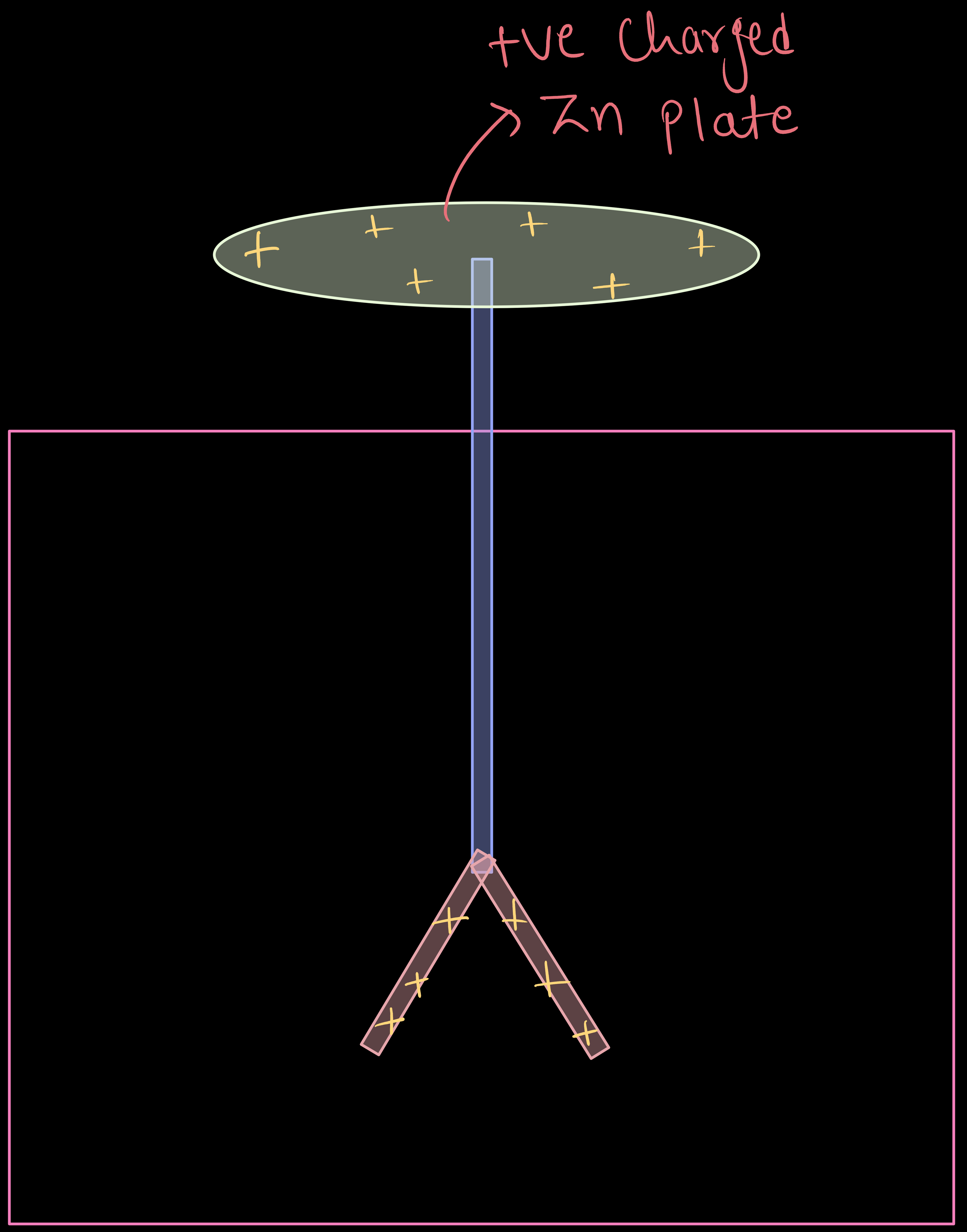
(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* He observed that the zinc plate lost its charge when it was illuminated by ultraviolet light.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.



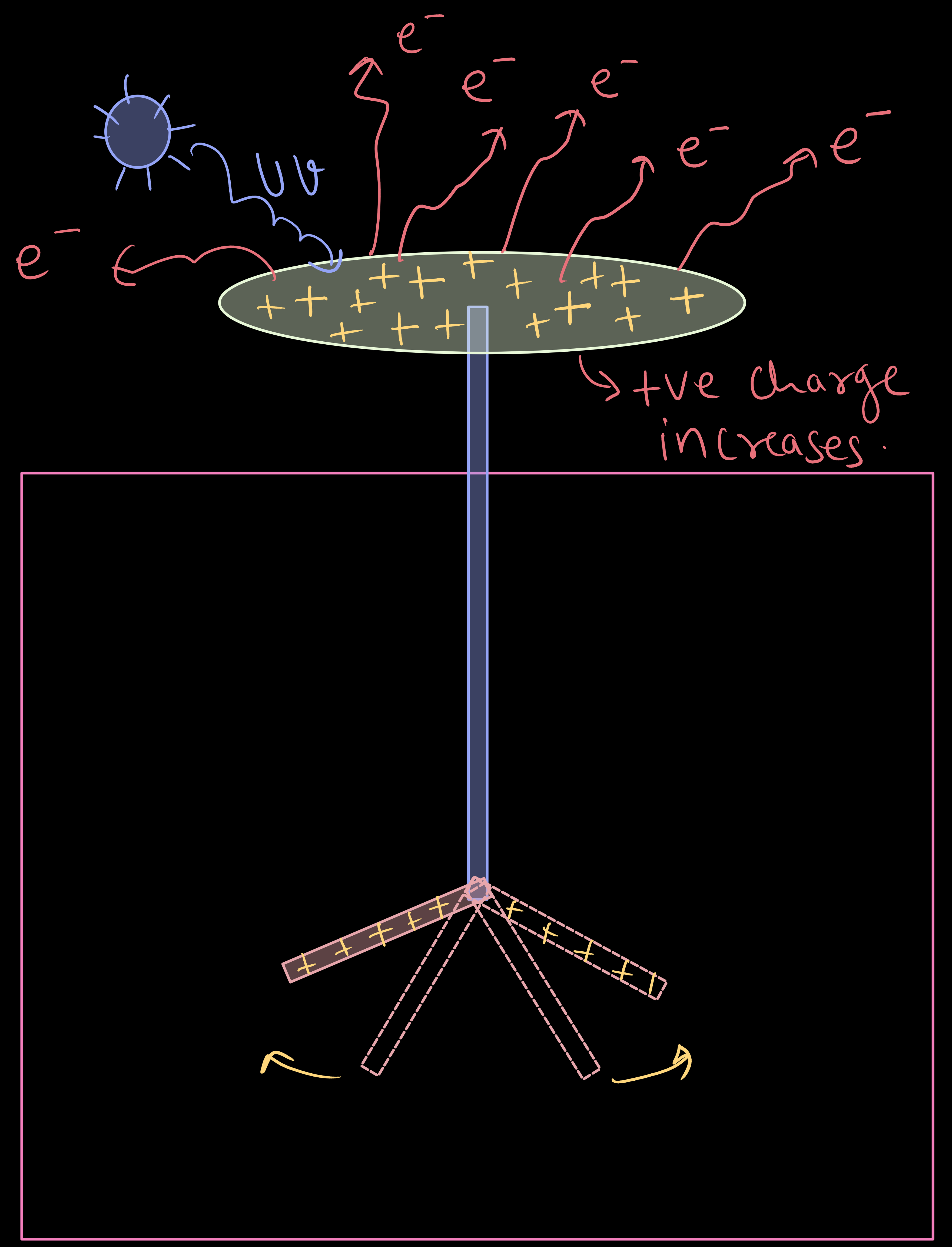
PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.



PHOTOELECTRIC EFFECT

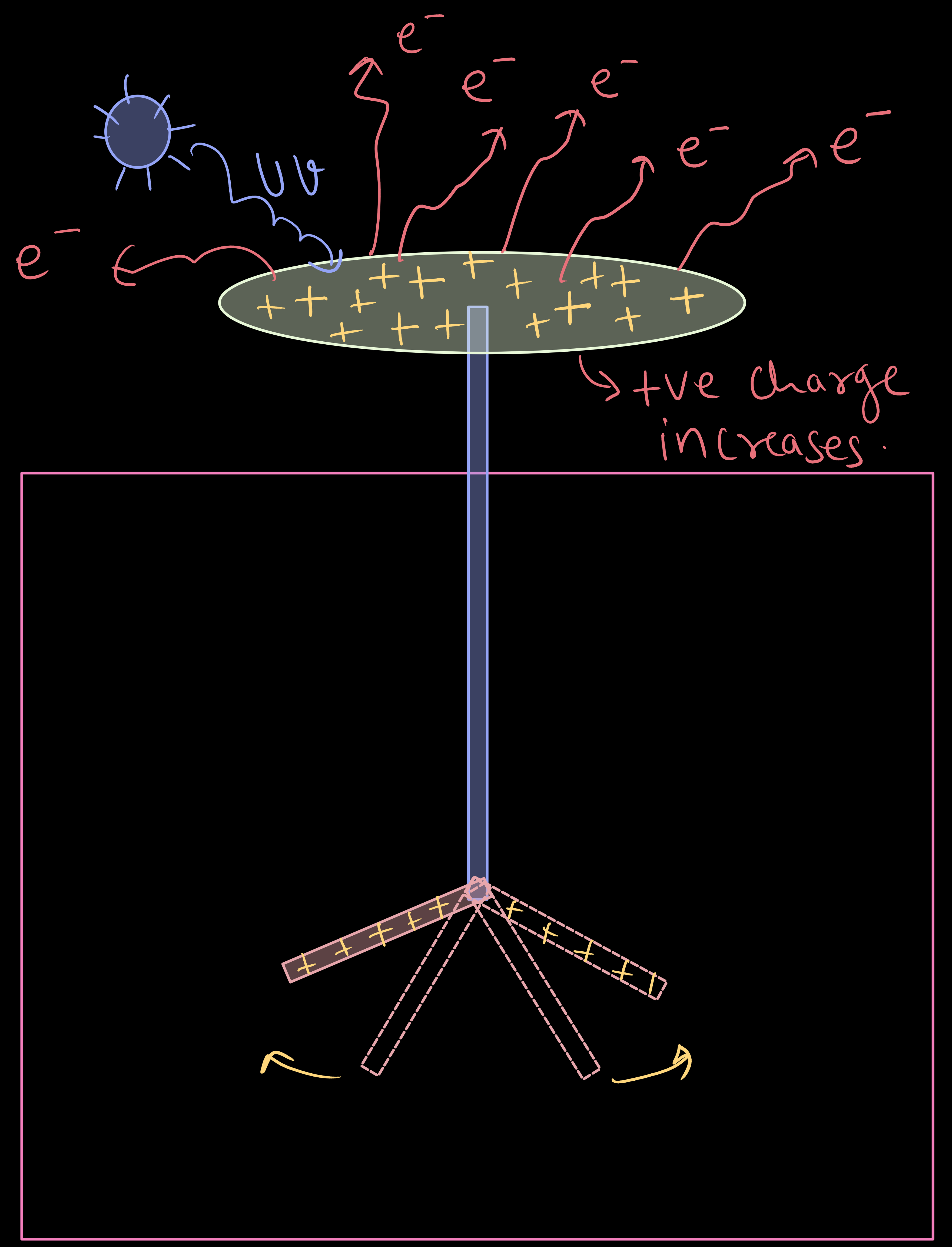
(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.

* From these observations he concluded that negatively charged particles were emitted from the zinc plate under the action of ultraviolet light.



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

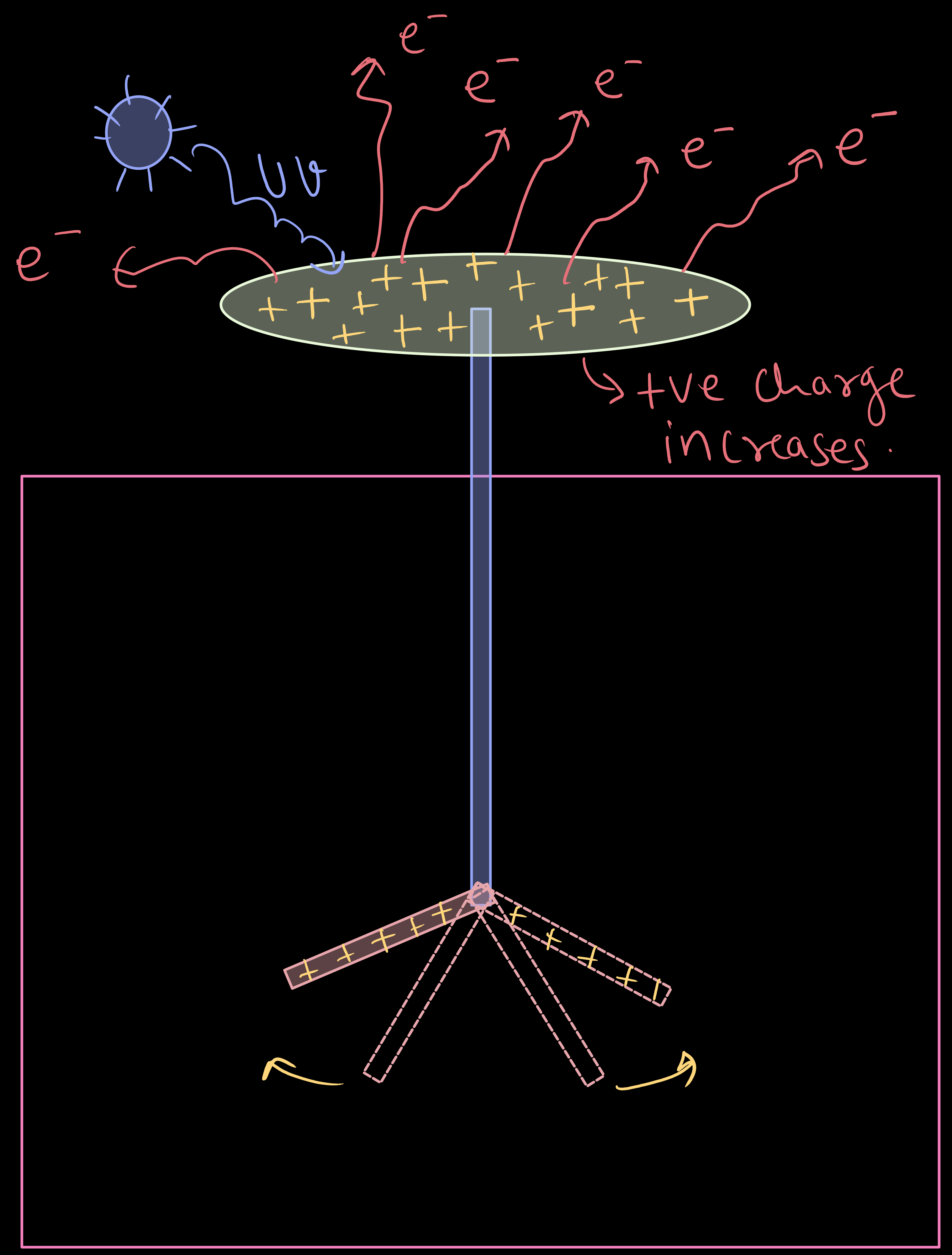
* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.

* From these observations he concluded that negatively charged particles were emitted from the zinc plate under the action of ultraviolet light.

* After the discovery of the electron in 1897, it became clear that these emitted particles were electrons.



PHOTOELECTRIC EFFECT

(2) Hallwachs' and Lenard's observations

* Hallwachs, in 1888, undertook the study further and connected a negatively charged zinc plate to an electroscope.

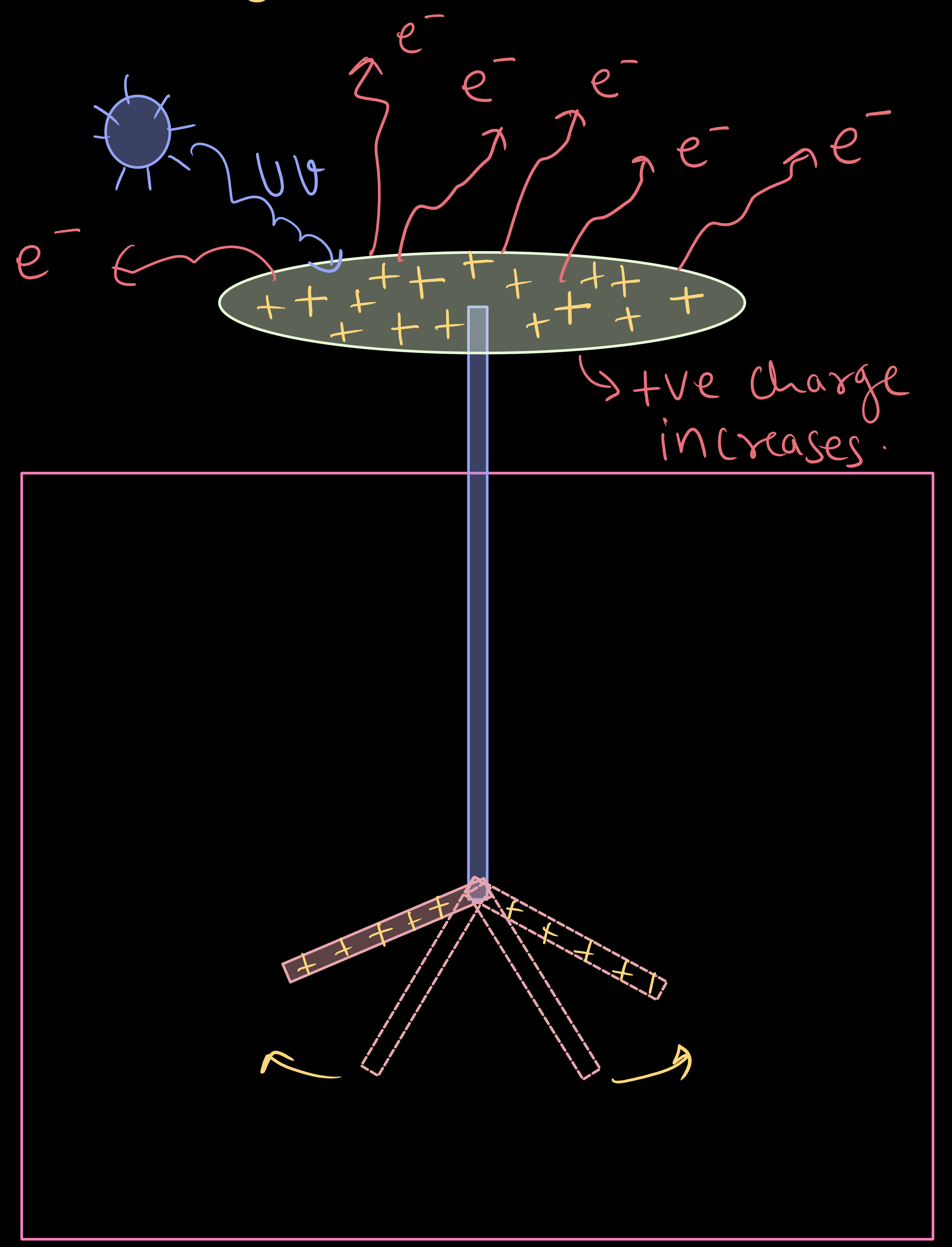
* He observed that the zinc plate lost its charge when it was illuminated by ultraviolet light.

* Further, the uncharged zinc plate became positively charged when it was irradiated by ultraviolet light.

* Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light.

* From these observations he concluded that negatively charged particles were emitted from the zinc plate under the action of ultraviolet light.

* After the discovery of the electron in 1897, it became clear that these emitted particles were electrons.



EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

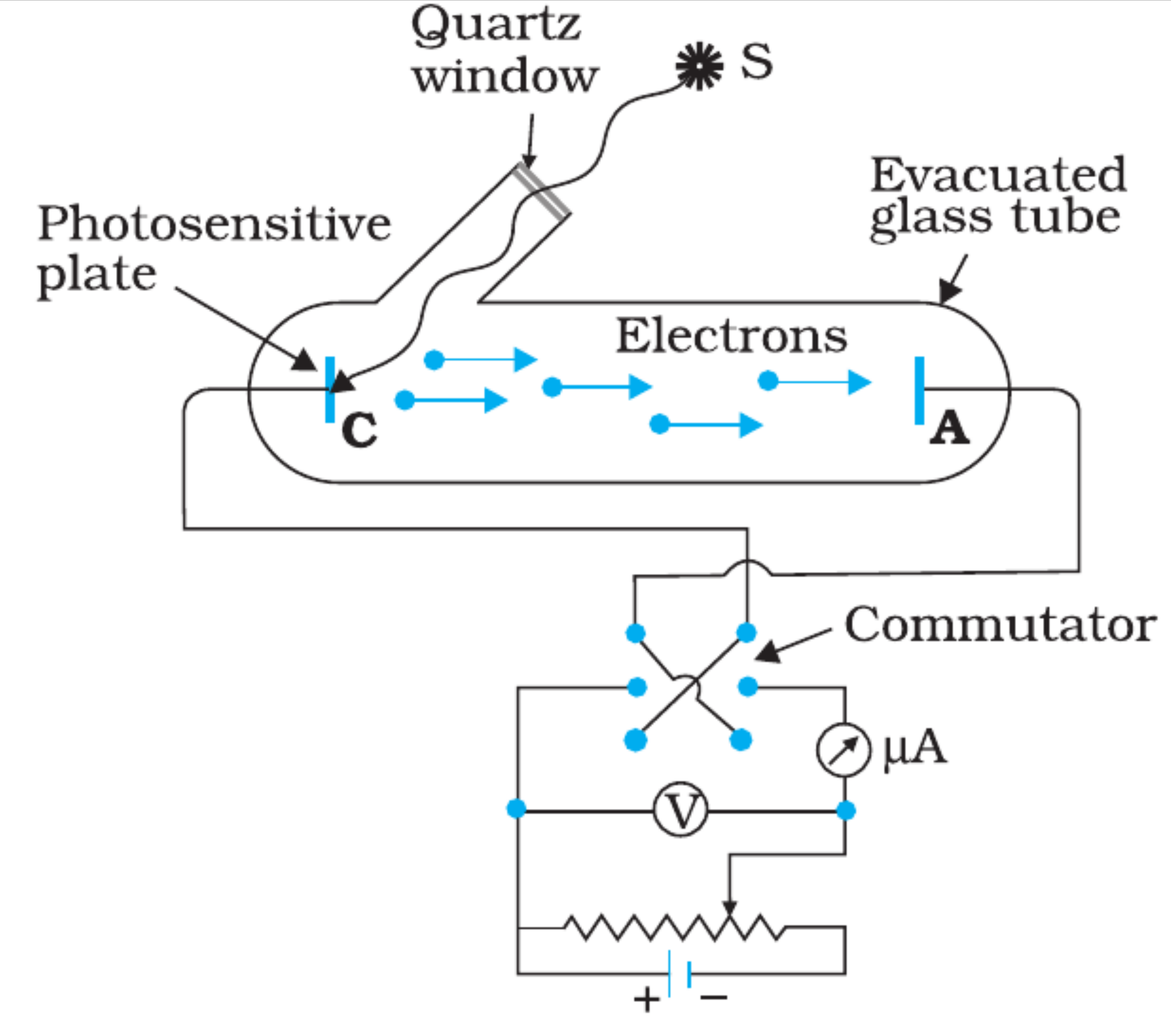


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

* Monochromatic light from the source S of sufficiently short wavelength passes through the quartz window W (Allows ultraviolet radiation to pass into the tube (glass absorbs UV)) and falls on the photosensitive plate C (emitter).

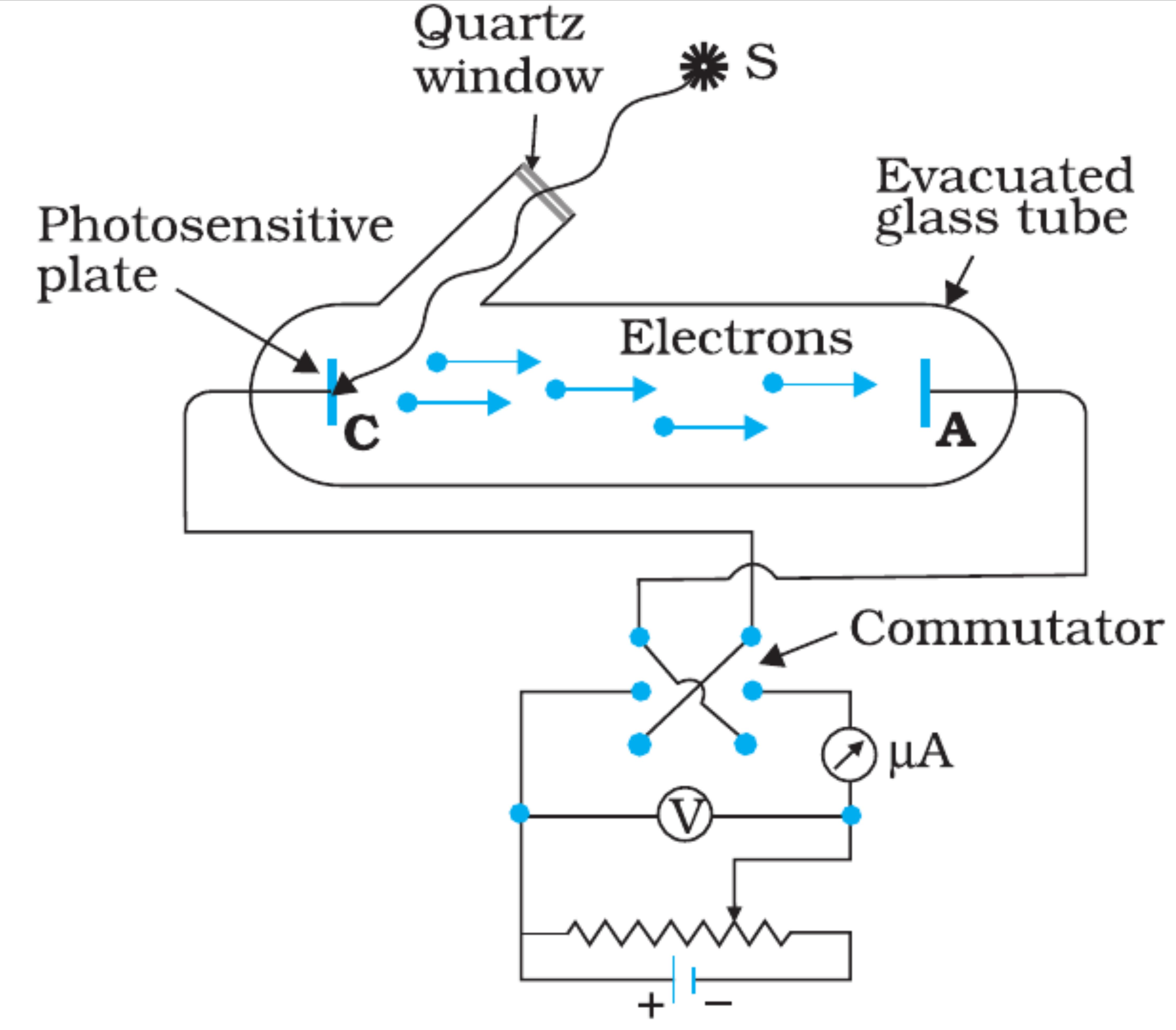


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

* Monochromatic light from the source S of sufficiently short wavelength passes through the quartz window W (Allows ultraviolet radiation to pass into the tube (glass absorbs UV)) and falls on the photosensitive plate C (emitter).

* The electrons are emitted by the plate C and are collected by the plate A (collector), by the electric field created by the battery.

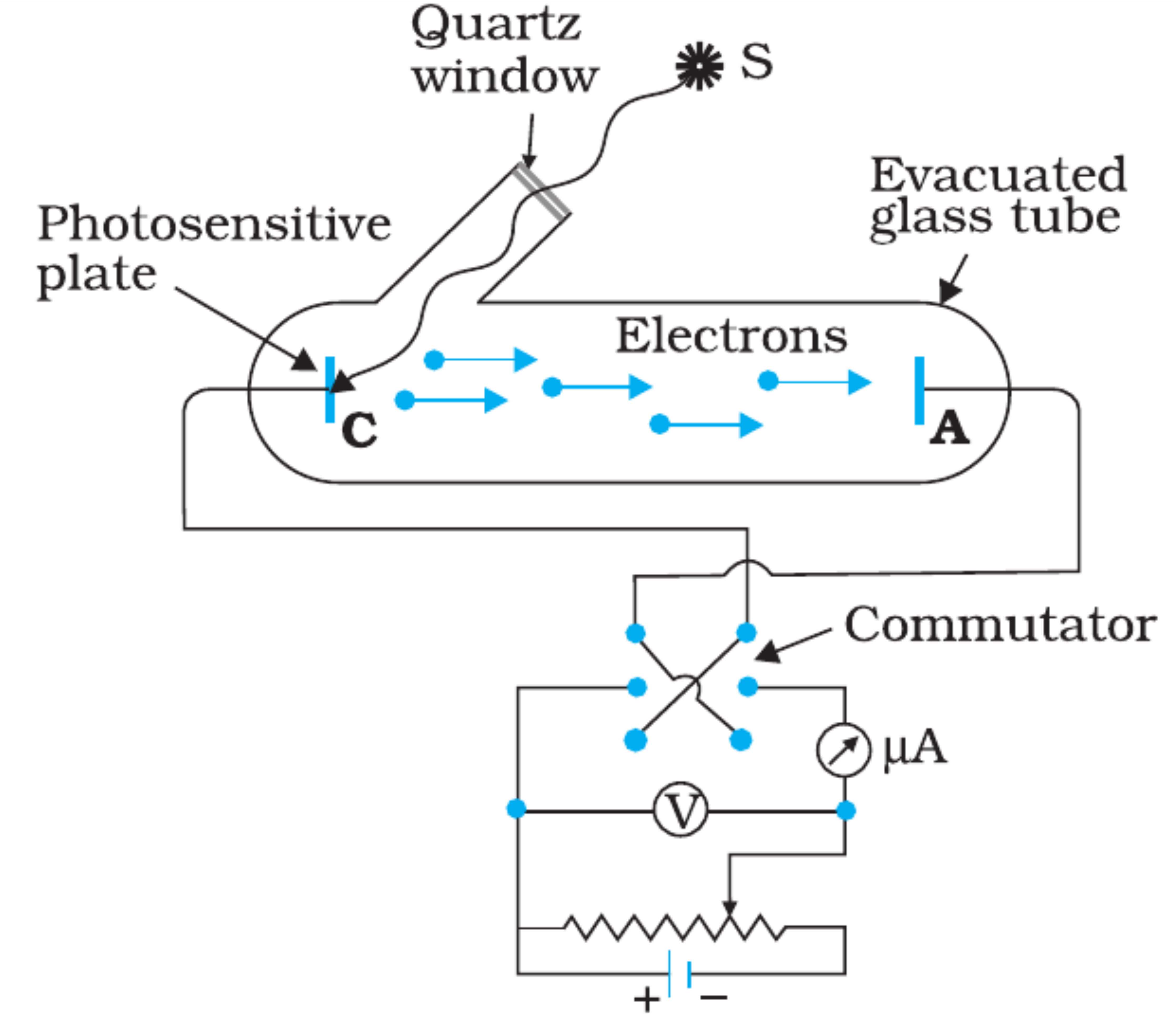


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

* Monochromatic light from the source S of sufficiently short wavelength passes through the quartz window W (Allows ultraviolet radiation to pass into the tube (glass absorbs UV)) and falls on the photosensitive plate C (emitter).

* The electrons are emitted by the plate C and are collected by the plate A (collector), by the electric field created by the battery.

* The battery maintains the potential difference between the plates C and A, that can be varied. The polarity of the plates C and A can be reversed by a commutator.

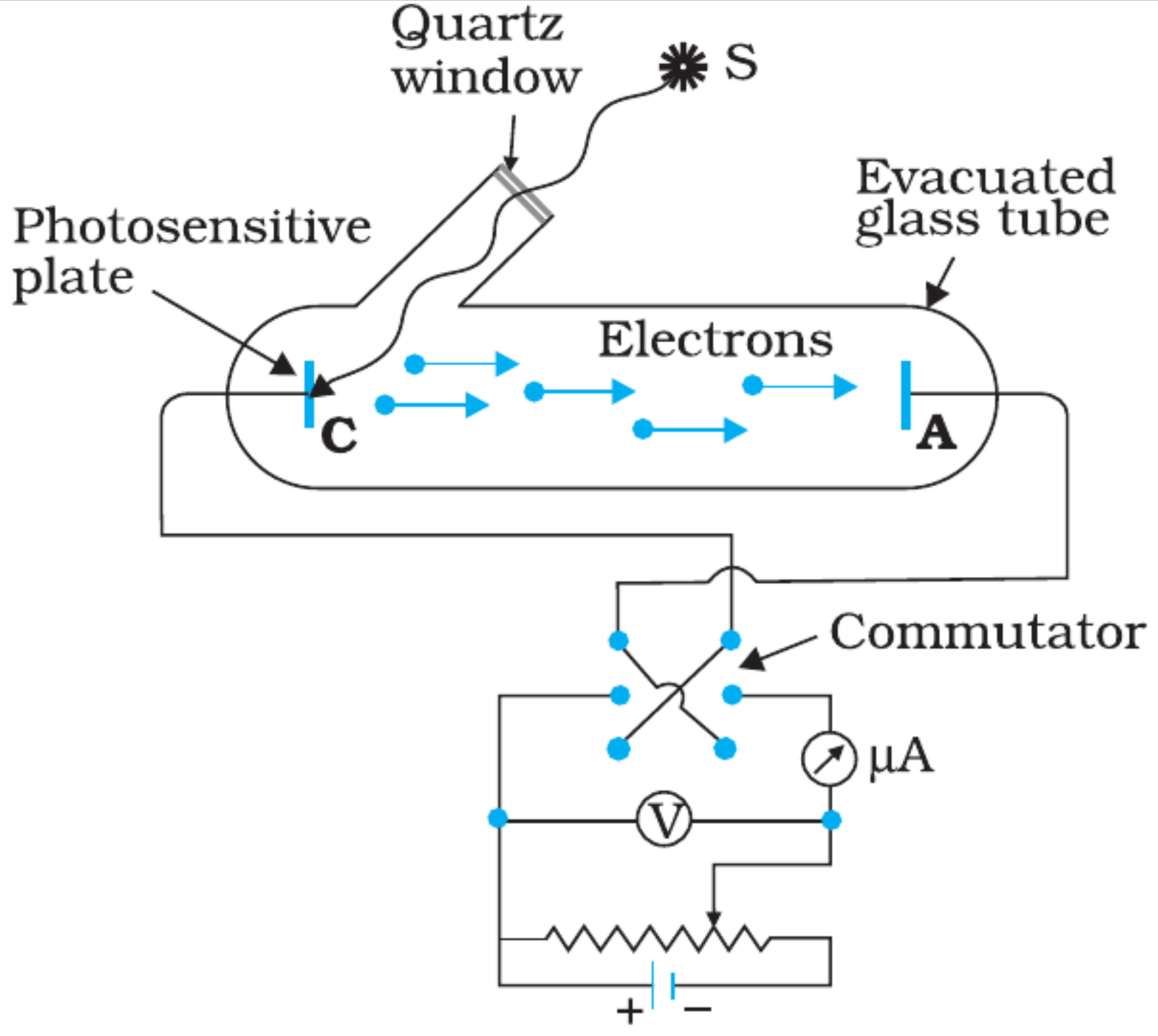


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

* Monochromatic light from the source S of sufficiently short wavelength passes through the quartz window W (Allows ultraviolet radiation to pass into the tube (glass absorbs UV)) and falls on the photosensitive plate C (emitter).

* The electrons are emitted by the plate C and are collected by the plate A (collector), by the electric field created by the battery.

* The battery maintains the potential difference between the plates C and A, that can be varied. The polarity of the plates C and A can be reversed by a commutator.

* The potential difference between the emitter and collector plates is measured by a voltmeter (V) whereas the resulting photo current flowing in the circuit is measured by a microammeter (μA).

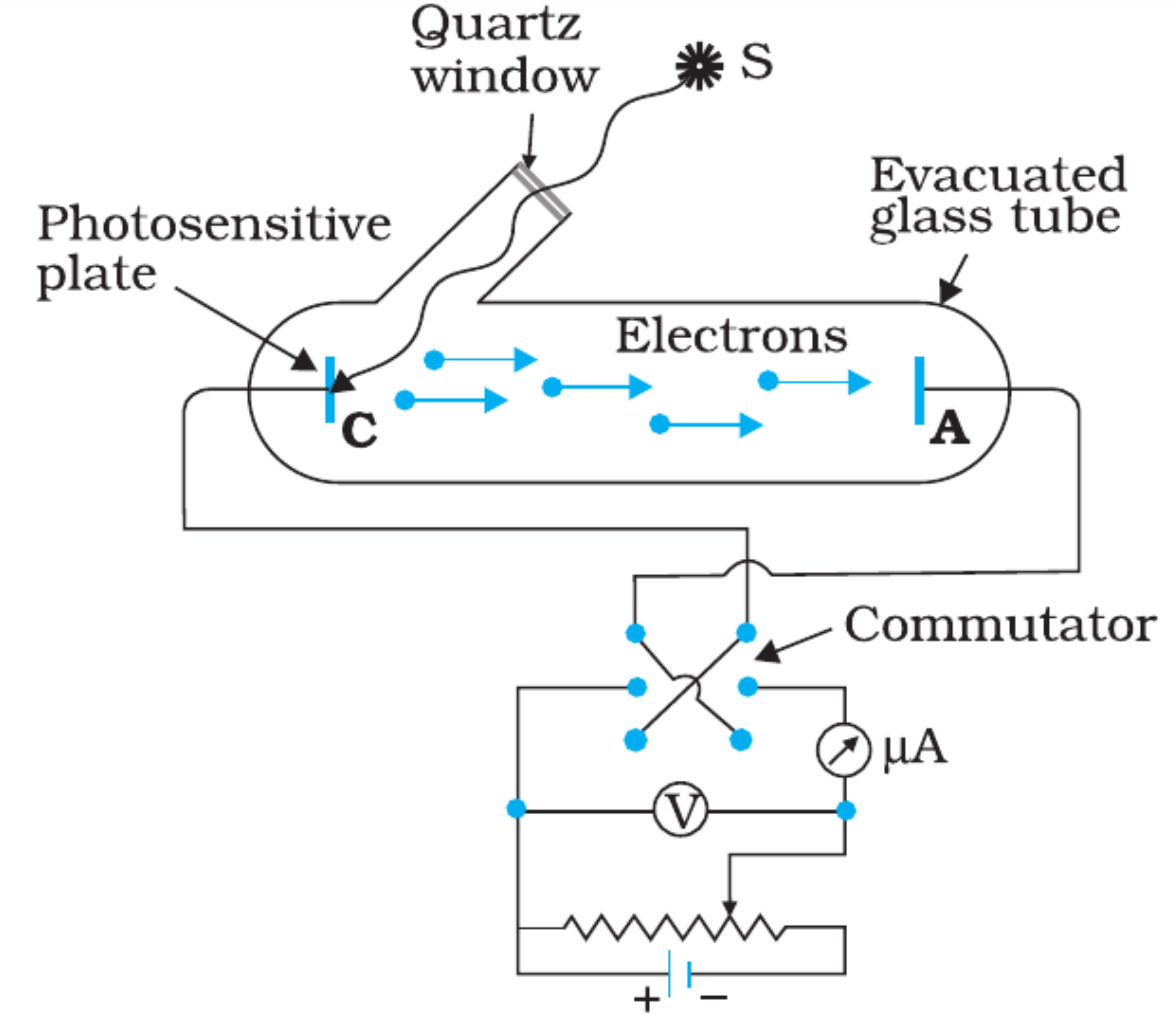


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

* The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.

* Monochromatic light from the source S of sufficiently short wavelength passes through the quartz window W (Allows ultraviolet radiation to pass into the tube (glass absorbs UV)) and falls on the photosensitive plate C (emitter).

* The electrons are emitted by the plate C and are collected by the plate A (collector), by the electric field created by the battery.

* The battery maintains the potential difference between the plates C and A, that can be varied. The polarity of the plates C and A can be reversed by a commutator.

* The potential difference between the emitter and collector plates is measured by a voltmeter (V) whereas the resulting photo current flowing in the circuit is measured by a microammeter (μA).

* We can use the experimental arrangement of to study the variation of photocurrent with (a) intensity of radiation, (b) frequency of incident radiation, (c) the potential difference between the plates A and C, and (d) the nature of the material of plate C

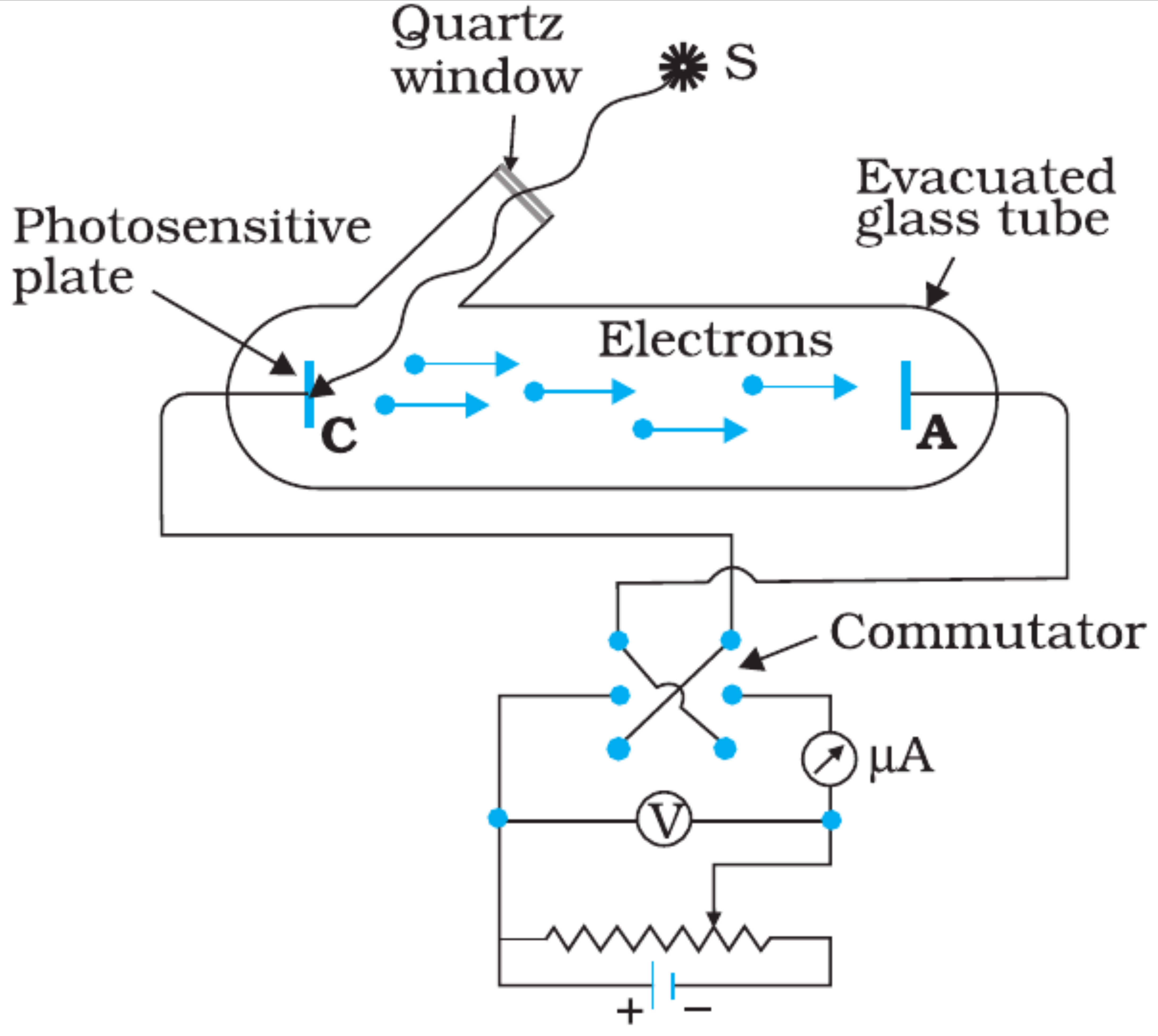


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(1) Effect of intensity of light on photocurrent

(Keep frequency and potential constant, vary only intensity.)

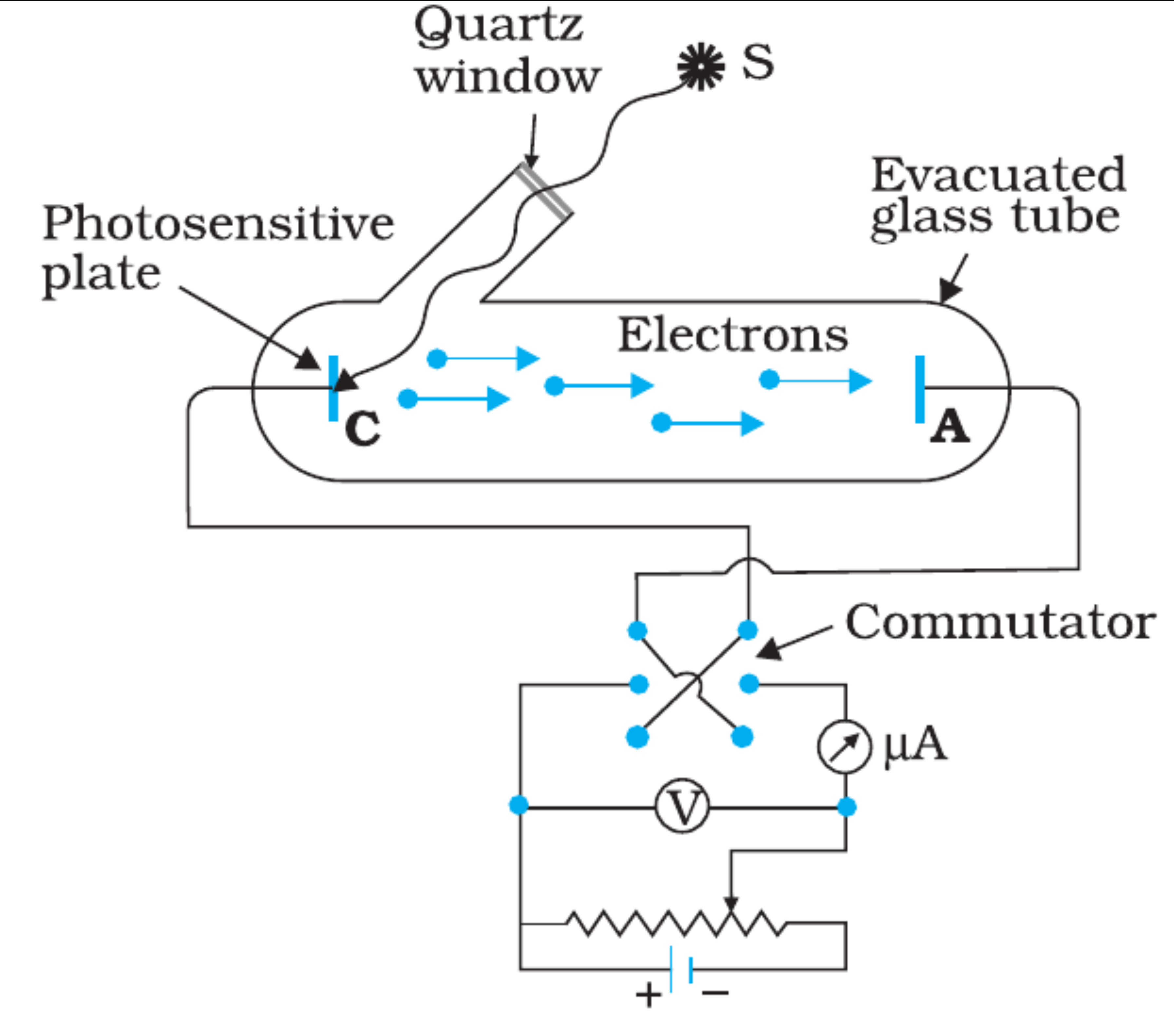


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(1) Effect of intensity of light on photocurrent

(Keep frequency and potential constant, vary only intensity.)

* It is found that the photocurrent increases linearly with intensity of incident light as shown graphically in Figure.

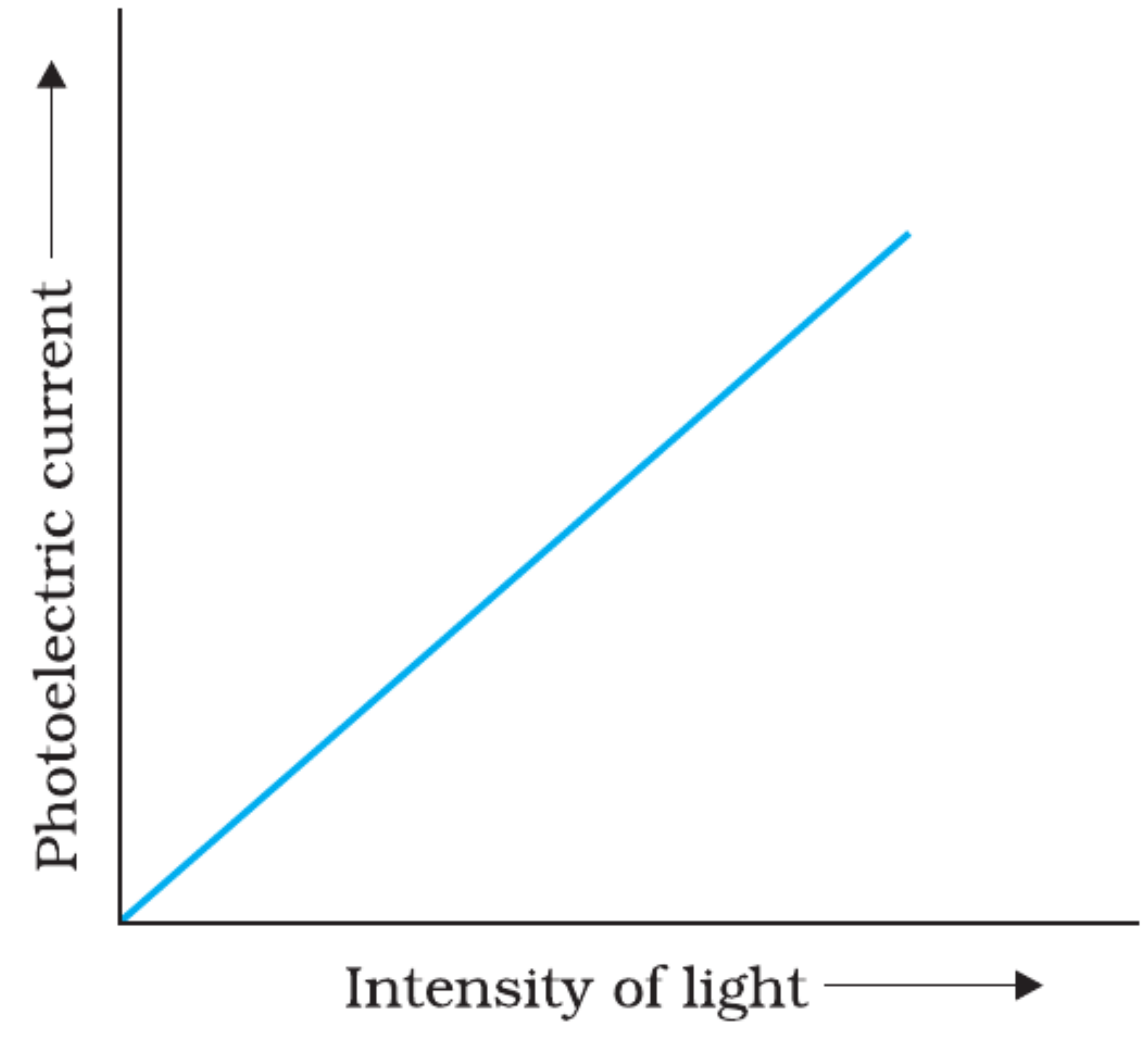


FIGURE 11.2 Variation of Photoelectric current with intensity of light.

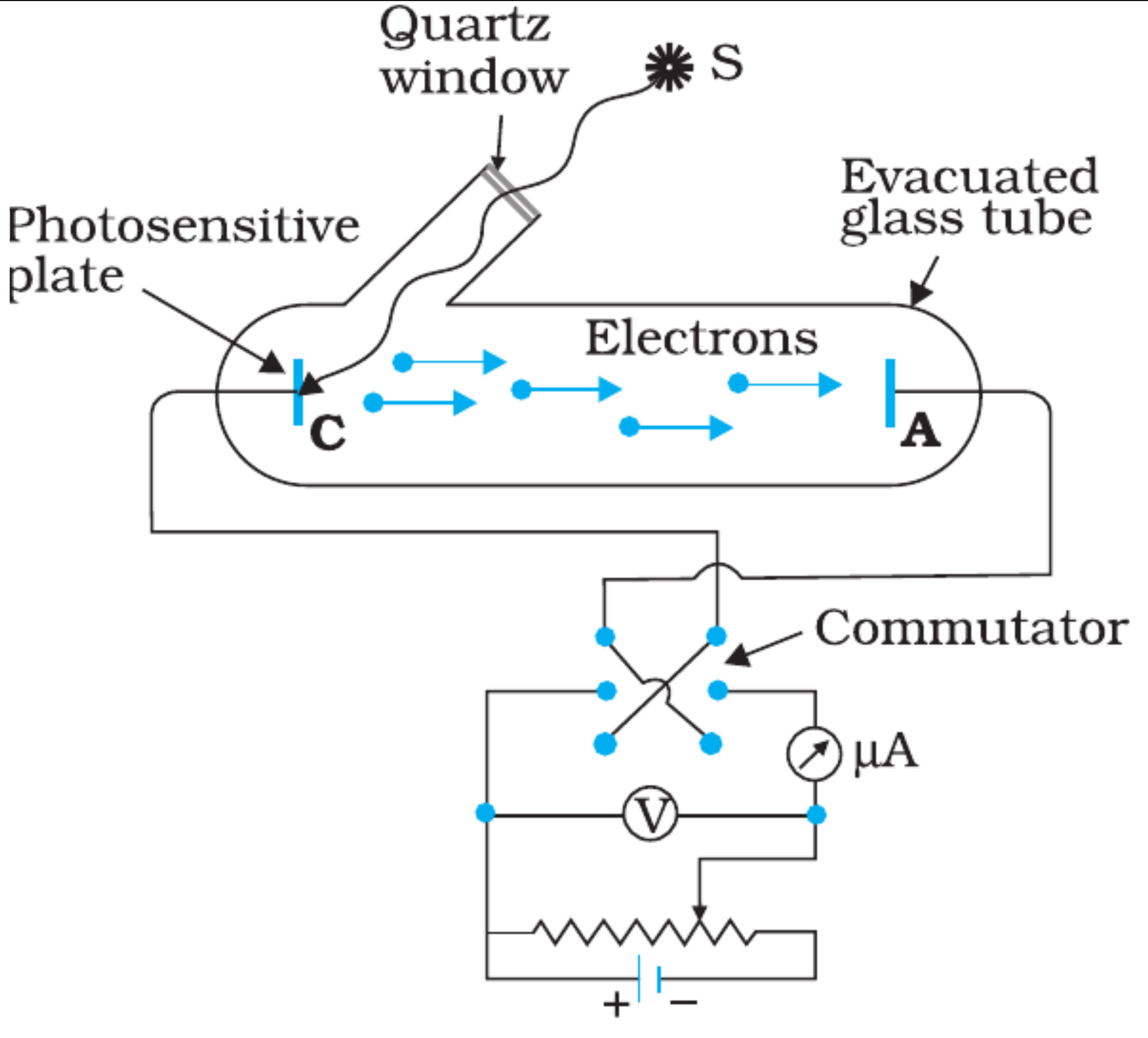


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(1) Effect of intensity of light on photocurrent

(Keep frequency and potential constant, vary only intensity.)

* It is found that the photocurrent increases linearly with intensity of incident light as shown graphically in Figure.

* The number of photoelectrons emitted per second (or photocurrent) is directly proportional to the intensity of incident radiation.

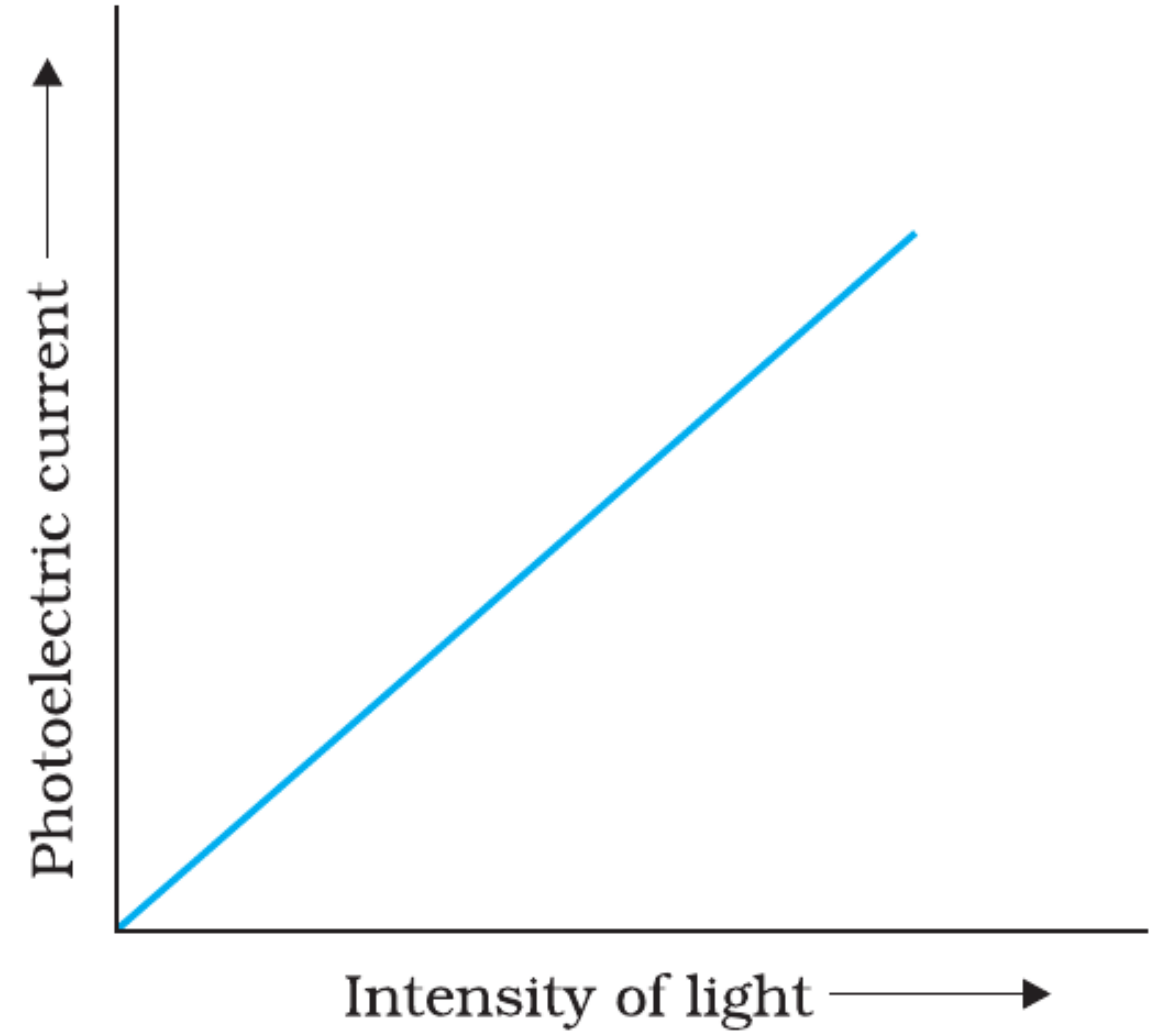


FIGURE 11.2 Variation of Photoelectric current with intensity of light.

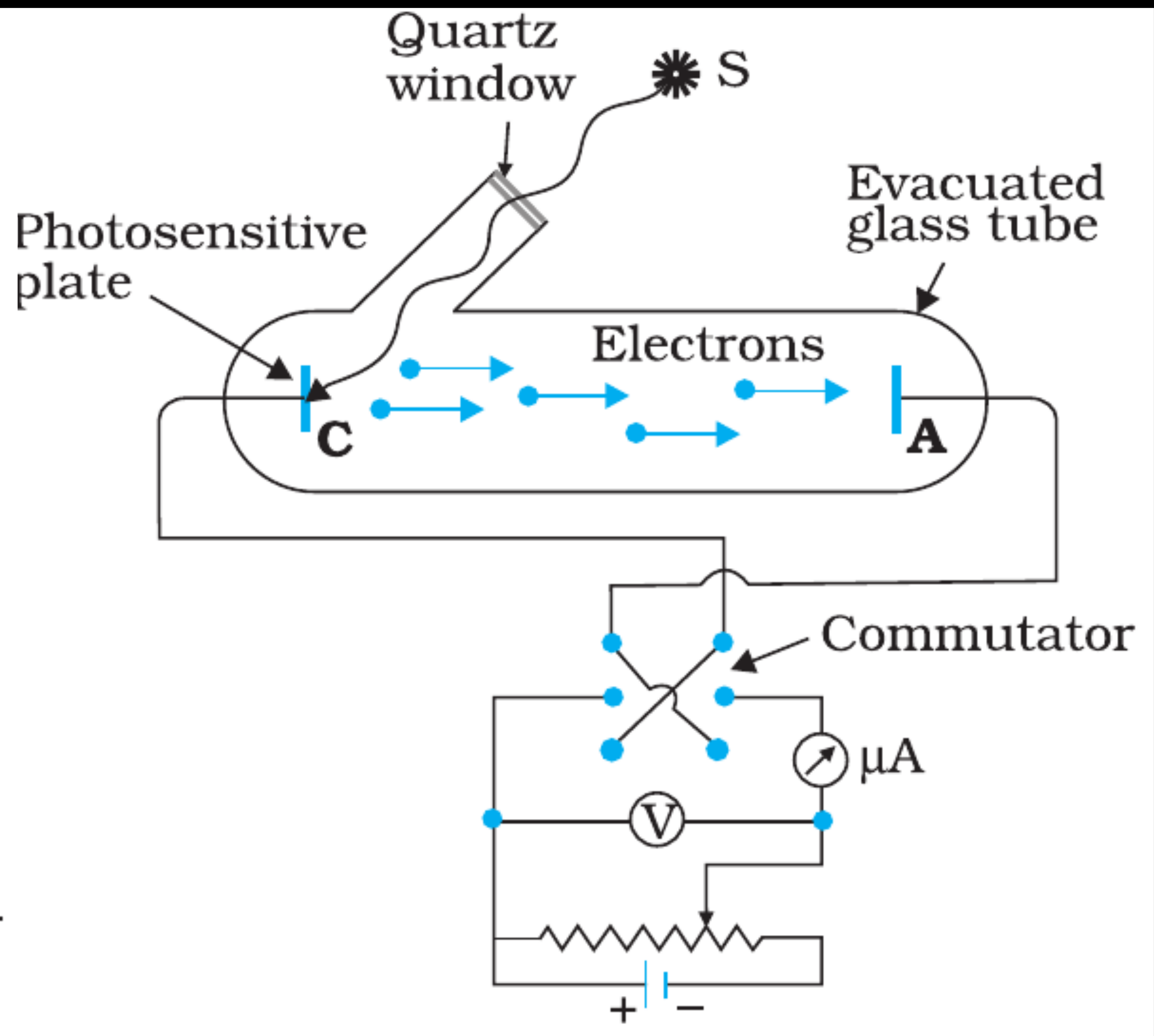


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

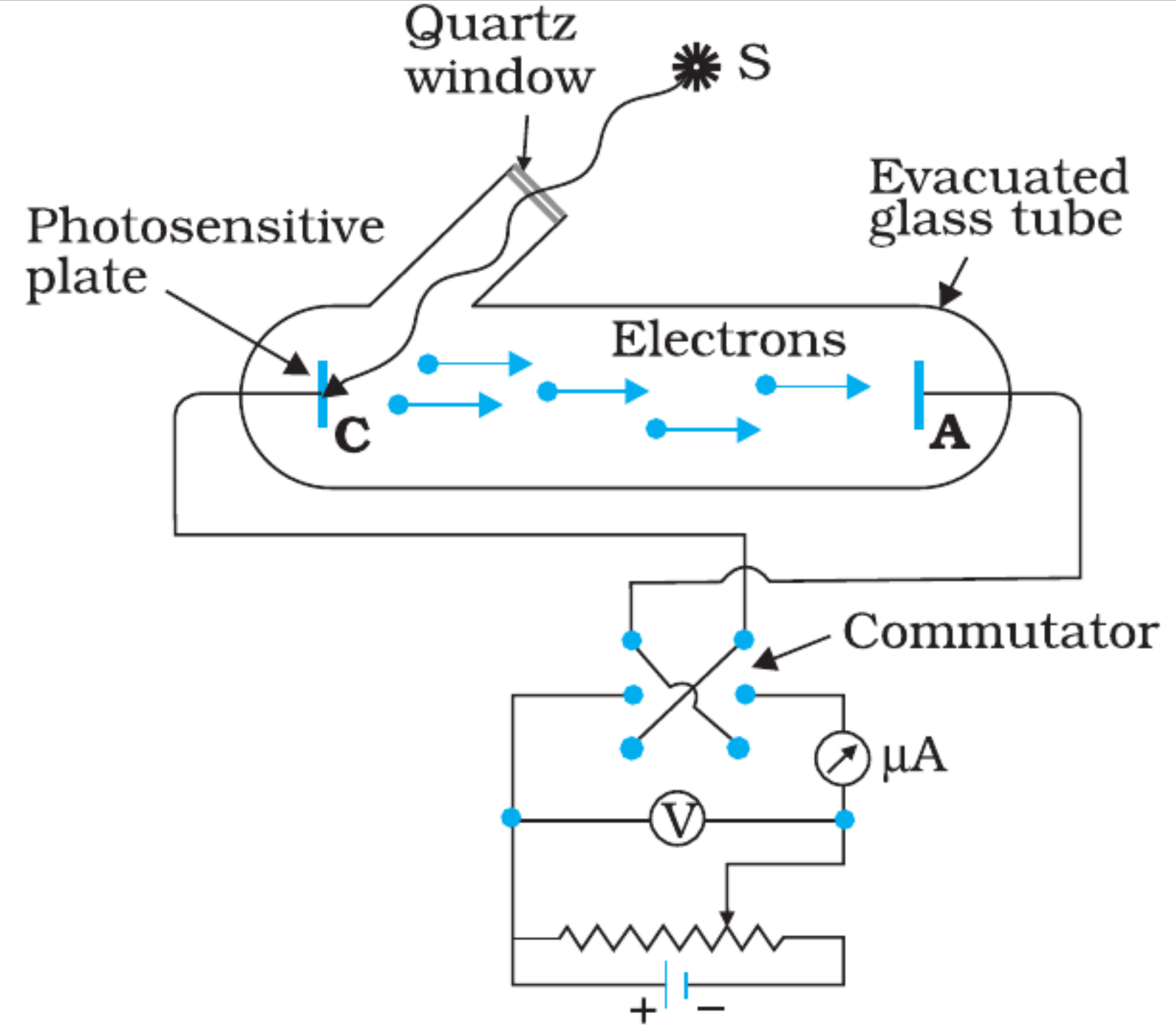


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

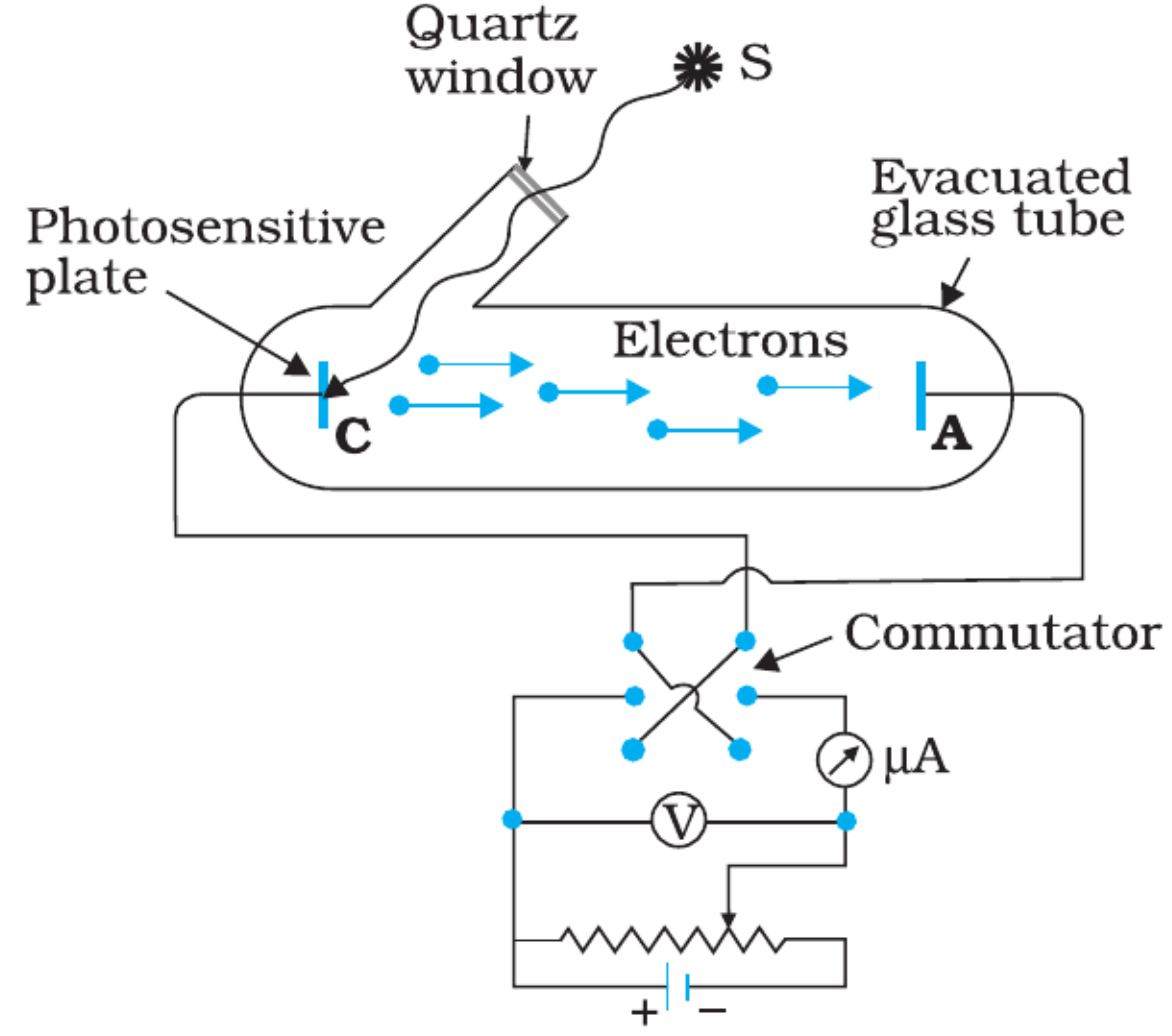


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

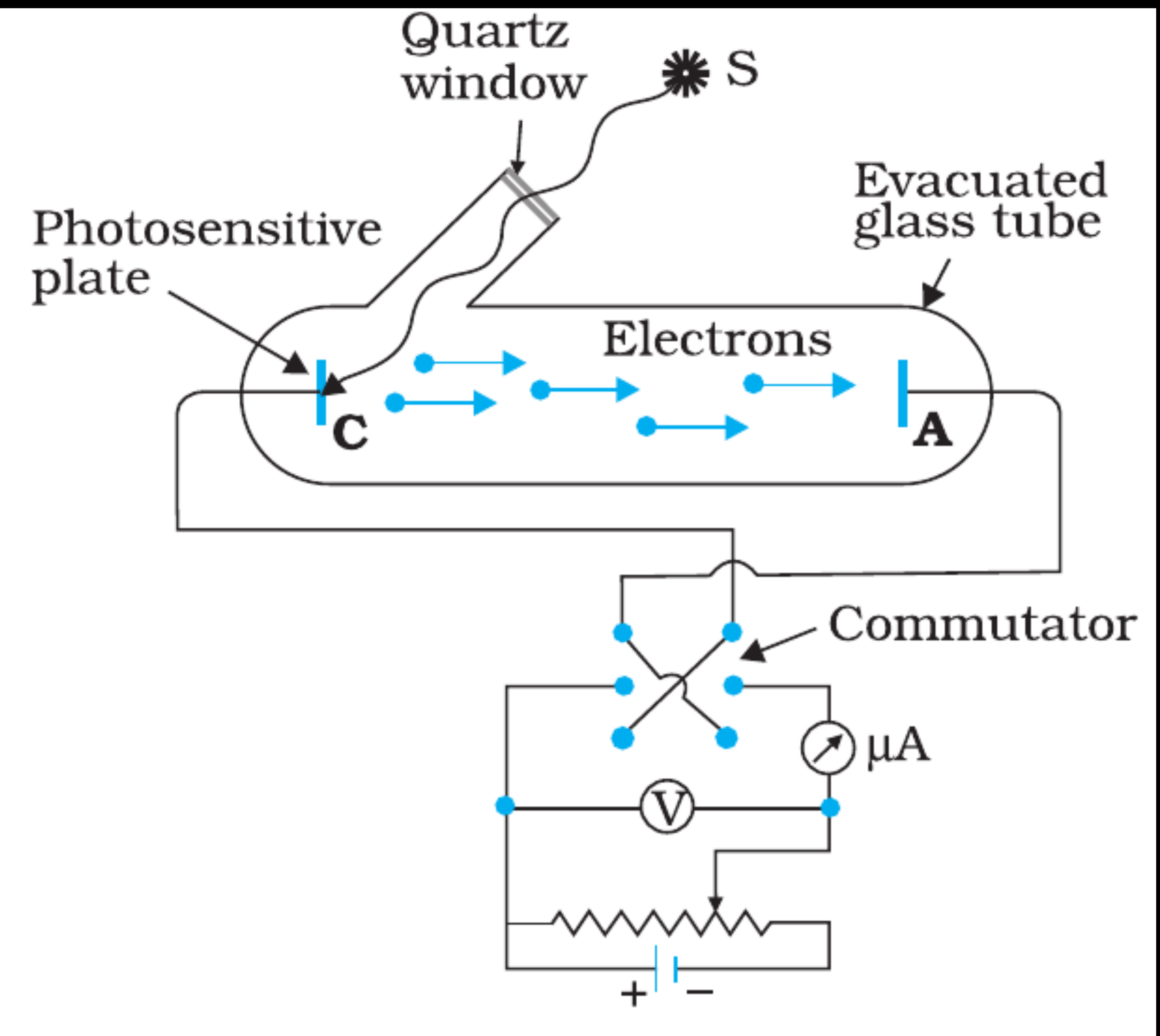


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

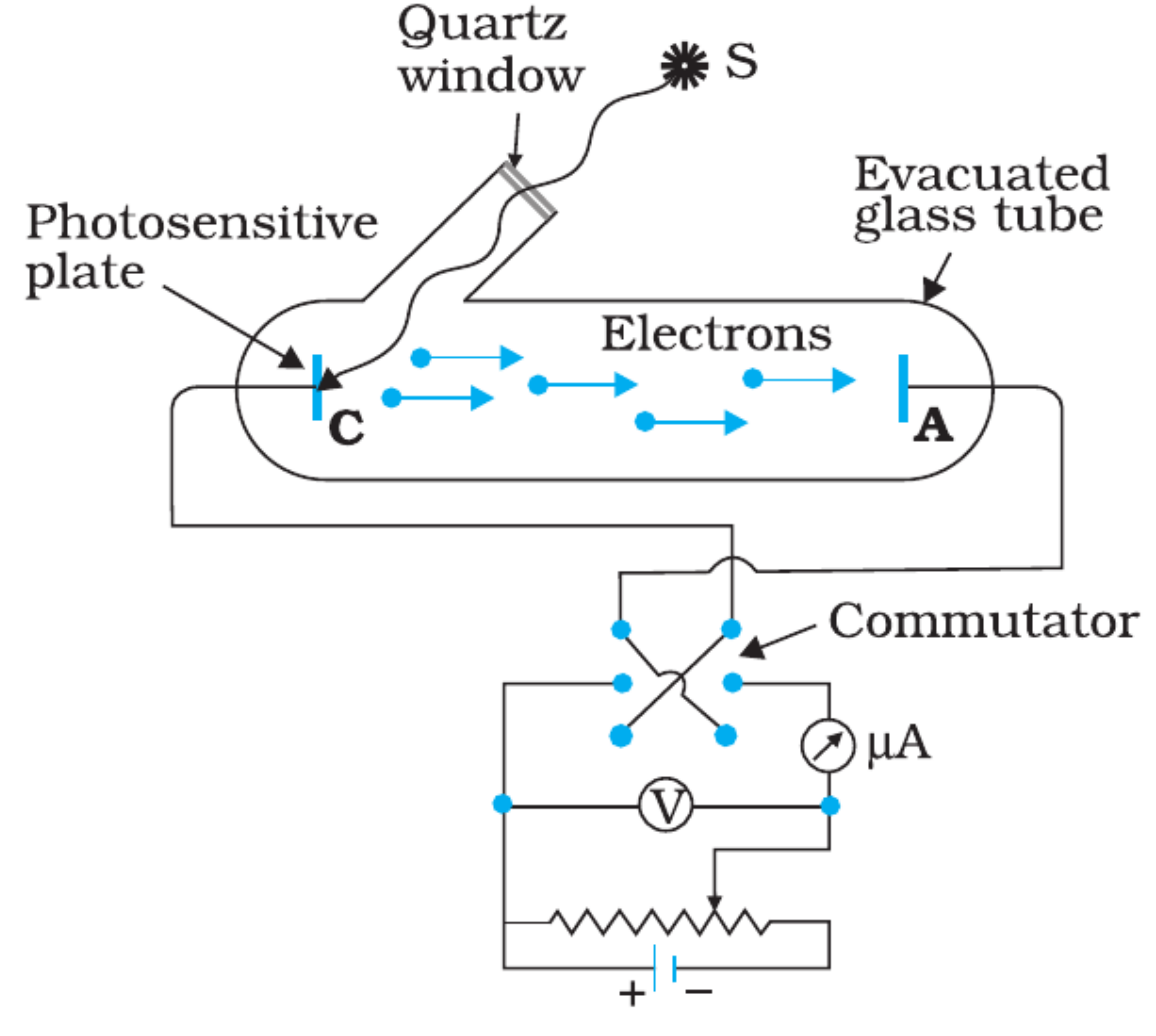


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

* At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

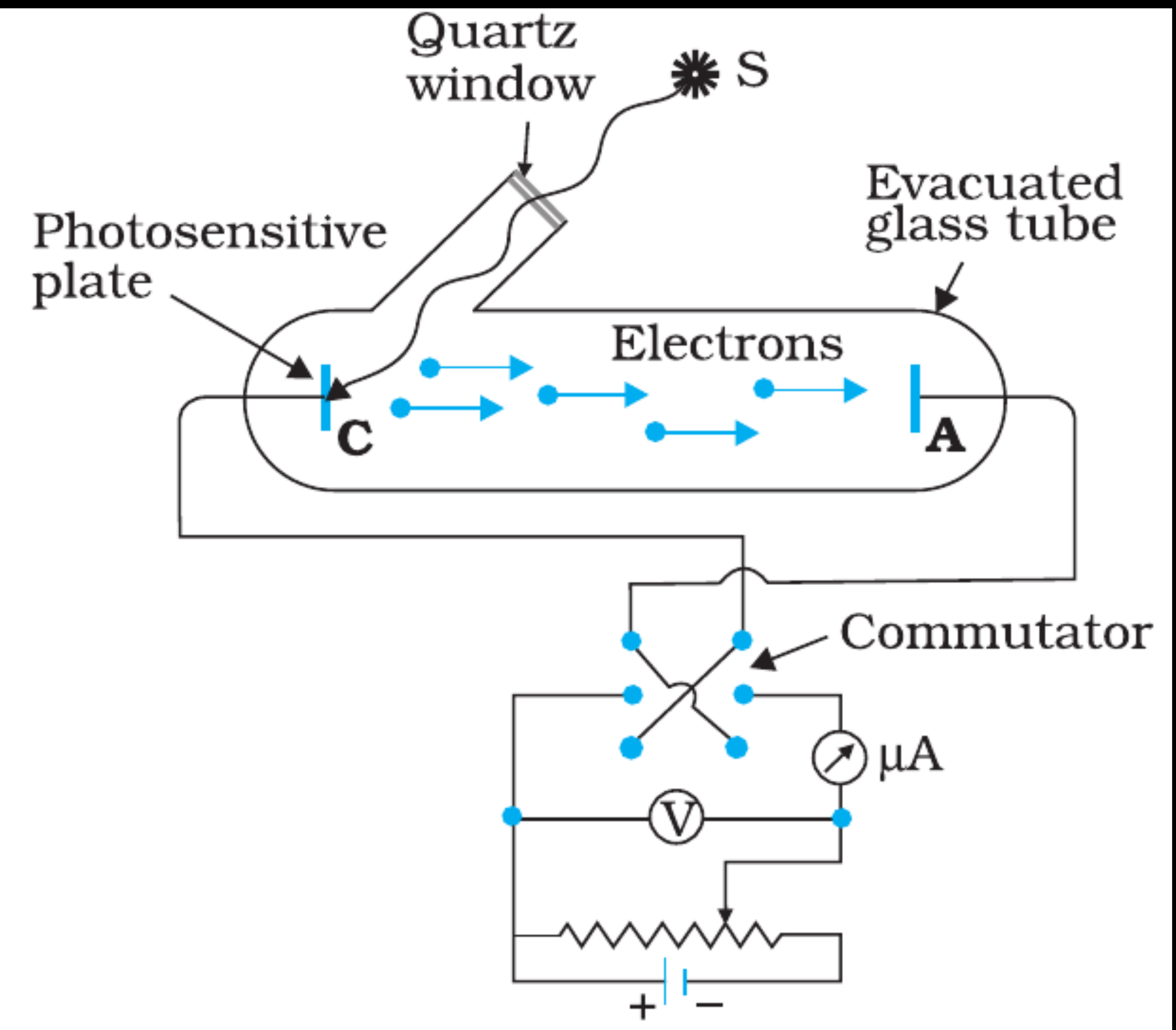


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

* At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

* This maximum value of the photoelectric current is called saturation current.

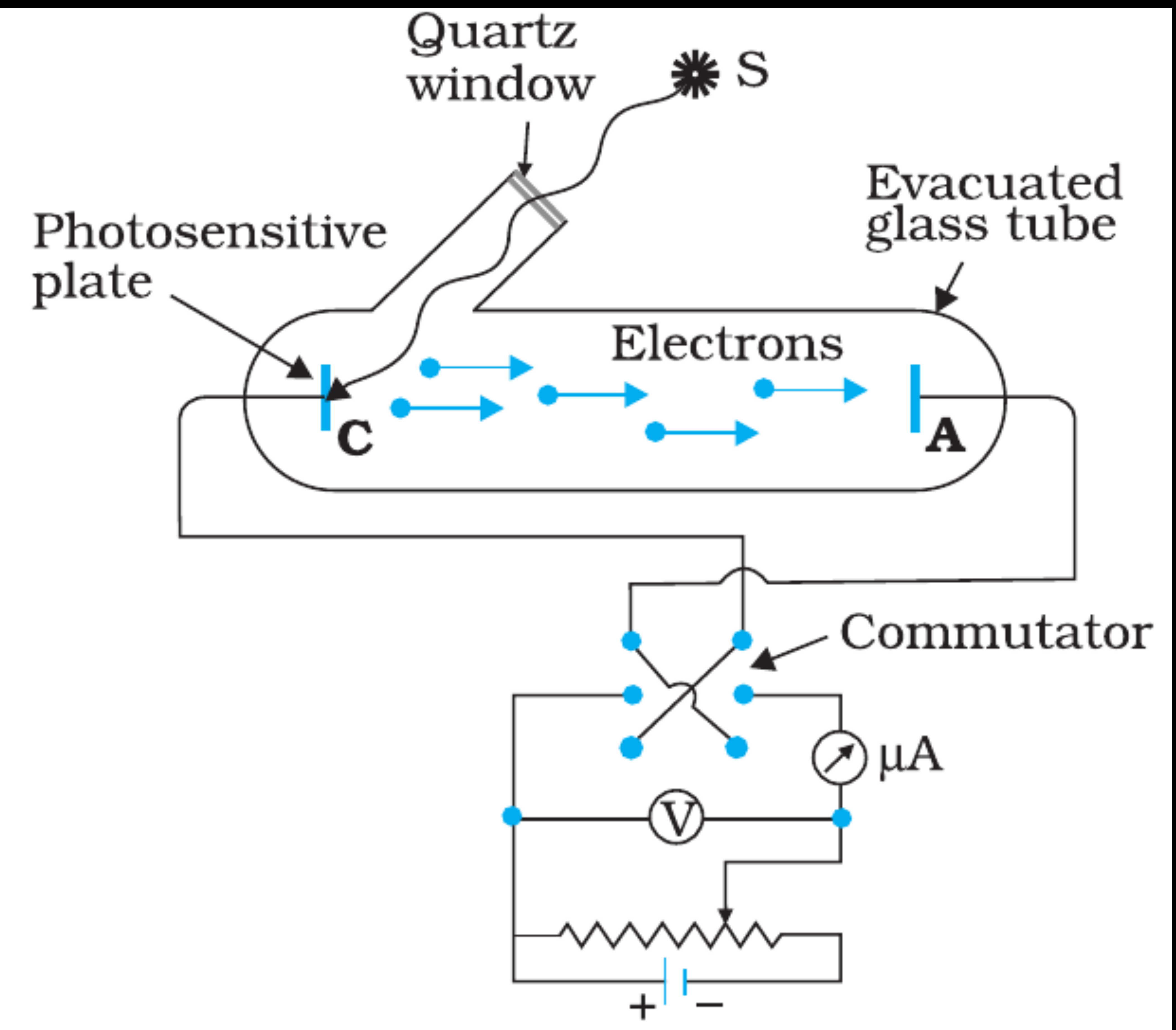


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

* At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

* This maximum value of the photoelectric current is called saturation current.

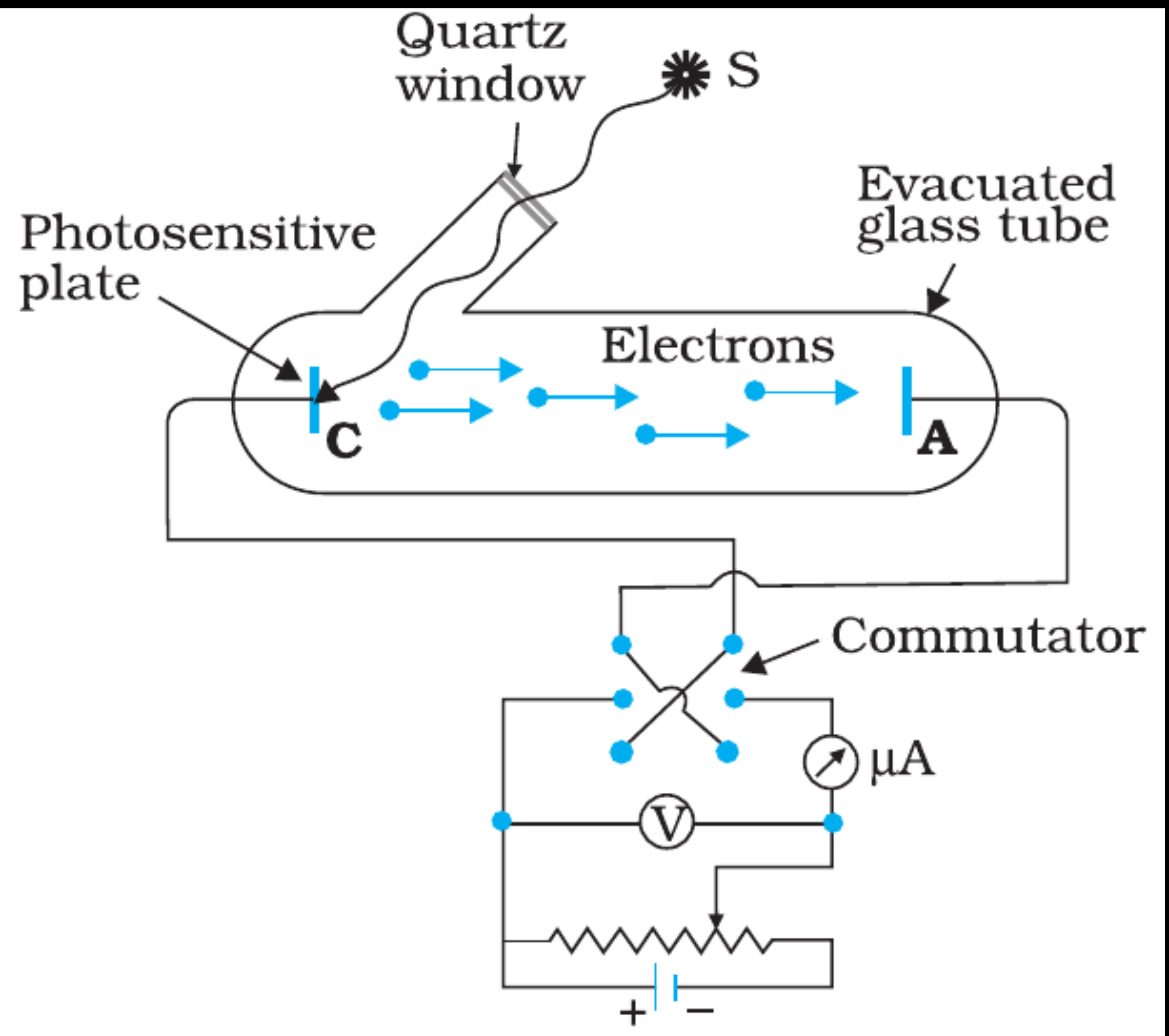


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

(ii) Negative Potential (Retarding Potential) and make it increasingly negative gradually.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

* At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

* This maximum value of the photoelectric current is called saturation current.

(ii) Negative Potential (Retarding Potential) and make it increasingly negative gradually.

* When the polarity is reversed, the electrons are repelled and only the sufficiently energetic electrons are able to reach the collector A.

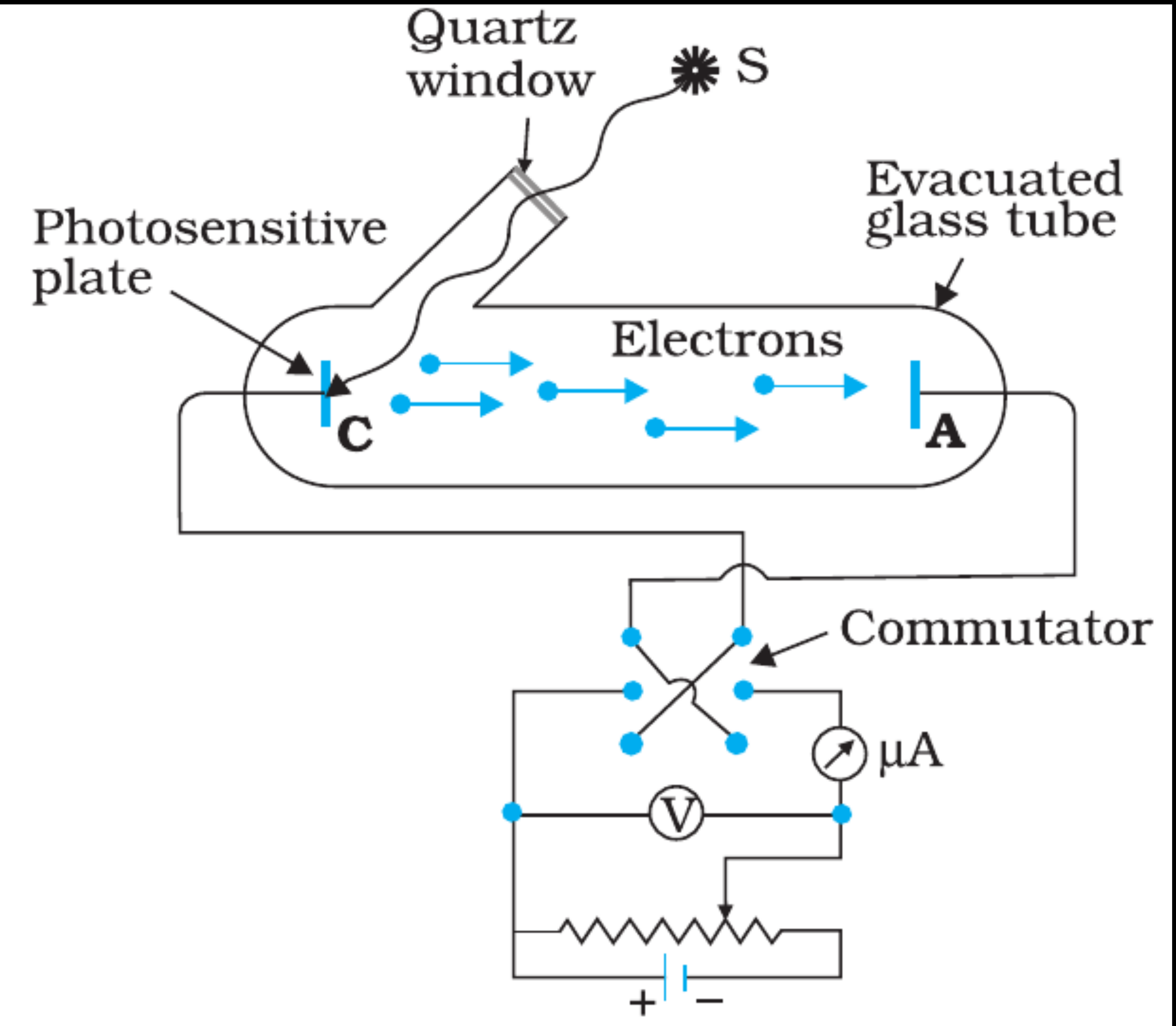


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

* If A is positive w.r.t. C and illuminate the plate C with light of fixed frequency ν and fixed intensity I_1.

* It is found that the photoelectric current increases with increase in positive (accelerating) potential.

* At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

* This maximum value of the photoelectric current is called saturation current.

(ii) Negative Potential (Retarding Potential) and make it increasingly negative gradually.

* When the polarity is reversed, the electrons are repelled and only the sufficiently energetic electrons are able to reach the collector A.

* At a certain negative voltage (V_0) called stoping potential:
Even the most energetic electrons cannot reach A. Photocurrent becomes zero.

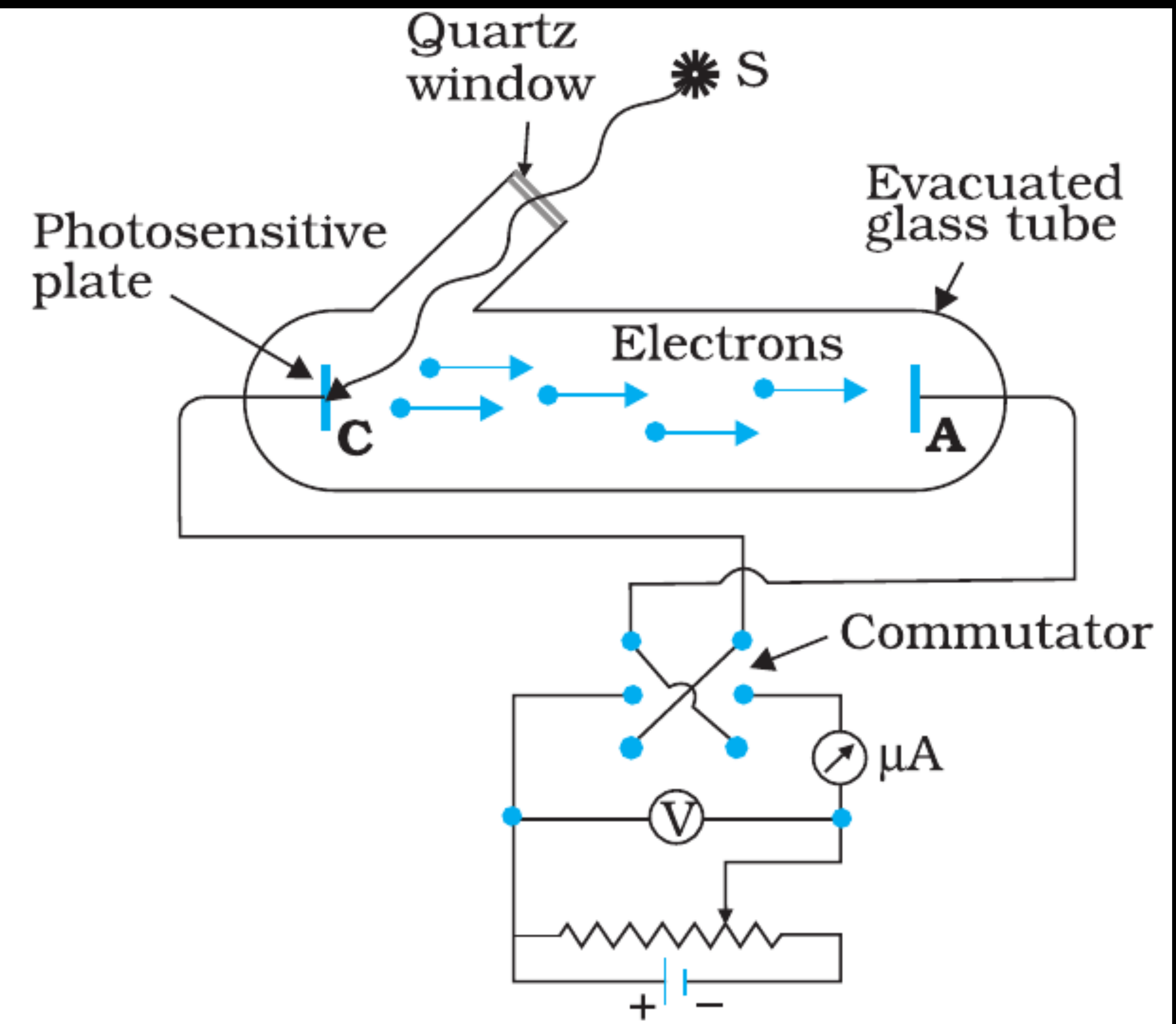


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

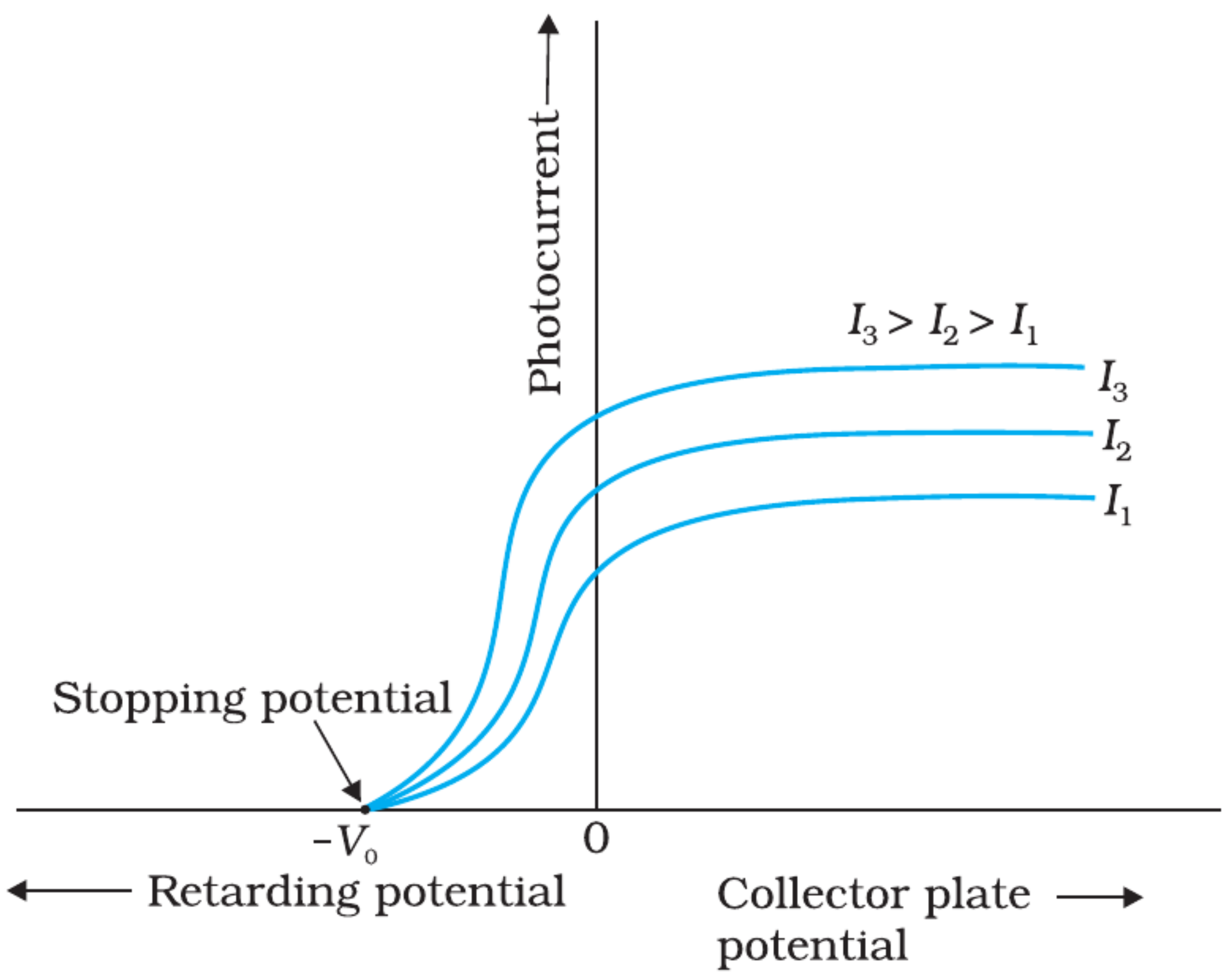


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

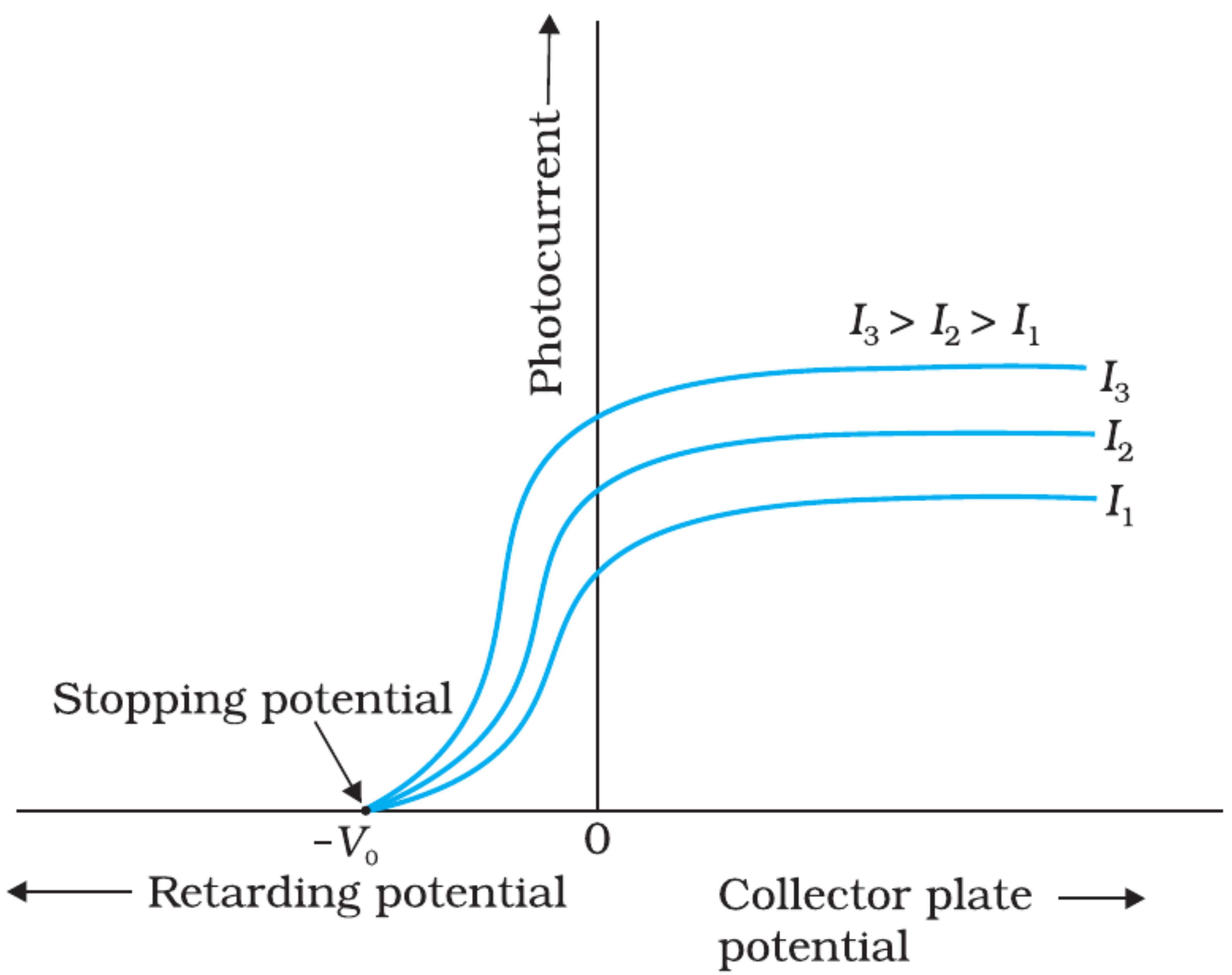


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

* Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy (K_{max}), so that

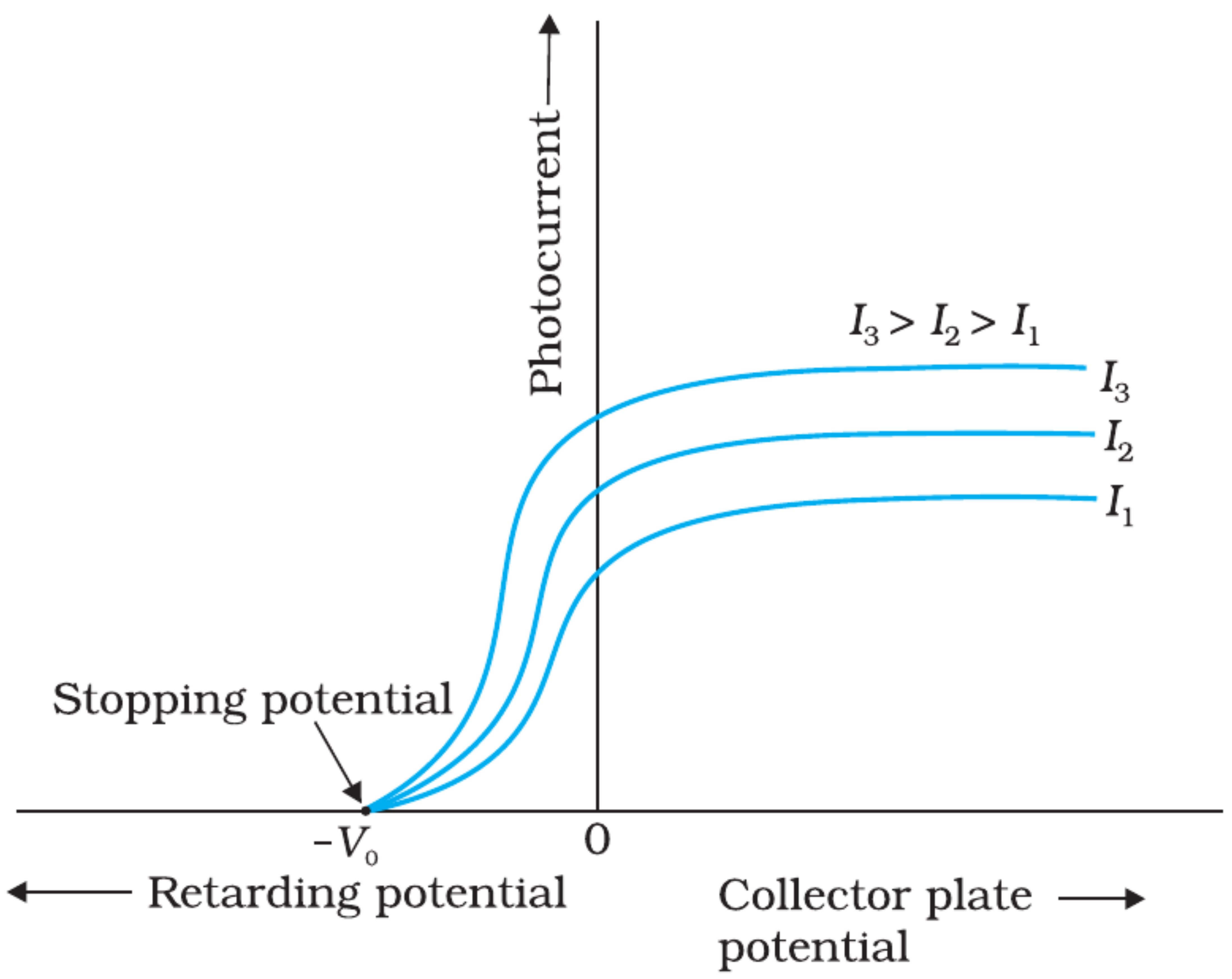


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

* Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy (K_{max}), so that

$$\boxed{K_{\text{max}} = eV_0}$$

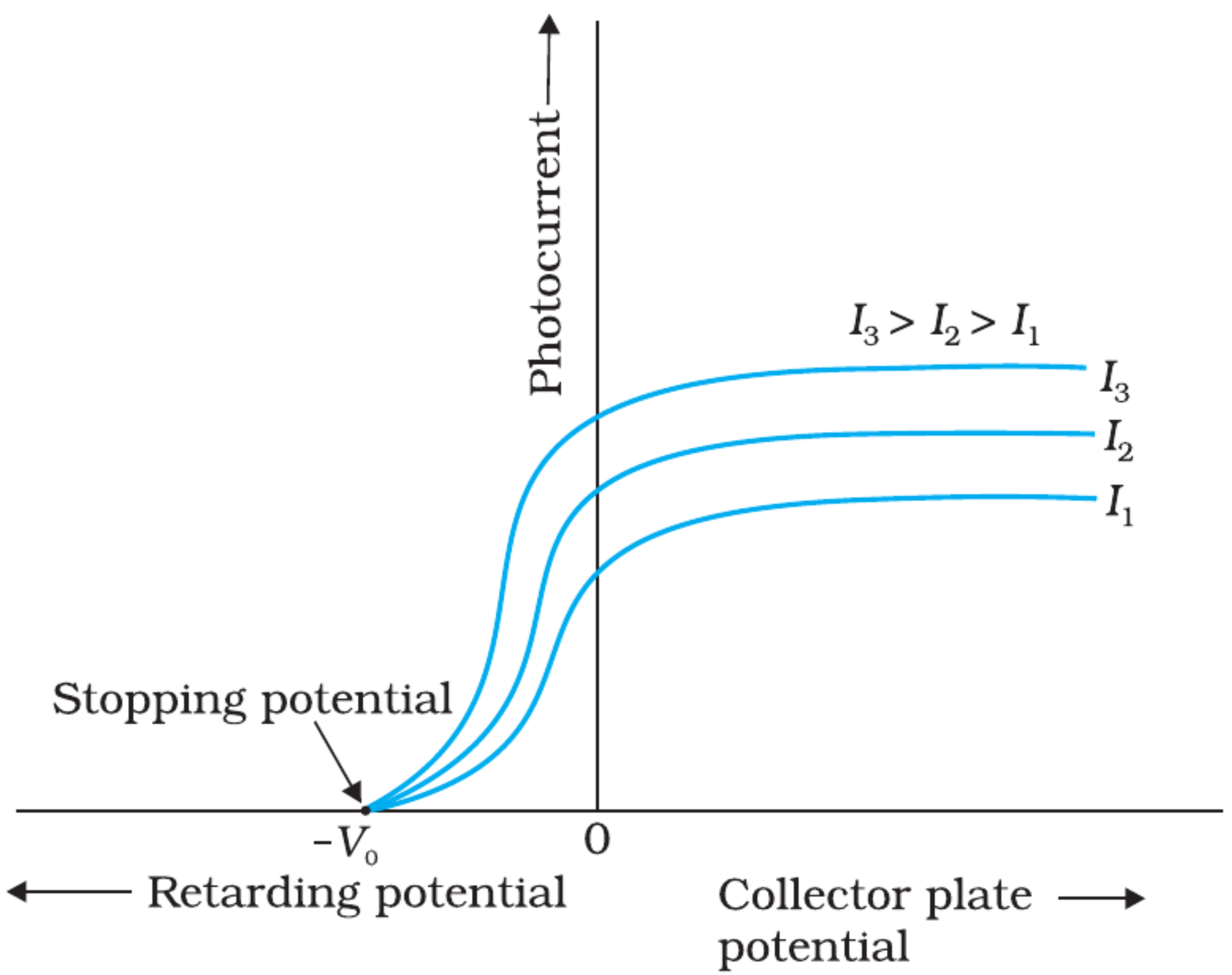


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

* Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy (K_{max}), so that

$$\boxed{K_{\text{max}} = eV_0}$$

* For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.

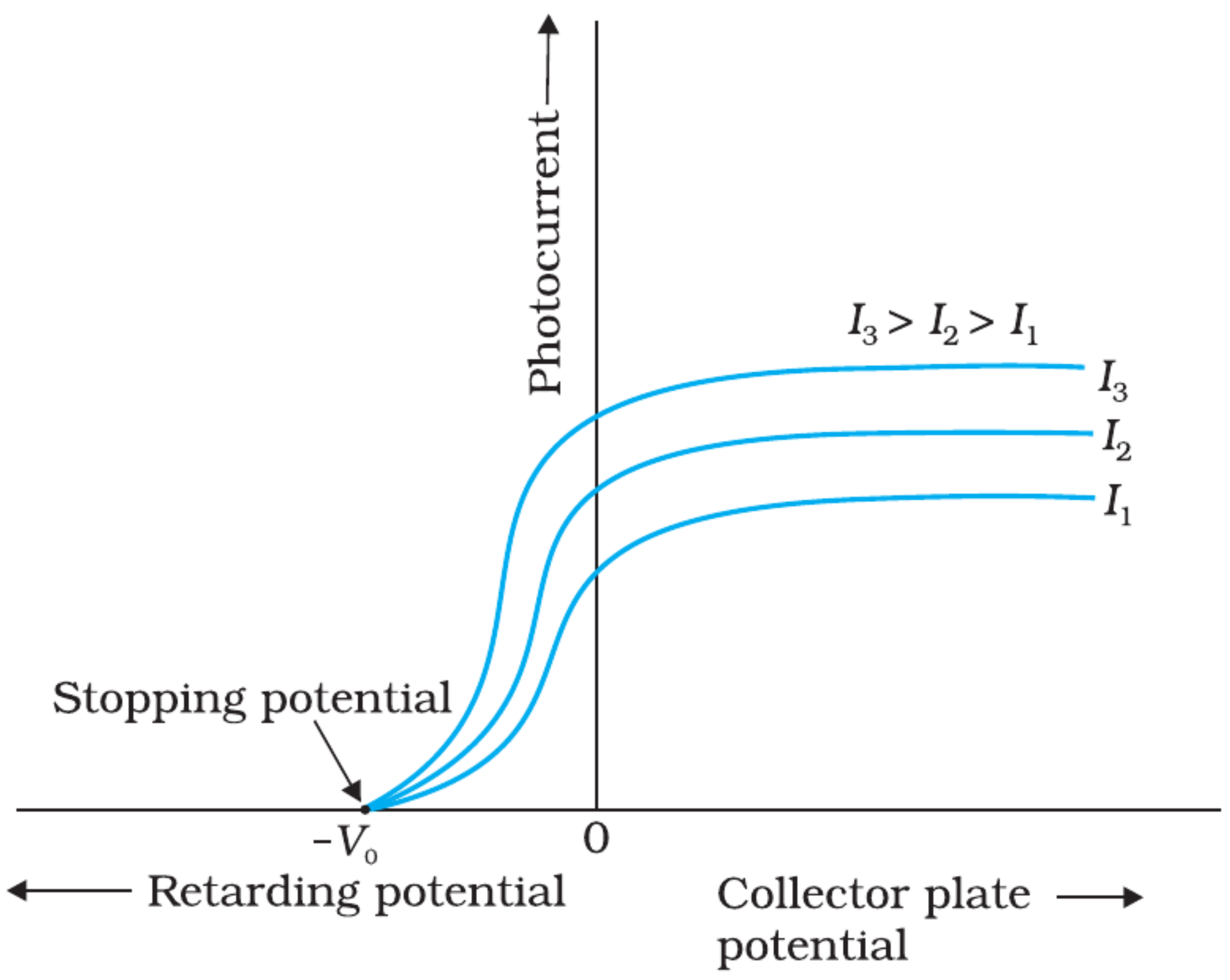


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

* Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy (K_{max}), so that

$$\boxed{K_{\text{max}} = eV_0}$$

* For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.

* For a given frequency of the incident radiation, the stopping potential is independent of its intensity.

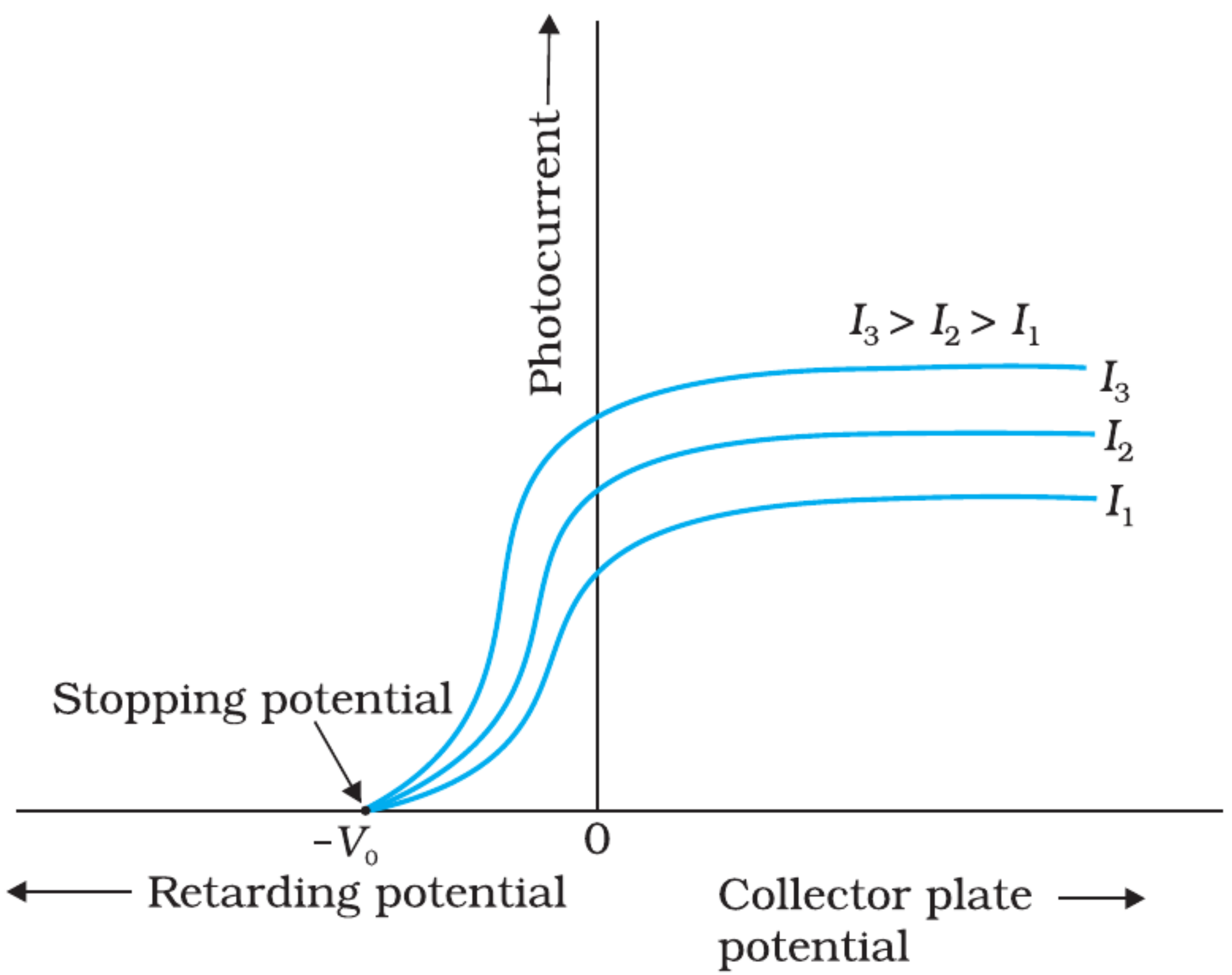


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stoping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

* All the photoelectrons emitted from the metal do not have the same energy.

* Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy (K_{max}), so that

$$\boxed{K_{\max} = eV_0}$$

* For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.

* For a given frequency of the incident radiation, the stopping potential is independent of its intensity.

* In other words, the maximum kinetic energy of photoelectrons depends on the light source and the emitter plate material, but is independent of intensity of incident radiation.

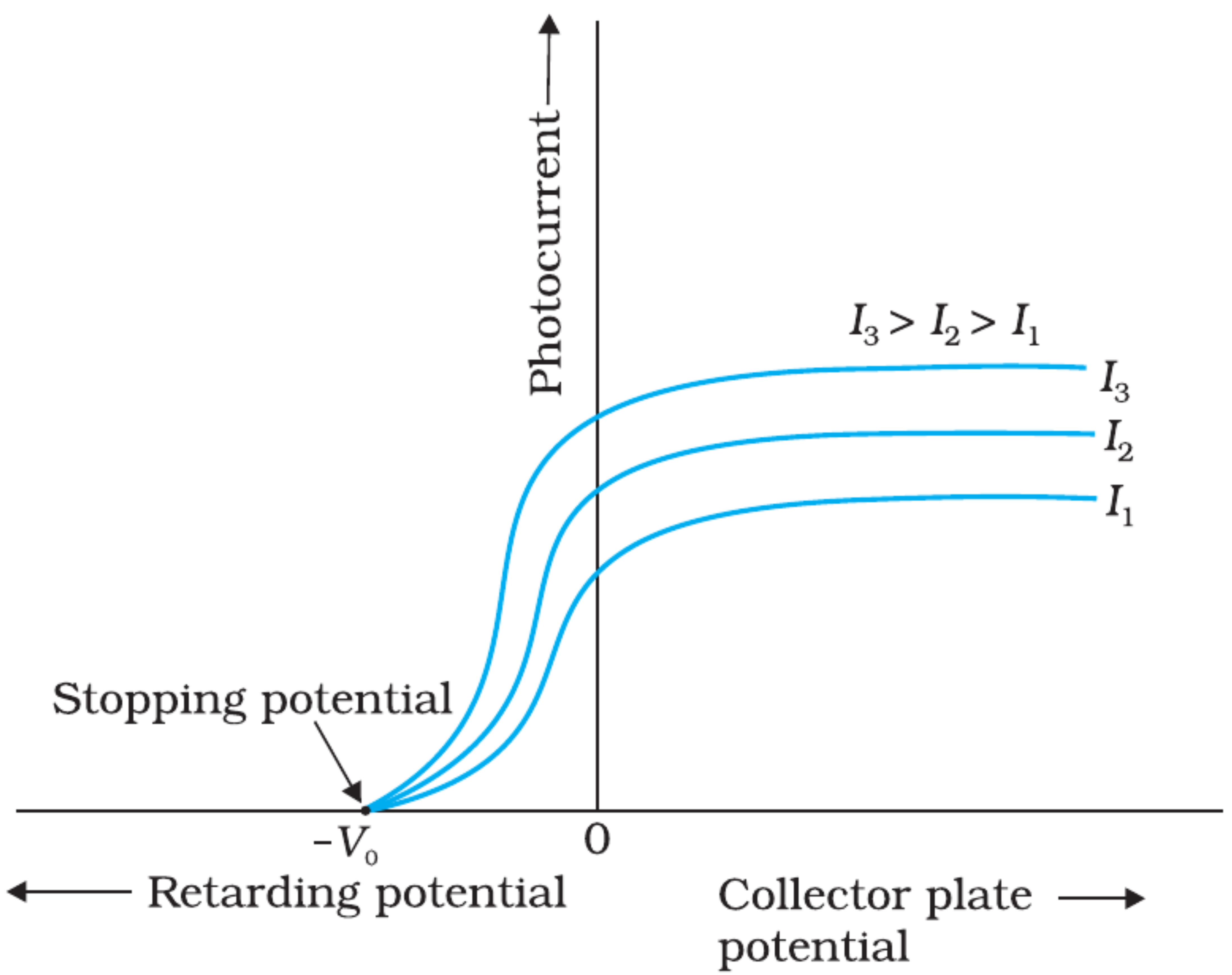


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

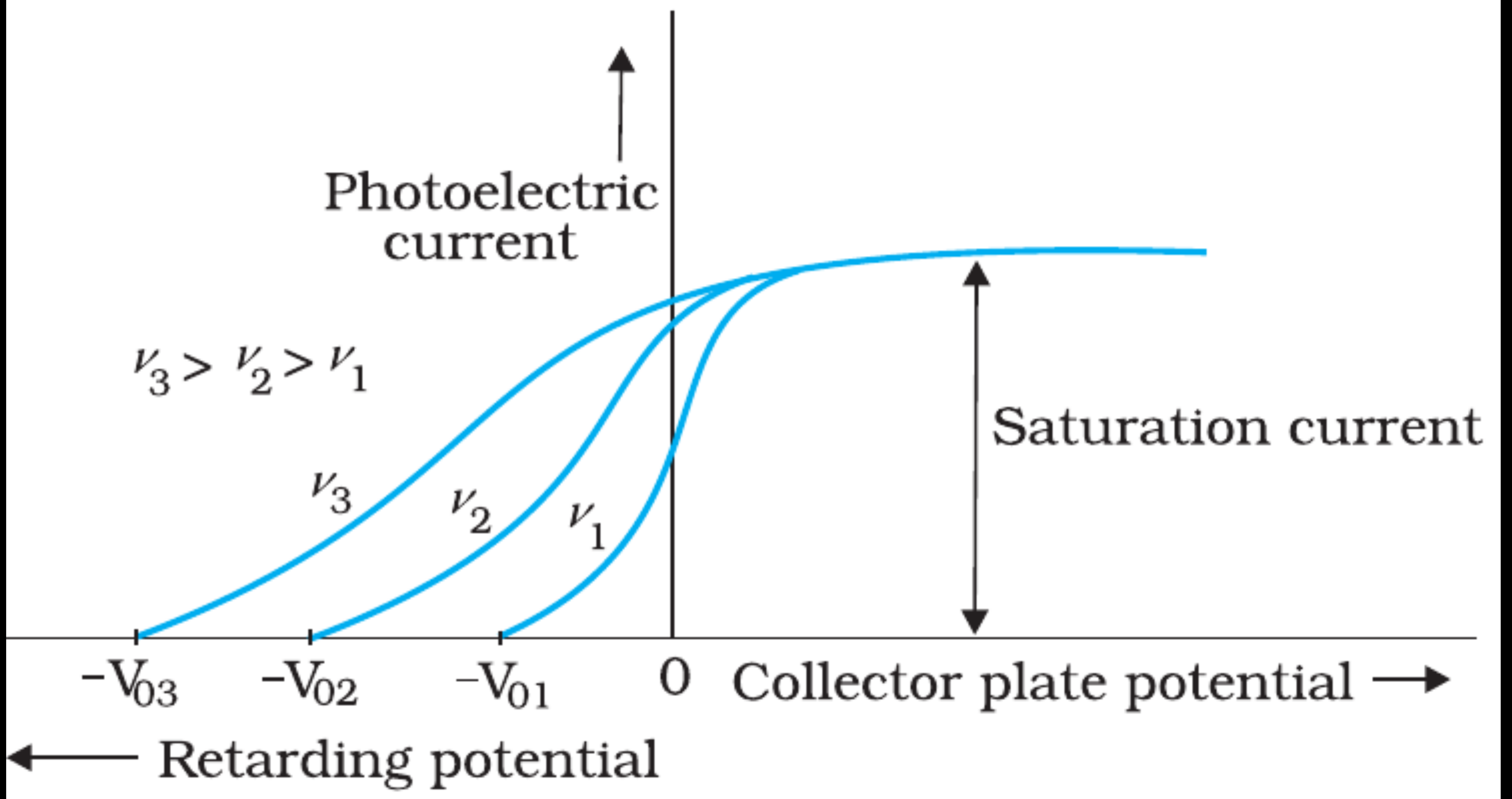


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

* We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

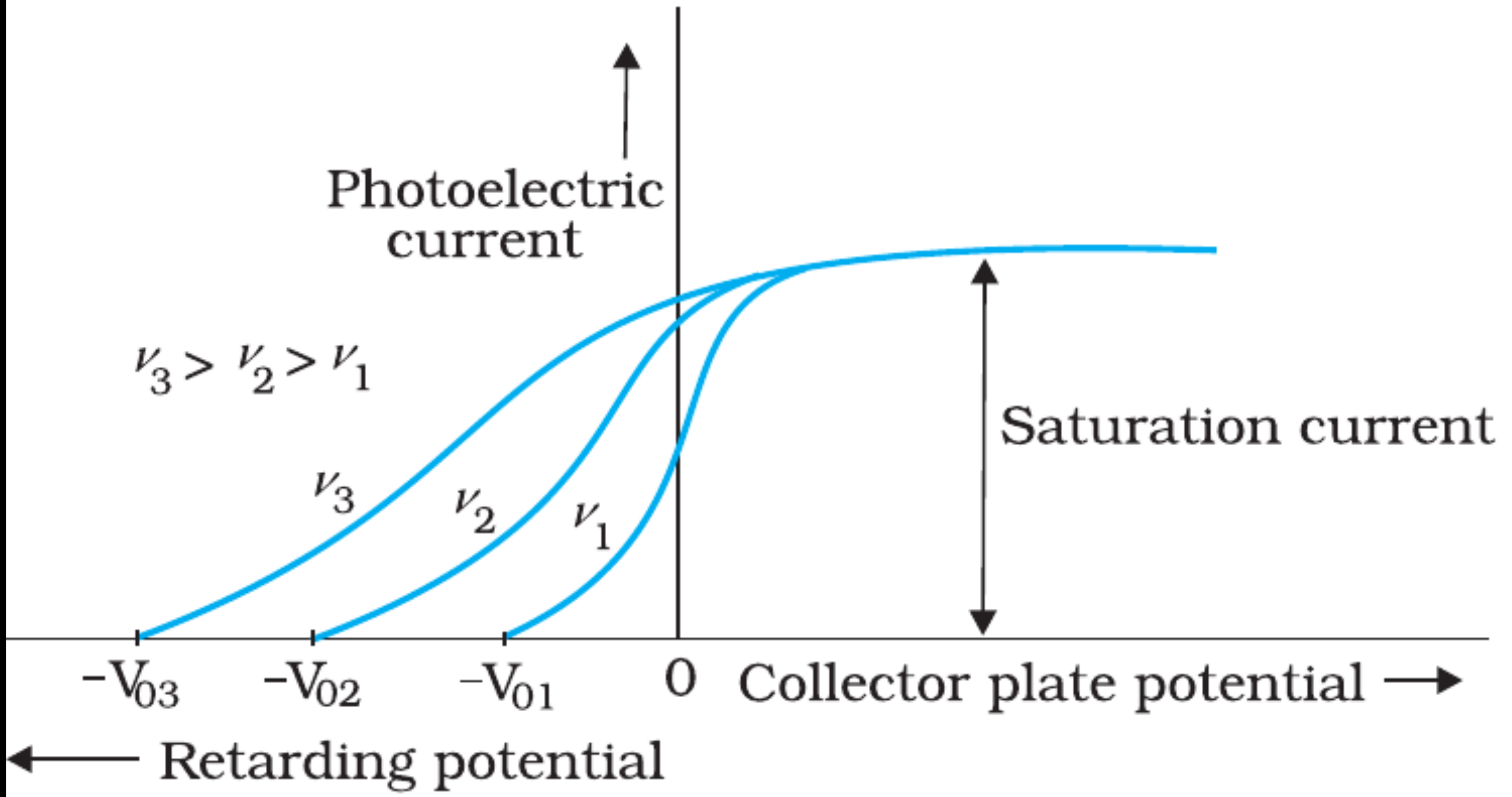


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

* We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

* The energy of the emitted electrons depends on the frequency of the incident radiations.

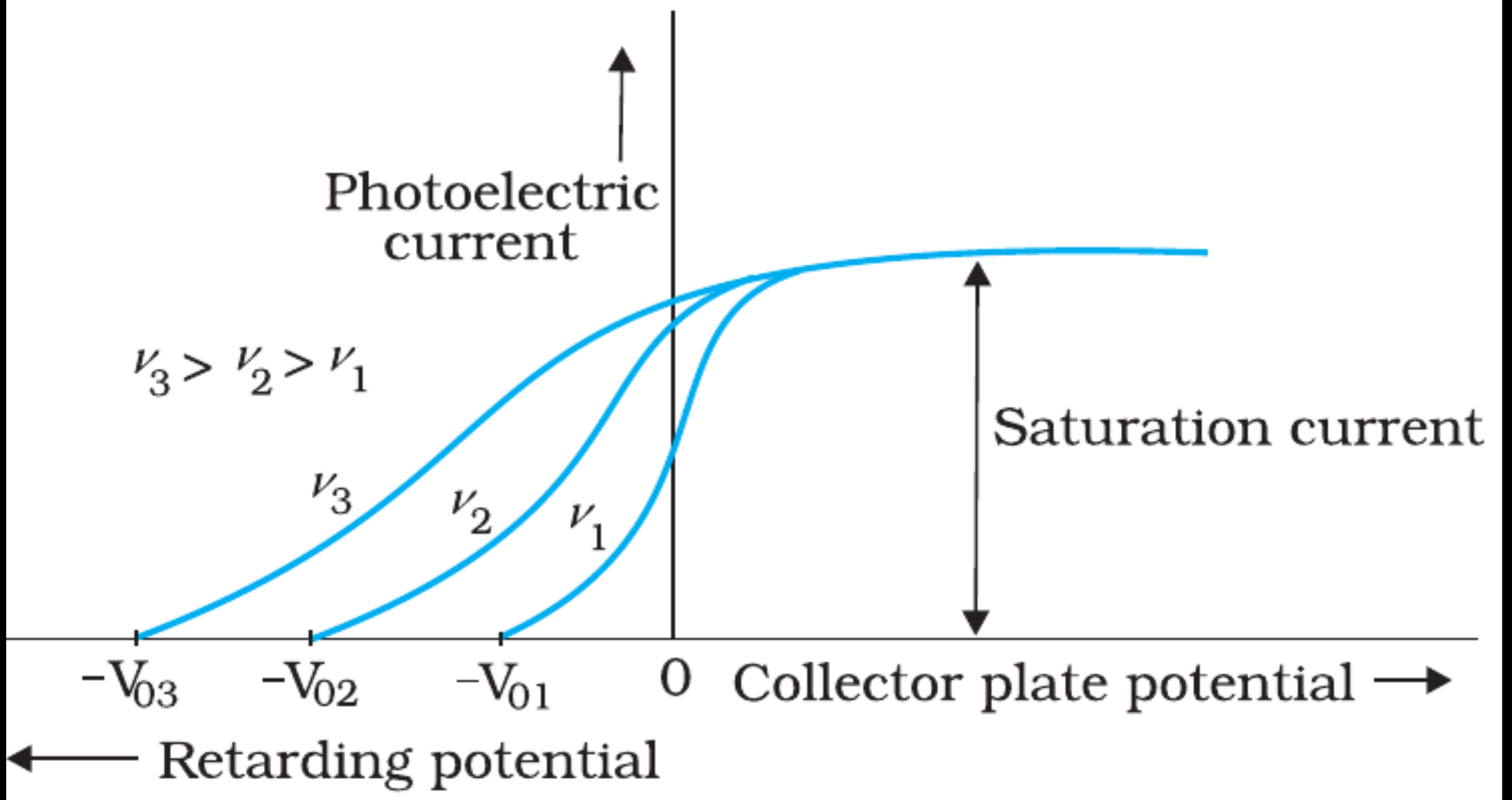


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

* We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

* The energy of the emitted electrons depends on the frequency of the incident radiations.

* The stopping potential is more negative for higher frequencies of incident radiation.

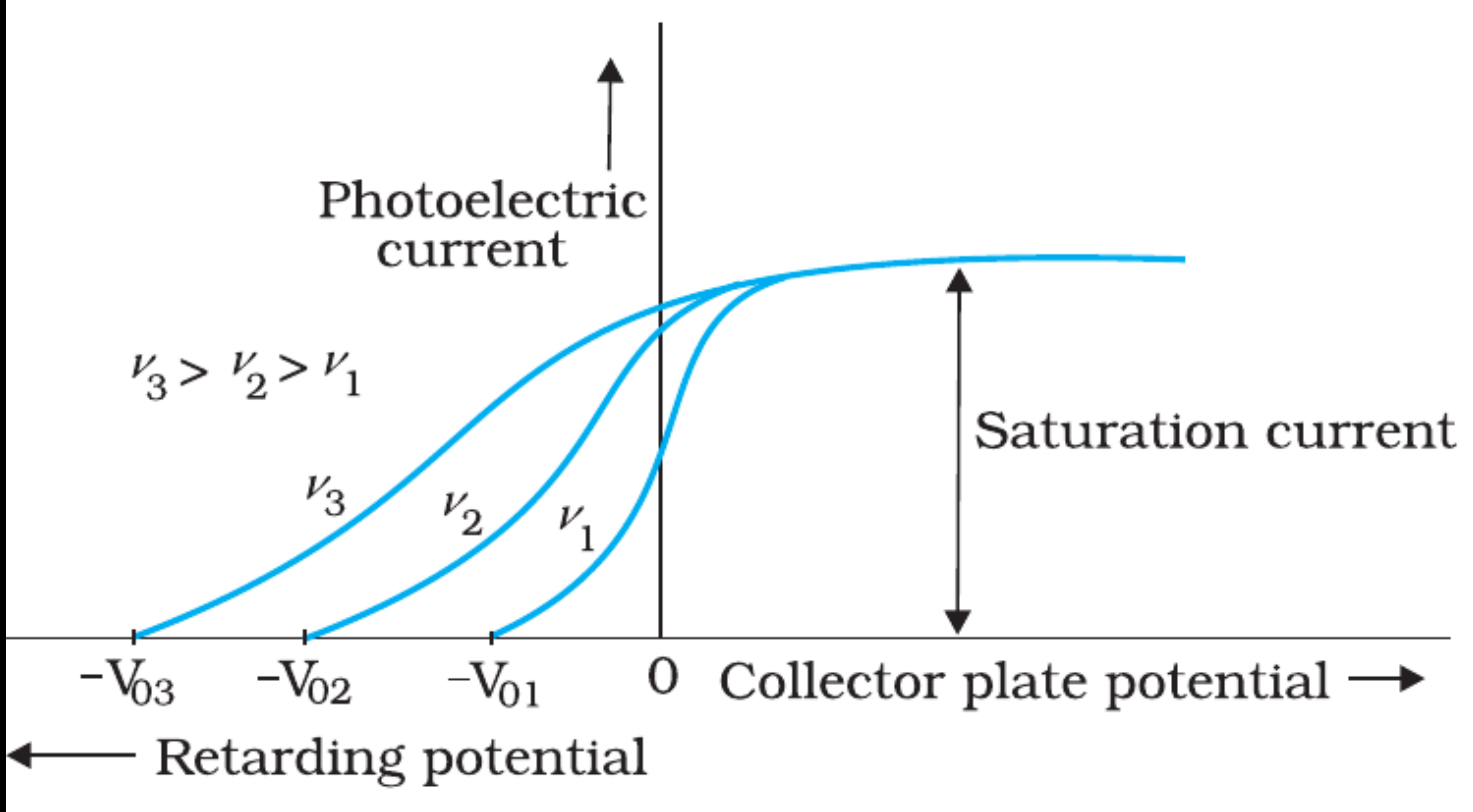


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

* We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

* The energy of the emitted electrons depends on the frequency of the incident radiations.

* The stopping potential is more negative for higher frequencies of incident radiation.

* This implies that greater the requency of incident light, greater is the maximum kinetic energy of the photoelectrons. Consequently, we need greater retarding potential to stop them.

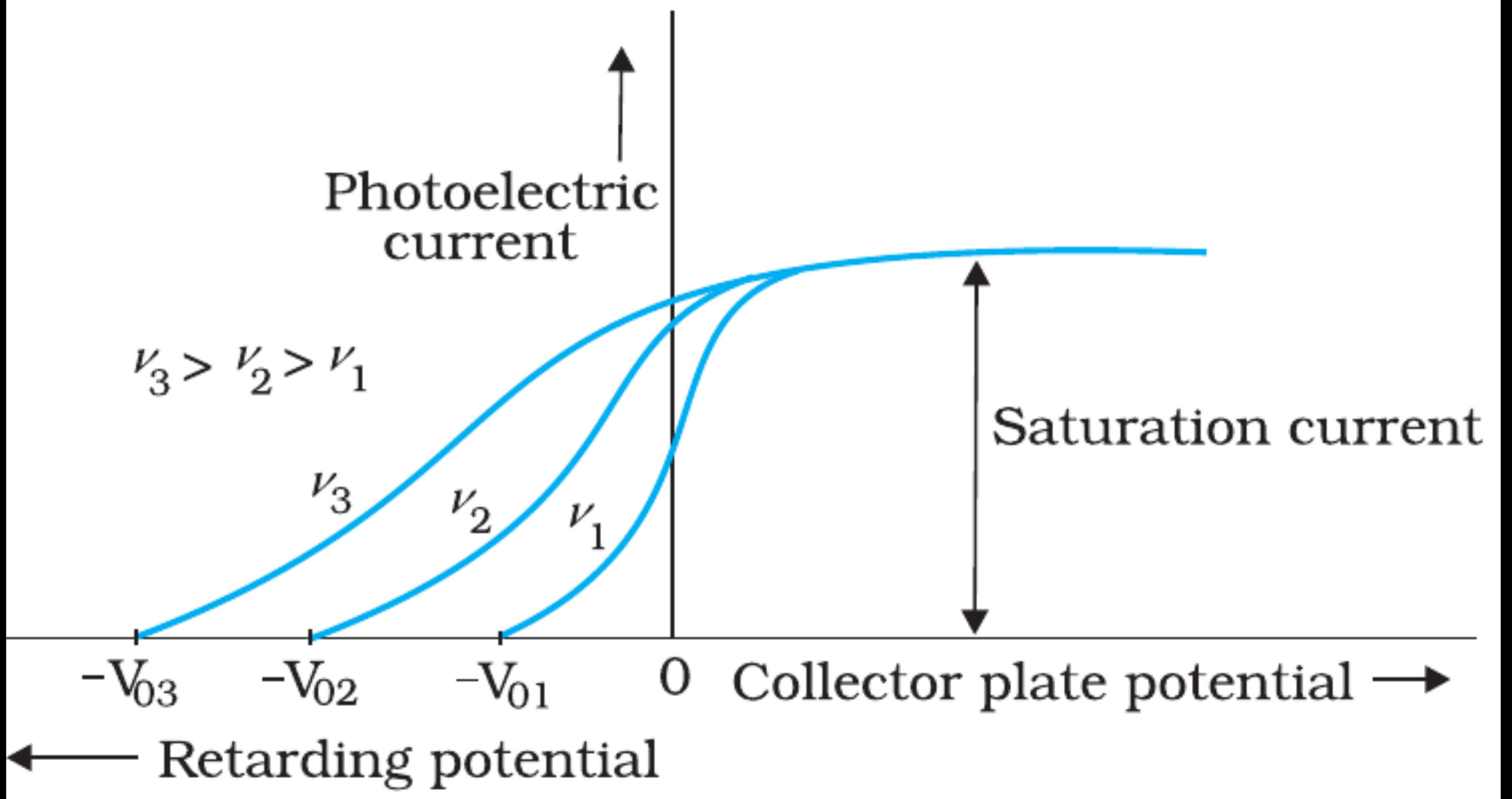


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

* adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

* We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

* The energy of the emitted electrons depends on the frequency of the incident radiations.

* The stopping potential is more negative for higher frequencies of incident radiation.

* This implies that greater the requency of incident light, greater is the maximum kinetic energy of the photoelectrons. Consequently, we need greater retarding potential to stop them.

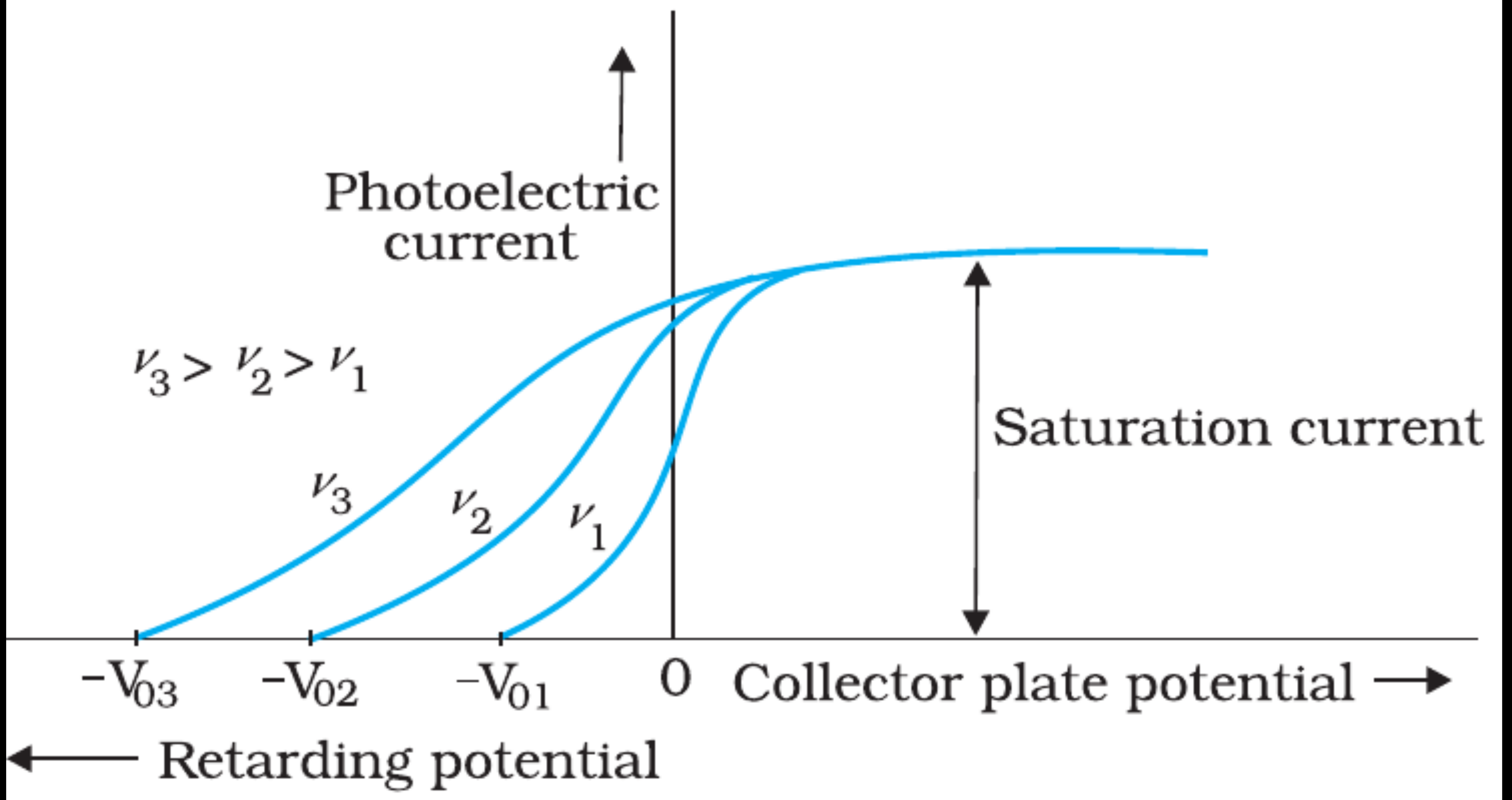


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

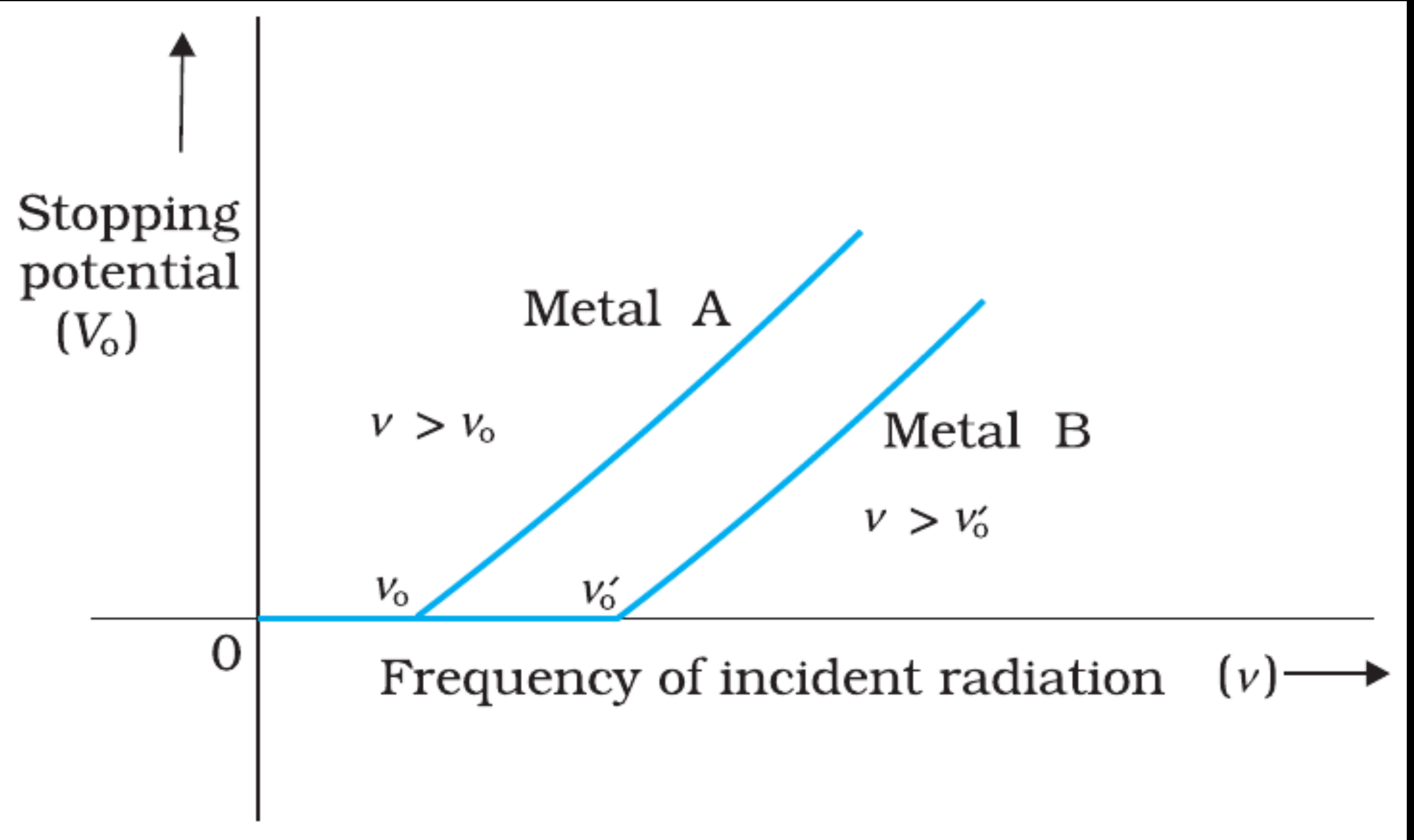


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

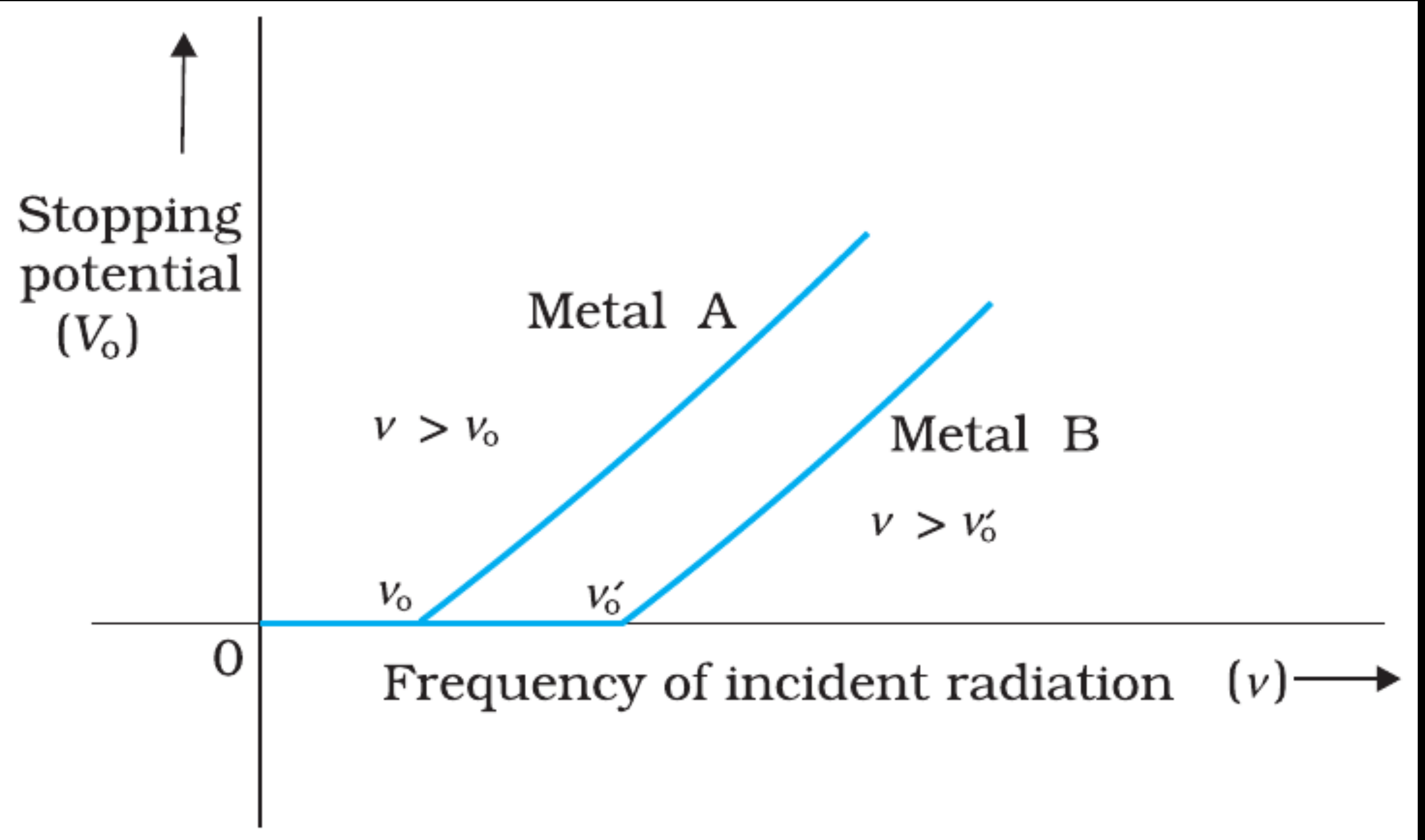


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

(ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

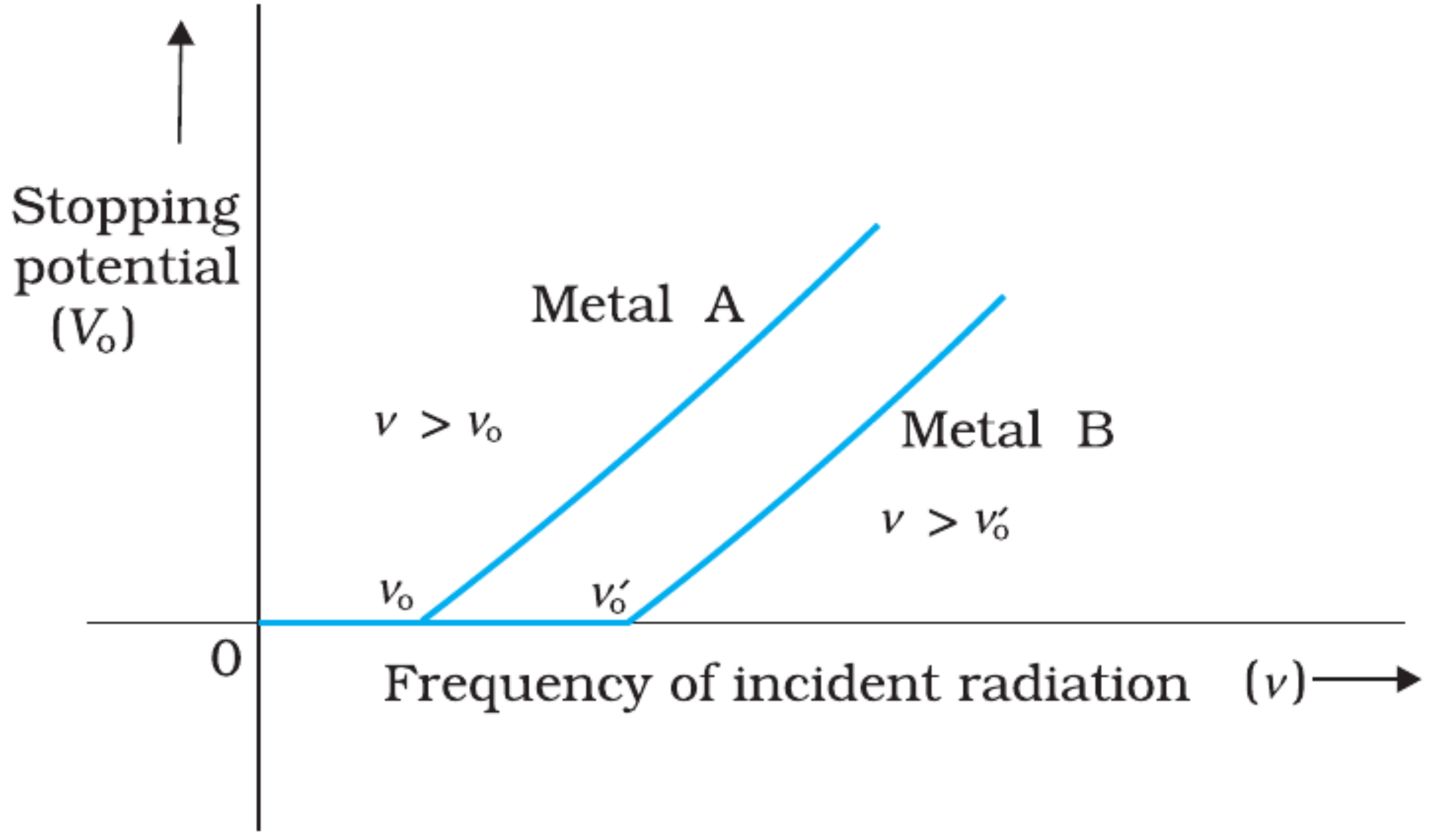


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

(ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

* These observations have two implications:
  (i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

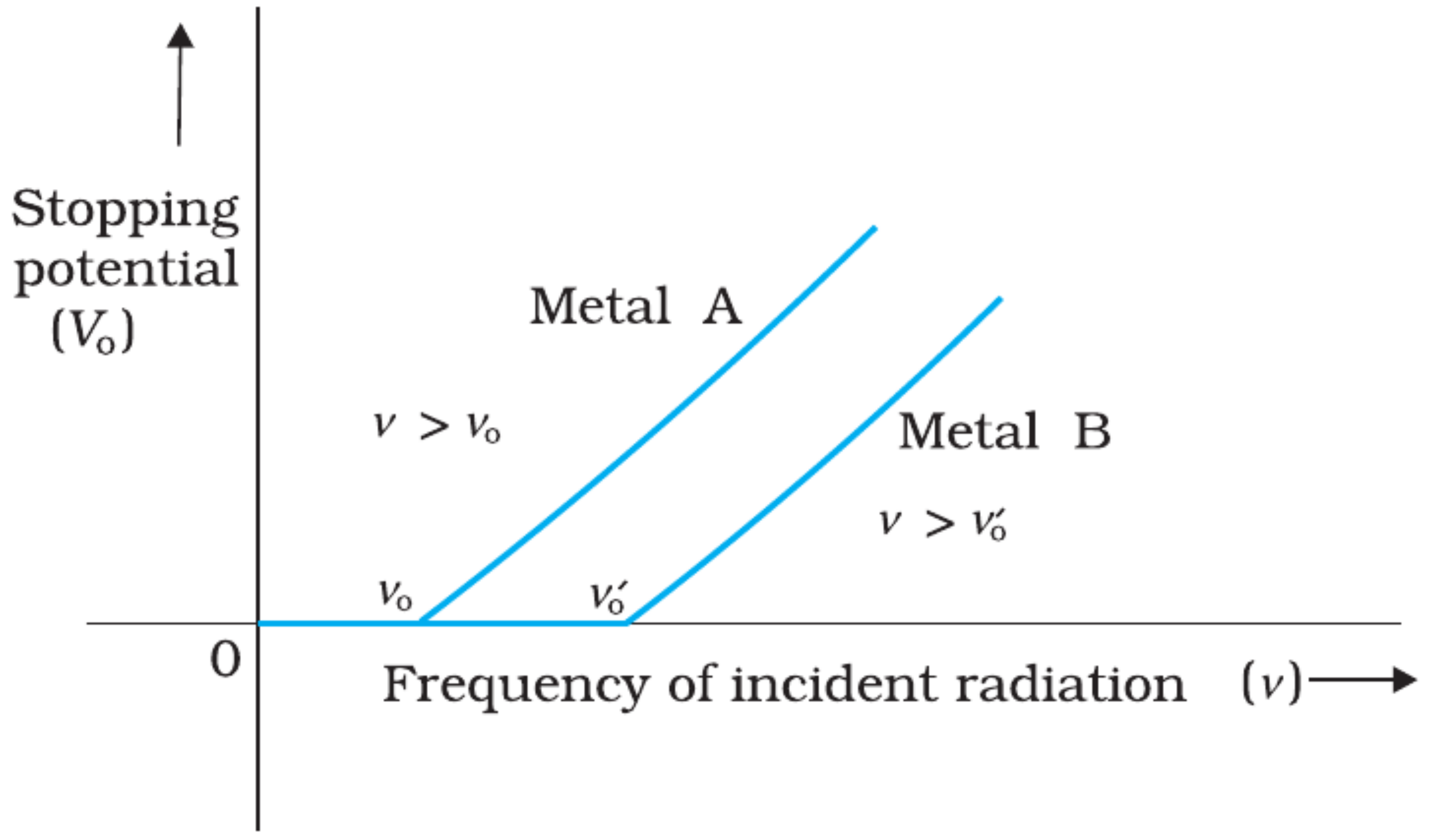


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

(ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

* These observations have two implications:

(i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

(ii) For a frequency ν of incident radiation, lower than the cut-off frequency ν_0, no photoelectric emission is possible even if the intensity is large.

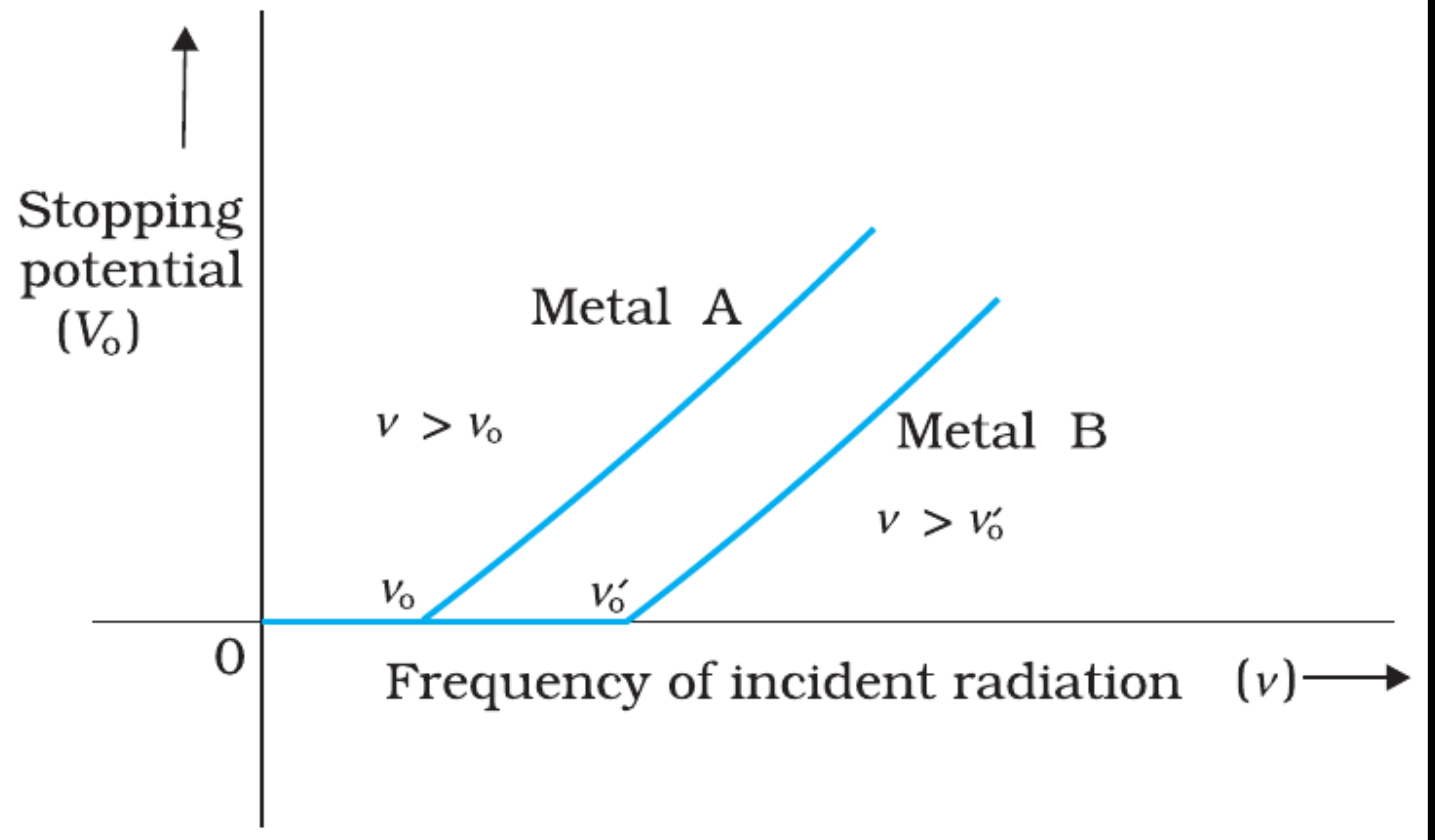


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

(ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

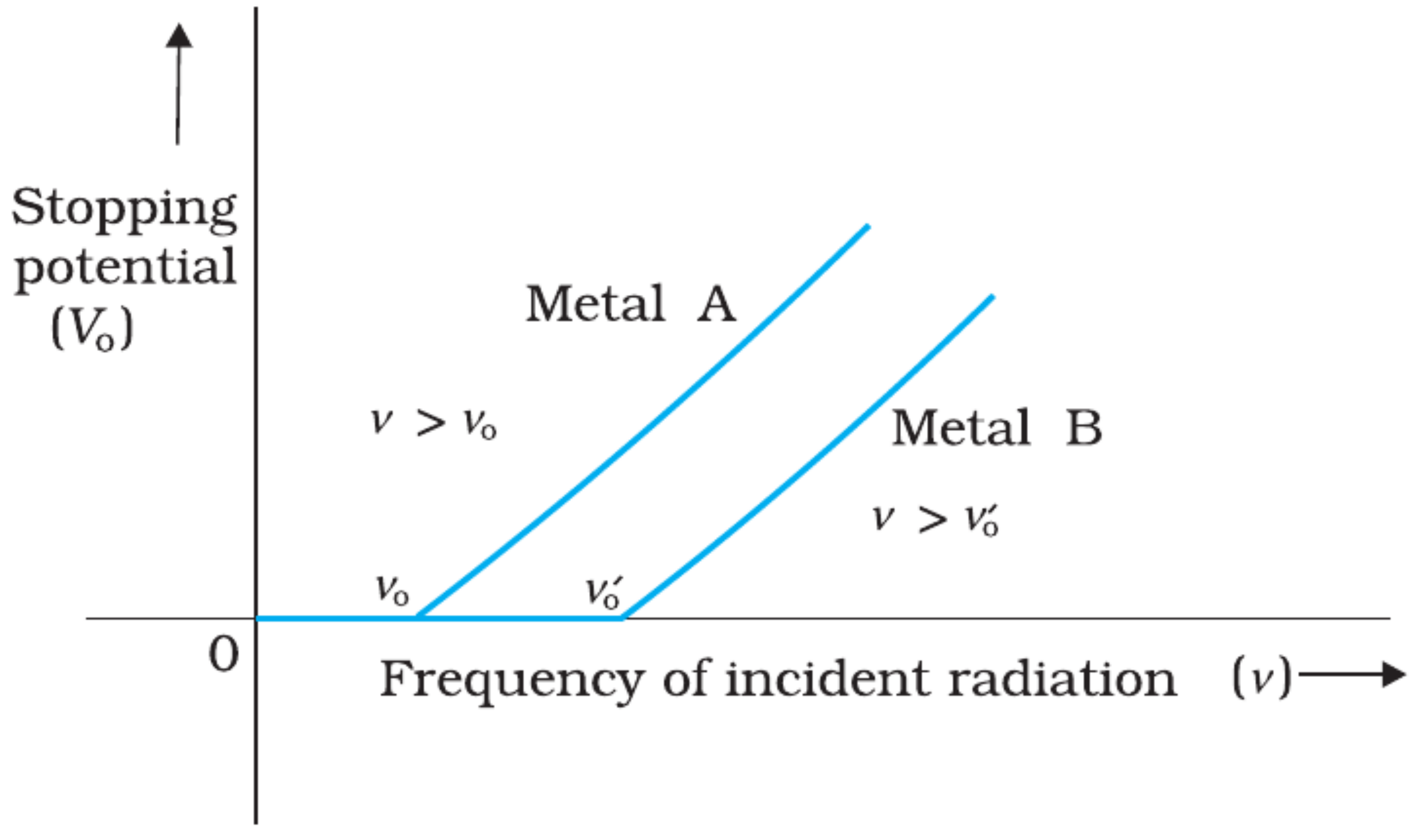


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

* These observations have two implications:

(i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

(ii) For a frequency ν of incident radiation, lower than the cut-off frequency ν_0, no photoelectric emission is possible even if the intensity is large.

* This minimum, cut-off frequency ν_0, is called the threshold frequency. It is different for different metals.

(4) Summary

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

(i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

(i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.

(ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

(i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.

(ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.

(iii) For a given photosensitive material, there exists a certain minimum cut-off frequency of the incident radiation, called the threshold frequency, below which no emission of photoelectrons takes place, no matter how intense the incident light is. Above the threshold frequency, the stopping potential or equivalently the maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident radiation, but is independent of its intensity.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

(i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.

(ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.

(iii) For a given photosensitive material, there exists a certain minimum cut-off frequency of the incident radiation, called the threshold frequency, below which no emission of photoelectrons takes place, no matter how intense the incident light is. Above the threshold frequency, the stopping potential or equivalently the maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident radiation, but is independent of its intensity.

(iv) The photoelectric emission is an instantaneous process without any apparent time lag ($\sim 10^{-9}$ s or less), even when the incident radiation is made exceedingly dim.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

* By the late 19th century, the wave nature of light was well established.

* Wave theory successfully explained:
    * Interference   * Diffraction   *Polarisation

* According to Maxwell, light is an electromagnetic wave with energy spread continuously over the wavefront.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

* By the late 19th century, the wave nature of light was well established.

* Wave theory successfully explained:
    * Interference   * Diffraction   *Polarisation

* According to Maxwell, light is an electromagnetic wave with energy spread continuously over the wavefront.

* The classical wave picture assumes:

✔ (i) Light energy is distributed uniformly across the wavefront

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

* By the late 19th century, the wave nature of light was well established.

* Wave theory successfully explained:
    * Interference * Diffraction *Polarisation

* According to Maxwell, light is an electromagnetic wave with energy spread continuously over the wavefront.

* The classical wave picture assumes:

✔ (i) Light energy is distributed uniformly across the wavefront

* The energy of electromagnetic waves spreads continuously.
* All electrons on the surface should receive energy gradually.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

* By the late 19th century, the wave nature of light was well established.

* Wave theory successfully explained:
    * Interference   * Diffraction   *Polarisation

* According to Maxwell, light is an electromagnetic wave with energy spread continuously over the wavefront.

* The classical wave picture assumes:

✔ (i) Light energy is distributed uniformly across the wavefront

* The energy of electromagnetic waves spreads continuously.
* All electrons on the surface should receive energy gradually.

✔ (ii) Higher intensity → higher energy per electron

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

* By the late 19th century, the wave nature of light was well established.

* Wave theory successfully explained:
    * Interference   * Diffraction   *Polarisation

* According to Maxwell, light is an electromagnetic wave with energy spread continuously over the wavefront.

* The classical wave picture assumes:

✔ (i) Light energy is distributed uniformly across the wavefront

    * The energy of electromagnetic waves spreads continuously.
    * All electrons on the surface should receive energy gradually.

✔ (ii) Higher intensity → higher energy per electron

    * Intensity corresponds to the square of amplitude.
    * So wave theory predicts:
        * Higher intensity should increase the maximum kinetic energy of emitted electrons.
        * More intense light should eject electrons with larger energies.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

✔ (iii) Frequency should not affect electron energy and no threshold frequency should exist.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

- ✔ (iii) Frequency should not affect electron energy and no threshold frequency should exist.

  * A light beam of any frequency, if intense enough and given enough time, should eject electrons.
  * A threshold frequency, therefore, should not exist.

- ✔ (iv) A time delay should exist.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

✔ (iii) Frequency should not affect electron energy and no threshold frequency should exist.

* A light beam of any frequency, if intense enough and given enough time, should eject electrons.
* A threshold frequency, therefore, should not exist.

✔ (iv) A time delay should exist.

* In the wave picture, the absorption of energy by electron takes place continuously over the entire wavefront of the radiation.
* Since a large number of electrons absorb energy, the energy absorbed per electron per unit time turns out to be small.
* Explicit calculations estimate that it can take hours or more for a single electron to pick up sufficient energy to overcome the work function and come out of the metal.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

✔ (iii) Frequency should not affect electron energy and no threshold frequency should exist.

* A light beam of any frequency, if intense enough and given enough time, should eject electrons.
* A threshold frequency, therefore, should not exist.

✔ (iv) A time delay should exist.

* In the wave picture, the absorption of energy by electron takes place continuously over the entire wavefront of the radiation.
* Since a large number of electrons absorb energy, the energy absorbed per electron per unit time turns out to be small.
* Explicit calculations estimate that it can take hours or more for a single electron to pick up sufficient energy to overcome the work function and come out of the metal.

* Experimental Observations (Contradictions):

(i) Kinetic energy depends only on frequency, not intensity.
(ii) Threshold frequency exists ($\nu \geq \nu_0$ required).
(iii) No time lag — emission is instantaneous ($< 10^{-9}$ s).
(iv) Low-frequency light cannot eject electrons, no matter the intensity.
(v) Intensity only changes photocurrent, not electron energy.

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

⭐ Basis of Einstein's Explanation

* Einstein used Planck's quantum theory (1900), which states:
    * Light consists of discrete packets of energy called photons (or quanta).
    * Each photon has energy: $\boxed{E = h\nu}$

      where, h = Planck's constant, ν = frequency of light

* Photons interact with electrons one-to-one (a single photon gives its entire energy to a single electron).

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

⭐ Basis of Einstein's Explanation

* Einstein used Planck's quantum theory (1900), which states:
    * Light consists of discrete packets of energy called photons (or quanta).
    * Each photon has energy: $\boxed{E = h\nu}$

      where, h = Planck's constant, ν = frequency of light

* Photons interact with electrons one-to-one (a single photon gives its entire energy to a single electron).

⭐ When a photon strikes an electron in the metal:

* Energy of photon = $h\nu$
* This energy is used in two parts:
    (i) To overcome the work function (ϕ_0).The minimum energy needed to free an electron from the metal surface.
    (ii) The electron is emitted with maximum kinetic energy = Remaining energy

* Thus, $\boxed{K_{max} = h\nu - \phi_0}$ → Einstein's photoelectric equation.

* More tightly bound electrons will emerge with kinetic energies less than the maximum value.

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

* Since, $\phi_0 = h\nu_0$ (ν_0 = threshold frequency),

* $K_{max} = h\nu - \phi_0$ becomes

$$\boxed{K_{max} = h\nu - h\nu_0}$$

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

* Since, $\phi_0 = h\nu_0$ (ν_0 = threshold frequency),

* $K_{max} = h\nu - \phi_0$ becomes

$$\boxed{K_{max} = h\nu - h\nu_0}$$

* Stopping Potential (V_0) Relation

The most energetic electrons are stopped by a retarding potential

$$\boxed{K_{max} = e\,V_0}$$

Substitute into Einstein's equation:

$$e\,V_0 = h\nu - h\nu_0 \qquad \text{Or} \qquad V_0 = h/e\,(\nu - \nu_0)$$

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

* Since, $\phi_0 = h\nu_0$ (ν_0 = threshold frequency),

* $K_{max} = h\nu - \phi_0$ becomes

$$\boxed{K_{max} = h\nu - h\nu_0}$$

* Stopping Potential (V_0) Relation

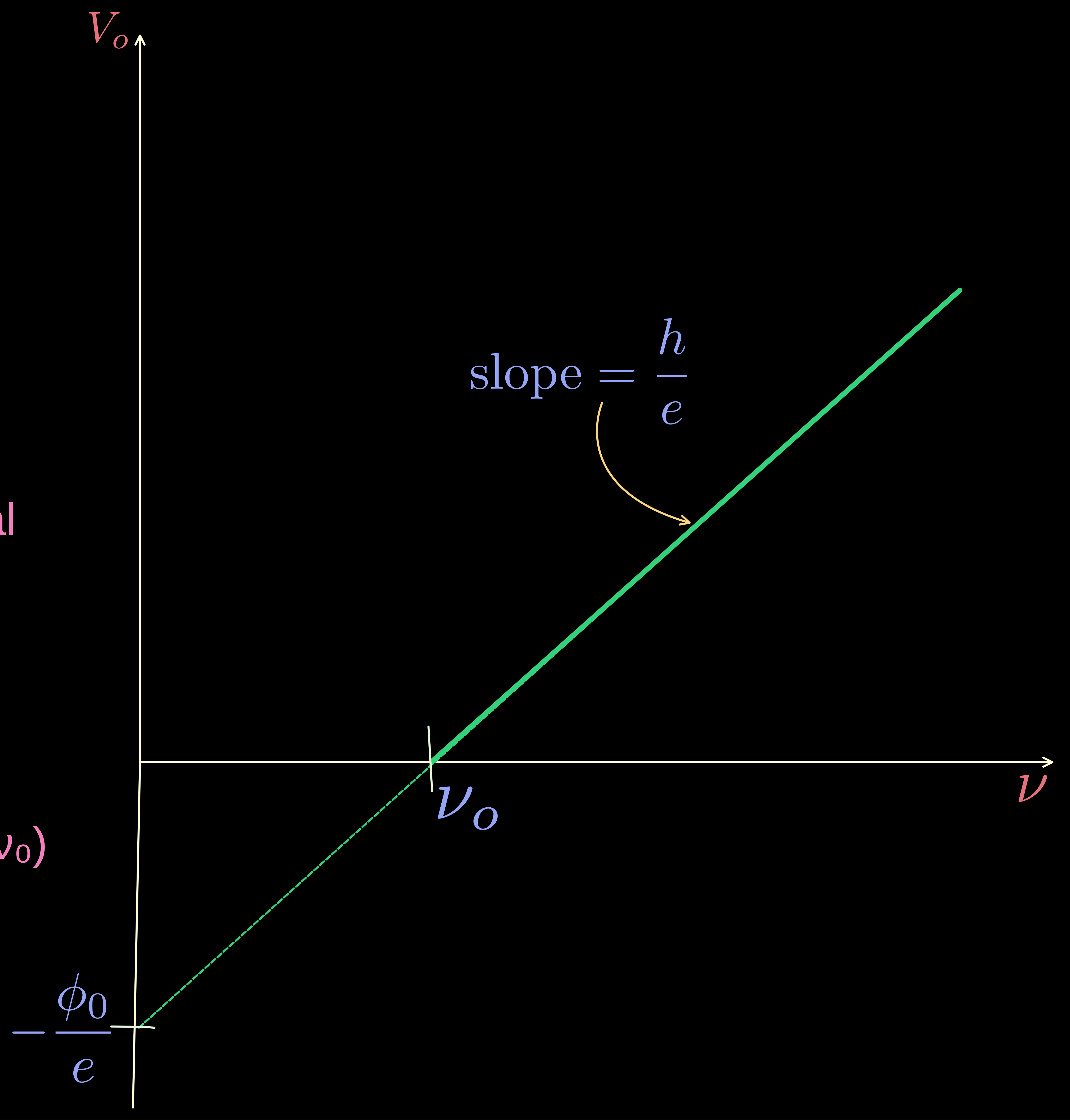
The most energetic electrons are stopped by a retarding potential

$$\boxed{K_{max} = e\,V_0}$$

Substitute into Einstein's equation:

$$e\,V_0 = h\nu - h\nu_0 \qquad \text{Or} \qquad V_0 = h/e\,(\nu - \nu_0)$$

* It predicts that the V_0 versus ν curve is a straight line with
* slope = (h/e), independent of the nature of the material.
* y-Intercept = $-h\nu_0/e = -\phi_0/e$



EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

* Millikan performed a series of experiments on photoelectric effect, aimed at disproving Einstein's photoelectric equation.

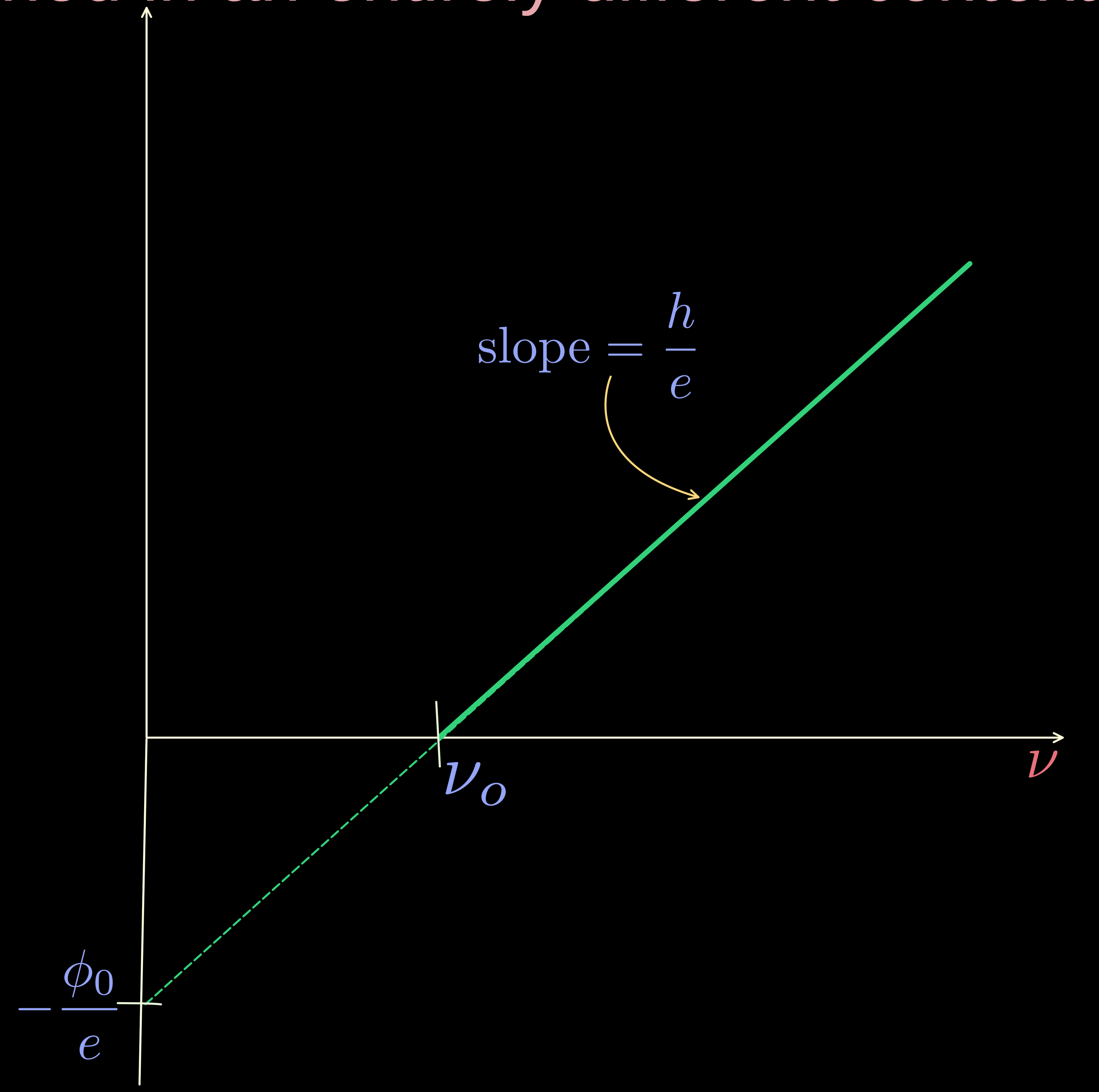
EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

* Millikan performed a series of experiments on photoelectric effect, aimed at disproving Einstein's photoelectric equation.

* He measured the slope of the straight line obtained for sodium, similar to that shown in Fig.

* Using the known value of e, he determined the value of Planck's constant h.

* This value was close to the value of Planck's constant (= 6.626 × 10^{-34} J s) determined in an entirely different context.



EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

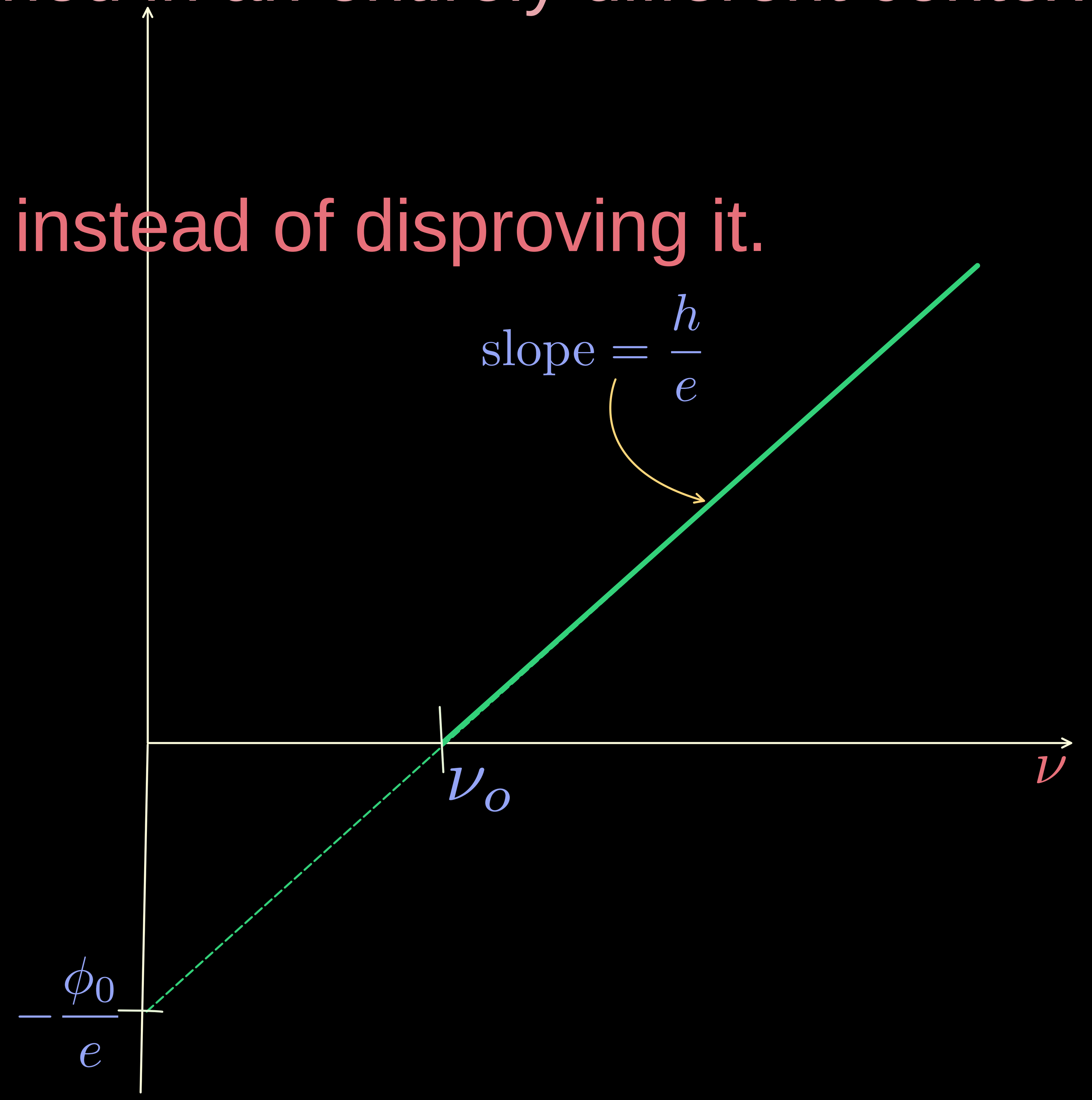
* Millikan performed a series of experiments on photoelectric effect, aimed at disproving Einstein's photoelectric equation.

* He measured the slope of the straight line obtained for sodium, similar to that shown in Fig.

* Using the known value of e, he determined the value of Planck's constant h.

* This value was close to the value of Planck's constant (= 6.626×10^{-34} J s) determined in an entirely different context.

* In this way, in 1916, Millikan proved the validity of Einstein's photoelectric equation, instead of disproving it.



PARTICLE NATURE OF LIGHT: THE PHOTON

WAVE NATURE OF MATTER: DE-BROGLIE HYPOTHESIS